

PCR-5E

Joint Standing-Carrier Closure for the Neutrino Sector

Target-indexed simultaneous closure across kinematic mass, lepton-number violation, active-sector completion, cosmological transport, and symmetry constraints



Scope of the result

PCR-5E constructs the simultaneous target, source, and physical realization problem for the completed neutrino target-extension programme. The D3-1.1 and D3-1.3 programmes remain distinct; the D4 public-summary and complete cosmological targets remain distinct; and all seven D5 symmetry/charge targets retain their target tags. Two D3-1.3/public-summary vectorlike-joint comparison classes are closed exactly: the bounded-accidental and exact-global branches have proper-continuum base images, continuum complements, continuum enriched fibres, and non-singleton joint quotients. The unresolved parent branches retain their explicit NotReady or ROOS status and are not treated as eliminated.

Abstract

PCR-5B identifies the standing-carrier quotient of the fixed neutrino-oscillation target as an exact mixed moduli object with at least continuum many Standing Carrier Identity Closure (SCIC) classes, and PCR-5C supplies the functorial calculus for target extension, connected lift fibres, no merger, composition, path equivalence, source sufficiency, and joint obstruction. PCR-5D1 through PCR-5D5 then determine how direct kinematic mass, lepton-number-violating response, heavy-state and active-sector closure, cosmological population and transport, and symmetry/charge conditions act on that base. PCR-5E asks the corresponding simultaneous question: which of these marginally lawful structures can coexist on one source-coherent physical realization?

The joint target is necessarily target-indexed. The original D3 programme THUCL-D3-1.1 and its quantitative successor THUCL-D3-1.3 remain distinct; the D4 public-95 summary target is not identified with the complete cosmological likelihood target; and PCR-5D5 contributes seven non-equivalent symmetry/charge targets. The resulting family therefore contains twenty-eight parent branches

$$\tau = (\tau_3, \tau_4, \tau_5) \in \{1.1, 1.3\} \times \{\text{pub95, complete}\} \times \mathcal{F}_{D5}.$$

For each branch we construct a typed target colimit T_E^τ , a conservative source amalgamation S_E^τ , an activity-reduced common-source category, a one-apex physical groupoid N_E^τ , a joint restriction functor R_J^τ , connected homotopy fibres $\mathcal{L}_J^\tau(q)$, a complete obstruction normal form, a standing-coordinate incidence relation, a no-repair source-sufficiency structure, and the quotient $Q_J^\tau = \pi_0 N_E^\tau$. Marginal agreement on the oscillation base is necessary but not sufficient: all shared-parameter, response, anomaly, vacuum, background, matching, continuation, and higher-arity residuals must vanish on the same physical apex.

A sequence of auxiliary closure results fixes the remaining simultaneous-compatibility burdens. The D3 successor branch is admitted as an additional target-tagged PCR-5E branch without altering D3-1.1. Semantic joint uses are defined before the finite cross-relation presentation and are shown to factor through the complete target/source incidence trace. The source-realized vectorlike Dirac family used by D5 is then shown to land in the complete D3-1.3 H/U/L and process target on a uniform positive product region, with exact heavy-sector CLFV cancellation, retained-oscillation transparency, and the required matching and continuation data. The corresponding source-qualified comparison classes are fixed before the genuinely joint D1/D2/D4 calculation. The D1/D4 common response is represented by a canonical complete-trace quotient with a universal factor property. Finally, the frozen PCR-5A/PCR-5B matching generators factor through the exact composite base reduct, so variation of the independently authorized common D1/D4 scale parameter τ preserves all seven base SCIC coordinates while changing the enriched D1/D4 response.

The positive result occurs in two D3-1.3/public-summary comparison classes. For bounded accidental $B - L$, the D2 amplitude retains a nondegenerate accepted interval; for exact global $B - L$, the physical D2 amplitude is restricted to its defined zero subfibre. In both cases the vectorlike-joint base image is a proper subset of $Q_{\text{osc}}^{\text{SCIC}}$, contains at least continuum many classes, has a continuum complement supplied by the D3-1.3 finite-unitary deficit family, and supports at least continuum many enriched lifts over each fixed vectorlike-joint base class. Hence neither completed joint quotient is singleton, and the one-role terminal Role-Occupancy Closure diagnosis is over-residue.

The standing-coordinate analysis further shows that D1 and D4 descend through a canonical minimal common mass-response generator at this scope, that D2 and exact-global $B - L$ are obstruction-coupled rather than commonly generated, and that the decided D3 and D5 coordinates are independent. No all-phenomenon common generator and no emergent mixed-carrier selector is established. At parent scope, complete-D4 branches remain NotReady_{JLR-D4-1}, D3-1.1 branches remain NotReady_{GFC-D3-1.1}, and unresolved D3/D5 sectors retain their typed ROOS or NotReady status. Thus PCR-5E closes two exact target-and-comparison-qualified joint sectors while preserving the unresolved parent theory as an explicit open boundary.

Keywords. AASC; neutrino oscillation; standing carrier; SCIC; target extension; homotopy fibre; joint constraint closure; one-apex realization; vectorlike Dirac source; lepton-number violation; cosmological transport; $B - L$; Role-Occupancy Closure.

Principal theorem

Target-indexed joint closure and exact readiness boundary

Let $\mathcal{F}_{D5}$ be the seven-member PCR-5D5 target family and let

$$\mathfrak{B}_E = \{1.1, 1.3\} \times \{\text{pub95}, \text{complete}\} \times \mathcal{F}_{D5}.$$

For every $\tau \in \mathfrak{B}_E$, there is a canonically formed parent branch certificate

$$\mathcal{H}_E^\tau = (T_E^\tau, S_E^\tau, \text{PhysMod}_E^\tau, R_J^\tau, \rho_J^\tau, \{\mathcal{L}_J^\tau(q)\}_{q \in \mathcal{Q}_{\text{osc}}^{\text{SCIC}}}, I_\nu^\tau, N_E^\tau, \partial\text{Fail}_E^\tau, \text{Cert}_E^\tau),$$

with the following properties.

- (i) ARC-U1 authorizes the target-indexed D3 input family $\{\text{THUCL-D3-1.1}, \text{THUCL-D3-1.3}\}$ before branch formation; the 1.3 branches are additional successor branches and do not complete, supersede, or strengthen 1.1.
- (ii) T_E^τ is the typed colimit of the marginal targets in the slice over T_{osc} ; target-version coercions are excluded.
- (iii) The semantic joint trace is formed before the finite cross presentation, and E-S1 proves that X1–X9 represent every target-active marginal, pairwise, and finite higher-arity use at the declared source scope.
- (iv) S_E^τ is generated before joint evaluation from the B15 source theory and the five marginal source extensions.
- (v) PhysMod_E^τ consists exactly of activity-reduced source objects whose complete inherited, marginal, cross, background, matching, continuation, and common-cone residual family vanishes on one physical apex.
- (vi) Joint restriction descends to SCIC quotients and satisfies no merger:

$$[x] \neq [y] \text{ in } \mathcal{Q}_{\text{osc}}^{\text{SCIC}} \implies \rho_J^{-1}([x]) \cap \rho_J^{-1}([y]) = \emptyset.$$

- (vii) For every base class q , the joint fibre is the obstruction equalizer

$$\mathcal{L}_J^\tau(q) \simeq \text{Eq}\left(\mathcal{P}_{\text{marg}}^\tau(q) \rightrightarrows \mathcal{O}_E^\tau(q)\right).$$

- (viii) The global quotient is the codescent union

$$\mathcal{Q}_J^\tau \cong \coprod_{q \in \mathcal{Q}_{\text{osc}}^{\text{SCIC}}} \pi_0 \mathcal{L}_J^\tau(q),$$

with every unresolved complement retained as `Undef`, `ROOS`, `SU`, `TU`, or `NotReady`, never as an empty fibre.

At parent scope,

$$\tau_4 = \text{complete} \implies \text{Status}_E(\tau) = \text{NotReady}_{\text{JLR-D4-1}},$$

$$\tau_4 = \text{pub95}, \tau_3 = 1.1 \implies \text{Status}_E(\tau) = \text{NotReady}_{\text{GFC-D3-1.1}},$$

$$\tau = (1.3, \text{pub95}, \tau_5), \tau_5 \in \{T_{\text{acc}}^{(4)}, T_{\text{glob}}\} \implies \text{Status}_E(\tau) = \text{ROOS}_{\text{unclosed parent}} \oplus \text{NotReady}_{\text{parent charts}},$$

$$\tau = (1.3, \text{pub95}, \tau_5), \tau_5 \notin \{T_{\text{acc}}^{(4)}, T_{\text{glob}}\} \implies \text{Status}_E(\tau) = \text{NotReady}_{\text{D3D5}(\tau_5)}.$$

For the independently frozen vectorlike-joint target-comparison certificates,

$$\text{Status}_E(T_{E, \text{VJ}}^{\tau_5}) = \text{TargetComparisonQualifiedProperContinuum}_{\text{VJ}}, \quad \tau_5 \in \{T_{\text{acc}}^{(4)}, T_{\text{glob}}\},$$

and

$$|\text{im } \rho_{J, \text{VJ}}^{\tau_5}| \geq \mathfrak{c}, \quad |\mathcal{Q}_{\text{osc}}^{\text{SCIC}} \setminus \text{im } \rho_{J, \text{VJ}}^{\tau_5}| \geq \mathfrak{c}, \quad |\mathcal{Q}_{J, \text{VJ}}^{\tau_5}| \geq \mathfrak{c}.$$

For every fixed $(m_0, \lambda) \in I_m^\circ \times I_E$, E-S7 gives a single base class $q_{m_0, \lambda}$ carrying an injected I_τ° -continuum of absolute-scale lifts, with equality of $T^{\text{face}}, R^{\text{stand}}, \Delta^{\text{del}}, C^{\text{act}}, M^{\text{match}}, B^{\text{bdy}}, F^{\text{same}}$ throughout the leaf. The D1/D4 shared response has the canonical minimal generator $\mathbf{G}_{\beta\Sigma}$. Thus no completed parent or vectorlike-joint target is singleton, and the parent PCR-5E certificate does not license a universal PCR-6 physical-nature conclusion.

Scope and interpretation

Established results

The paper establishes the target-indexed simultaneous construction, semantic saturation of the cross-relation system, source-first D3-1.3/D5 vectorlike realization, exact one-apex joint physicality, seven-channel base conservativity of the common D1/D4 scale extension, proper-continuum joint images for the bounded-accidental and exact-global comparison classes, the canonical D1/D4 common-generator theorem, the standing-coordinate relations, the enriched singleton criterion, and the corresponding Role-Occupancy Closure conclusions.

Open boundaries

The paper does not complete D3-1.1, reconstruct the missing complete D4 cosmological likelihood, classify the full finite-operational D3-1.3 source universe, determine the five unresolved D5 joint comparisons, or infer terminal carrier identity from spectral, symmetry, or response multiplicity. Those questions remain outside the completed scope proved here.

The logical order is fixed by the target hierarchy: target authorization and source specification; semantic joint trace; D3/D5 common-source realization; simultaneous D1/D2/D4 evaluation on the same physical object; fixed-base restriction; fibre reconstruction; standing-coordinate analysis; SCIC, cardinality, and Role-Occupancy Closure. Supplementary formal records document provenance and reproducibility, but the scientific conclusions rest on the definitions and proofs in the manuscript.

Summary of results

Claim class	Status	Meaning
Joint target/source/physical construction	Established	E00–E06 construct the typed target family, common source theory, one-apex groupoid, obstruction normal form, restriction, fibres, and codescent object at each declared scope.
D3 successor-branch authority	Established	ARC-U1 admits D3-1.3 as an additional target-tagged PCR-5E branch while leaving D3-1.1 and all unqualified GFC burdens unchanged.
Semantic cross-relation saturation	Established	E-S1 derives the complete finite cross grammar from the semantic target/source trace rather than a prescribed finite list.
Joint source normal form	Established	E-S2 reconstructs every semantic joint source presentation after activity reduction.
Common vectorlike D3/D5 source	Established	E-S4 lands the D5 exact-response source in every applicable D3-1.3 H/U/L/process coordinate on one apex and retains the exact typed complement.
Vectorlike-joint accidental and exact-global sectors	Established at the stated target and comparison scope	Explicit one-apex continua have proper-continuum joint images, continuum complements, and continuum joint quotients.
D1/D4 minimal common generator	Established at vectorlike-joint scope	E-S6 proves the canonical complete-trace quotient and its universal factor property.
Absolute-scale base conservativity	Established at vectorlike-joint scope	E-S7 proves equality of all seven PCR-5B packet coordinates along the τ -leaf at fixed intrinsic vectorlike source, while the D1/D4 enriched signatures remain nonconstant.
Complete D4 parent branches	NotReady	The fresh correlated theory-to-likelihood evaluator required by JLR-D4-1 is absent.

Claim class	Status	Meaning
D3-1.1 parent branches	NotReady	The original Global Fibre Completion programme remains open.
D3-1.3 parent remainder	ROOS/NotReady	The vectorlike-joint chart is closed directly; the larger finite-operational/joint atlas is not.
Five unresolved D5 comparison rows	NotReady	One or more target-faithful deformation, matching, stabilizer, faithful-remnant, or cardinality obligations remains open.
Singleton joint carrier	Not established	Every positive completed target-comparison sector contains at least continuum many joint classes.
Common all-phenomenon generator	Not established	D1/D4 share a canonical mass generator, D2/D5 exact-global form an obstruction, and D3/D5 remain independent on the decided rows.
Downstream PCR-6 analysis	Target-and-comparison-qualified only	E-H1–E-H10 are complete for the two solved vectorlike-joint sectors; the parent neutrino sector remains NotReady.

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1 Introduction

1.1 From marginal target extensions to a simultaneous neutrino problem

The Physical Carrier Realization programme is organized as an inverse-realization analysis. PCR-1 fixes the physical work required by the observed oscillation interface before a microscopic occupant is named; PCR-2 reconstructs the complete target-facing interface; PCR-3 generates the source-first carrier universe; PCR-4 imposes simultaneous physical consistency; and PCR-5 through PCR-5B determine the standing-carrier quotient of the oscillation target. PCR-5C then supplies the functorial calculus by which that base can be extended without rewriting it. The D1–D5 papers apply this calculus to five distinct classes of neutrino observables.

Those marginal results do not by themselves define a simultaneous neutrino realization. D1 introduces direct-kinematic response, D2 an exact LNV amplitude, D3 heavy-state and active-closure structure, D4 population and background history, and D5 target-specific charge and symmetry data. Even when all five enriched objects restrict to the same oscillation SCIC class, they may disagree in source incidence, matching, vacuum, background, or continuation structure. Agreement on the base is therefore necessary but not sufficient for physical co-instantiation.

PCR-5E addresses this remaining problem. Its task is to form the simultaneous target, derive all target-authorized cross relations, construct a common source, require one physical apex, classify the resulting obstructions, and determine the induced joint standing-carrier quotient. The outcome is not assumed in advance: a joint branch may be empty, singleton, finite, continuous, functional, categorical, mixed, or remain outside the completed response scope.

1.2 Four distinctions that control the analysis

Three non-equivalences are immediate:

$$\begin{aligned} \text{marginal survival} &\not\Rightarrow \text{joint survival}, \\ \bigcap_{\alpha} \text{im } \rho_{\alpha} &\not\equiv \text{im } \rho_J \quad \text{without one-apex amalgamation}, \\ |\text{im } \rho_J| = 1 &\not\Rightarrow |Q_J| = 1. \end{aligned}$$

The first is physical: separately admissible processes need not coexist. The second is source-theoretic and categorical: a set-theoretic intersection forgets the common-base groupoid, source equations, isotropy, and cross residuals. The third is geometric: a single surviving base class can split into many enriched joint classes.

A fourth distinction separates structural closure from realized trace. A target-relative quotient is determined by the target and its standing-bearing continuation structure. A later empirical trace may characterize a surviving class, but it cannot retroactively change the earlier quotient unless that trace was already part of the target certificate.

1.3 Target indexing

PCR-5E is intrinsically target-indexed. The original D3 target THUCL-D3-1.1 and the quantitative successor THUCL-D3-1.3 differ in acceptance semantics and process scope and therefore remain distinct. The D4 public-summary target is likewise distinct from the complete cosmological likelihood target. Finally, D5 separates seven non-equivalent symmetry and charge targets,

$$\mathcal{F}_{D5} = \left\{ T_{\text{acc}}^{(4)}, T_{\text{bg}, \text{min}}, T_{\text{loc}, 3\text{N}}, T_{\text{glob}}, T_{\text{exp}}[O], T_{\text{sp}}[q\Phi], T_{\text{rem}}[\Gamma, q\Phi] \right\}.$$

The simultaneous family consequently contains twenty-eight branches, $\tau = (\tau_3, \tau_4, \tau_5)$, and every branch-sensitive statement below retains this tag.

1.4 Structure of the proof

The proof proceeds through the following stages.

1. **Target and import discipline.** Fix branch identity, applicability, uncertainty semantics, source scope, and continuation data before joint evaluation.

2. **Joint target formation.** Construct the typed colimit of marginal targets over the oscillation base and derive the complete cross-relation system from semantic joint uses.
3. **Common-base geometry.** Form the homotopy pullback of marginal lift groupoids over a concrete PCR-5B component.
4. **Common source and one-apex physicality.** Amalgamate the source theories, reduce source-null incidence, and impose the complete simultaneous residual system on one semantic physical object.
5. **Restriction and fibre reconstruction.** Prove base conservativity, SCIC descent, no merger, exact obstruction, connected fibres, and codescent.
6. **Source composition.** Determine no-repair minimal supports, redundancy, synergy, and higher-arity source dependence.
7. **Standing-coordinate structure.** Construct the phenomenon-to-coordinate incidence relation and distinguish common generation, obstruction, and independent standing coordinates.
8. **Selection and realization.** Determine base support, enriched singletonhood, quotient geometry, realized-trace status, and Role-Occupancy Closure.

1.5 Positive vectorlike-joint sectors

Two simultaneous sectors can be closed at the D3-1.3/public-summary target. The bounded-accidental sector retains a nondegenerate accepted D2 amplitude interval while preserving the $d \leq 4$ accidental- $B - L$ condition. The exact-global sector instead restricts the LNV amplitude to its defined zero subfibre while retaining absolute scale, cosmological mass sum, heavy matching threshold, and active-closure information.

Both sectors use the same positive light spectral object in D1 and D4 and the same source-realized vectorlike Dirac threshold family in D3 and D5. The common D3/D5 source is established on a uniform positive parameter region before the D1/D2/D4 simultaneous calculation. D4 heavy nonparticipation is imposed only through the frozen source-activity semantics, not through numerical smallness. A separate base-conservativity theorem then shows that the common D1/D4 scale direction lies inside one PCR-5B fibre: all seven base-packet coordinates remain fixed while the enriched D1/D4 response varies.

The resulting joint base images are proper continuum subsets of the oscillation quotient and have continuum complements. Each fixed vectorlike-joint base class also supports continuum many enriched joint lifts. The positive quotients are therefore non-singleton, although the larger parent atlases remain incompletely classified.

1.6 Interpretive boundary

PCR-5E determines joint target-relative carrier structure; it does not yet determine physical nature. Questions such as whether a surviving realization is particle-like, law-only, composite, topological, hidden, or a common carrier bearing several spectral sectors belong to the subsequent physical-identity analysis. Such a characterization may classify members of the joint residue but may not alter its membership, merge SCIC-distinct classes, or convert an unresolved branch into a theorem.

Nonclaim 1.1 (No Yukawa or metric backflow). Neither numerical masses, displayed mixing matrices, Yukawa coefficients, nor Higgs-scale representatives are used to form the PCR-5E carrier identity relation. They enter only through already-authorized response, matching, or projection records. AASC governance and carrier standing remain logically prior to those presentations.

2 Foundational dependencies and target separation

2.1 Fixed dependency spine

The scientific dependency order is

$$\text{PCR-1} \longrightarrow \text{PCR-2} \longrightarrow \text{PCR-3} \longrightarrow \text{PCR-4} \longrightarrow \text{PCR-5/5A/5B/5C} \longrightarrow \text{D1-D5} \longrightarrow \text{PCR-5E} \longrightarrow \text{PCR-6}.$$

Each arrow carries a prohibition as well as an import. Later stages may use the established object exported by an earlier stage; they may not reinterpret the earlier target, rebuild its quotient with a preferred representative, or

backfill a source constructor after seeing the endpoint.

The imported roles are as follows.

Stage	Imported object	Use and authority boundary
PCR-1	Target-fixed oscillation role complex ($r_{\triangleright\Delta}, r_{\Delta}; r_{\Delta} \succ r_{\triangleright\Delta}$)	Fixes physical work and empirical boundary without microscopic occupant or carrier count.
PCR-2	Complete carrier-neutral interface and semantic evaluator algebra	Supplies target-facing trace, shared-incidence requirements, lawful equivalence, and interface adequacy; does not select a carrier.
PCR-3	Source-first carrier theory S_{car} , activity-reduced trace χ_{car}	Supplies candidate-complete source grammar at the declared finite presentation scope; downstream survival cannot generate missing source families.
PCR-4	Simultaneous physical survivor groupoid	Supplies the one-object/common-cone rule and complete residual normal form; componentwise pass is insufficient.
PCR-5/5A	Internal occupancy and seven-channel SCIC packet	Separates factorization from terminal identity and fixes the standing packet Ξ_{osc} .
PCR-5B	Exact groupoid-valued oscillation atlas $\mathcal{A}_{\text{osc}}^{\text{exec}} \simeq \mathcal{N}_{\text{osc}}$	Freezes source strata, solution objects, lawful groupoids, overlaps, failure frontiers, and at least continuum many SCIC classes.
PCR-5C	Target-extension pseudofunctor and joint calculus	Supplies restriction, no merger, connected fibres, composition, path equivalence, joint pullback, obstruction, source sufficiency, and codescent.
D1–D5	Target-tagged marginal marginal data	Supply exact marginal source/physical/fibre data only at their declared targets.
SCIC	Complete continuation-signature quotient	Fixes the unique same-scope carrier identity relation after activation.
ROC	Compatible role-occupancy criterion	Licenses realization only from a target-derived role set and a singleton compatible assignment orbit.

2.2 E00: target identity and import discipline

Definition 2.1 (Branch index). The PCR-5E branch set is

$$\mathfrak{B}_E = \{1.1, 1.3\} \times \{\text{pub95, complete}\} \times \mathcal{F}_{D5}.$$

A branch is written $\tau = (\tau_3, \tau_4, \tau_5)$.

Definition 2.2 (Branch import certificate). The branch import certificate TauCert_E^τ is the tuple

$$\begin{aligned} \text{TauCert}_E^\tau = & (T_{\text{osc}}, \mathcal{N}_{\text{osc}}, \mathcal{S}_{B15}, \Xi_{\text{osc}}, \{H_{D\alpha}\}_{\alpha=1}^5, \\ & \tau_3, \tau_4, \tau_5, \text{App}^\tau, \text{Unc}^\tau, \text{U}^\tau, \text{Versions}^\tau, \text{Authority}^\tau). \end{aligned}$$

It records target fingerprints, source identifiers, applicable evaluators, uncertainty objects, continuation monoids, theorem versions, and every `NotReady`, `ROOS`, `source-update`, or `target-update` boundary.

Theorem 2.3 (E00 – target/version/import lock). *For every branch $\tau \in \mathfrak{B}_E$, the certificate TauCert_E^τ can be formed before the joint target and before any joint survivor is inspected. Every PCR-5E object constructed below factors through this certificate. If two proposed constructions disagree on a target fingerprint, source comparison class, applicability rule, uncertainty object, continuation monoid, or theorem version, they belong to different branch certificates and cannot be combined as same-scope data.*

Proof. The required fields are exported by the frozen base and the marginal handoff objects. Their values are logically prior to PCR-5E: D3 programme identity is fixed by its target certificate; D4 summary and complete

objects are separately named; D5 target identity is indexed by the seven target records. The branch certificate is therefore a product of already formed identifiers and authority tags, not a function of the joint residue.

Suppose two constructions disagree on one listed field. A disagreement in target fingerprint changes the object of the target category. A disagreement in source scope changes the candidate domain. A disagreement in applicability or uncertainty semantics changes the evaluated fragment. A disagreement in continuation changes the complete standing signature. By fixed-domain discipline, each is a scope change rather than a lawful representative change. Hence the two constructions cannot inhabit one same-scope branch. Conversely, every later symbol in the manuscript is defined with an explicit τ tag or is inherited from the common oscillation base, so it factors through TauCert_E^τ . $\square$

Corollary 2.4 (No target laundering). *The following substitutions are invalid:*

1. using the D3-1.3 proper-image theorem while reporting a D3-1.1 result;
2. using the D4 public mass-sum band as the complete cosmological likelihood;
3. replacing the seven D5 targets by an untyped generic $B - L$ condition;
4. treating a target-qualified subcategory as the parent comparison class.

2.3 D3 programme separation

The D3-1.1 target supplies a source-first prompt-HNL continuum and a marginal active-closure obstruction, but it does not provide the complete global fibre atlas, a proper-base-image theorem, or fixed-base splitting. D3-1.3 adds a universal process-tensor tester and a finite-operational comparison rule. It proves a proper-continuum base image at full finite-operational scope and closes the RCF-finite atlas, while retaining the non-RCF-finite remainder as ROOS.

Proposition 2.5 (Permanent D3 programme separation). *No theorem whose proof uses the D3-1.3 quantitative coverage rule, retained tester, threshold $c_e^{\min} = 0.987$, or finite-operational comparison class transports to D3-1.1. Conversely, incompleteness of D3-1.1 does not invalidate the target-qualified 1.3 theorems.*

Proof. The two targets differ in admitted records, acceptance semantics, and comparison class. These are target fields rather than presentations. The fixed-domain no-repair rule therefore classifies their relation as an explicit target update, not a lawful redescription. The two directions of the statement follow immediately. $\square$

2.4 ARC authorization of the D3 successor branch

The PCR-5D3 Programme Separation Memorandum correctly left the original Mini-ARC unchanged and required a separate explicit architecture update before the successor target could enter PCR-5E. The *PCR-5E D3 Successor-Branch ARC Target-Update Memorandum*, version 1.0.0, now supplies that authority [Mal26t; Mal26u]. It updates the D3 input of PCR-5E from one unindexed leg to the target-indexed family

$$\mathcal{F}_{D3}^E = \{\text{THUCL-D3-1.1}, \text{THUCL-D3-1.3}\},$$

without replacing either programme by the other.

Theorem 2.6 (ARC-U1 – successor-branch admission with permanent programme separation). *The twenty-eight-branch PCR-5E target family is research-architecture authorized. Every branch whose D3 field is THUCL-D3-1.3 is an additional successor branch, not a completion or supersession of the original THUCL-D3-1.1 Mini-ARC programme. In particular:*

1. the original 1.1 Global Fibre Completion target, labels, and readiness burden remain unchanged;
2. every 1.3 theorem retains its target tag, process-scope rule, continuation monoid, and comparison class;
3. no 1.3 result transports backward into a 1.1 branch;
4. no 1.1 incompleteness invalidates a target-qualified 1.3 branch;
5. the branch family is fixed before any PCR-5E survivor, image, fibre, cardinality, or PCR-6 endpoint is inspected.

Proof. The target-update memorandum records the provenance of the 1.3 certificate, the changed dependency graph, the revised D3 handoff, and endpoint independence. Its update is a coproduct of two separately tagged target objects, not a coercion between them. Every PCR-5E object below factors through the exact branch fingerprint of Theorem 2.3, so the added 1.3 authority cannot alter any 1.1 object or conclusion. Conversely, the 1.3 branch is now explicitly admitted by the ARC rather than merely present as a mathematically definable target. Since the update document is formed only from upstream target records and programme locks, it contains no joint endpoint backflow. $\square$

Corollary 2.7 (D3 branch citation and handoff rule). *A PCR-5E theorem using the retained 1.3 process tester or quantitative-coverage rule must cite both the successor-target manuscript and ARC-U1, and must report its result as THUCL-D3-1.3-qualified. Unqualified GFC labels remain reserved for the 1.1 programme.*

2.5 D4 two-level separation

The D4 public-summary target contains one exact released scalar band, a standard-thermal source compiler, a mass-indexed background family, and a one-apex transparency theorem. The complete cosmological target requires a total theory-to-observable-to-correlated-likelihood map that is not present.

Proposition 2.8 (Permanent D4 two-level separation). *For every branch τ , replacing T_{D4}^{complete} by T_{D4}^{pub95} changes T_E^τ . A theorem on the summary branch cannot discharge JLR-D4-1.*

Proof. The complete target contains an evaluator and covariance geometry absent from the summary object. The summary target contains only membership in the released scalar band together with source/physical/provenance conditions. Since target-admitted evaluators determine standing and failure, the objects are non-isomorphic in the target category. No same-scope substitution exists. $\square$

2.6 D5 target separation

Proposition 2.9 (Seven non-equivalent D5 targets). *The seven D5 targets are pairwise non-identical as target objects. Their overlaps are legitimate multi-target comparisons, but no one target can be coerced into another by dropping its action, anomaly, protector, breaking, stabilizer, remnant, matching, or incidence data.*

Proof. Each target contains a different target-defining predicate and witness packet. For example, accidental conservation through $d \leq 4$ neither supplies an exact global protector nor a local anomaly packet; spontaneous breaking contains a vacuum orbit and stabilizer absent from explicit breaking; a remnant target contains the faithful quotient and operator congruences. Deleting such fields weakens or changes the target, not merely its presentation. $\square$

2.7 Exact nonclaims

The following are fixed scope boundaries for the entire manuscript.

Nonclaim 2.10 (No marginal intersection theorem). The joint image is not defined as $\bigcap_\alpha \text{im } \rho_{D\alpha}$. The intersection is only a necessary base-support condition.

Nonclaim 2.11 (No null-to-zero inference). The D2 empirical null search does not by itself supply exact physical zero or exact lepton-number conservation. Exact zero appears only on a source-physical subfibre proved to have zero amplitude.

Nonclaim 2.12 (No ordering selection). The D4 public-summary pullback does not instantiate a pre-lock realized mass-ordering trace. Ordering-sensitive scalar-band disposition and realized branch selection remain distinct.

Nonclaim 2.13 (No spectral-to-terminal promotion). D5 connected spectral charge uniformity is a symmetry-bearer statement. It neither multiplies terminal carriers nor proves a one-carrier/multiple-sector endpoint.

Nonclaim 2.14 (No computational authority substitution). Hash checks, exact arithmetic scripts, formula validators, and JSON ledgers certify package identity and encoded consistency. They do not prove source completeness, one-apex physicality, or the semantic exhaustion of a branch.

3 Marginal target extensions and inherited results

3.1 Canonical marginal data

For each marginal leg $\alpha \in \{D1, D2, D3, D4, D5\}$, PCR-5E requires the reconstructive object

$$H_\alpha = (T_\alpha, S_\alpha, \text{PhysMod}_\alpha, R_\alpha, \rho_\alpha, \{\mathcal{L}_\alpha(q)\}_{q \in Q_{\text{osc}}^{\text{SCIC}}}, Q_\alpha, \text{Sep}_\alpha, \text{SourceComp}_\alpha, \text{Overlap}_\alpha, \partial\text{Fail}_\alpha, \text{Prov}_\alpha, U_\alpha, \text{Status}_\alpha).$$

A headline base image or cardinality statement is not a sufficient downstream interface. The source category, physical realization, restriction, fibres, overlaps, failure/update frontier, and continuation data must be present or explicitly marked NotReady.

3.2 D1: direct kinematic mass

D1 extends the oscillation target by the KATRIN endpoint response and the conservative source extension SOE-D1-1. Its source-authorized hyperprocess envelope preserves every PCR-3 incidence and has exact base reduct. The retained oscillation evaluator factors through the enriched object with the endpoint cells present.

The target-level conclusions are

$$\text{im } \rho_{D1} = Q_{\text{osc}}^{\text{SCIC}}, \quad |\pi_0 \mathcal{L}_{D1}(q)| \geq \mathfrak{c} \quad \forall q \in Q_{\text{osc}}^{\text{SCIC}}.$$

The minimal new standing coordinate is absolute electron-boundary spectral support. The endpoint-neutral marked family proves universal splitting, but does not replace the intrinsic discrete, continuous, hidden, composite, boundary, or hybrid endpoint strata.

The joint payload is

$$H_{D1} = (T_K^{BB}, \text{SOE-D1-1}, \text{PhysMod}_{D1}, R_{D1}, \rho_{D1}, \{\mathcal{L}_{D1}(q)\}, Q_{D1}, \mu_\beta, \Delta_{\text{del}}^{D1}, \partial\text{Fail}_{D1}, \text{Prov}_{D1}, U_{D1}).$$

3.3 D2: ^{76}Ge LNV search response

D2 uses the target-scale transition response rather than the label “Majorana neutrino”. Every B15 frame has an exact source amplitude A_H , a source-defined correction space, and a physical contraction locus. A correction C is attached to the retained physical carrier on one split-anchored apex with exact total amplitude $A_H + C$. The old oscillation evaluator is applied in situ and ignores the D2-only ideal.

The marked affine family

$$C_{H,z} = -A_H + zA_{76}^*$$

has total transition block zA_{76}^* , response rate $z^2\gamma_{76}^*$, and a nondegenerate accepted interval. In particular, $z = 0$ is a *defined zero-amplitude survivor*; it is not inferred from non-observation.

The marginal conclusions are

$$\text{im } \rho_{D2} = Q_{\text{osc}}^{\text{SCIC}}, \quad |\pi_0 \mathcal{L}_{D2}(q)| \geq \mathfrak{c} \quad \forall q \in Q_{\text{osc}}^{\text{SCIC}}.$$

The new coordinate is conjugation/LNV transition response. The null search neither forces nonzero LNV nor selects exact lepton-number conservation.

The joint payload includes the general amplitude object, the marked family, nuclear/detector/statistical frames, source-composition hypergraphs, exact zero subfibre, overlaps, and the total status partition.

3.4 D3: two permanently distinct programmes

3.4.1 The original D3-1.1 programme

The original target combines direct heavy-state response, active-sector closure/weak universality, and charged-lepton-flavour response. A source-first prompt-HNL family with heavy mass in a nondegenerate interval survives

on one apex and retains the oscillation standing core. Distinct heavy thresholds change retained matching data, hence

$$|\text{im } \rho_{D3}^{1.1}| \geq \mathfrak{c}, \quad |Q(T_{D3}^{1.1})| \geq \mathfrak{c}.$$

The finite-submatrix family supplies a marginal active-closure obstruction. It does not prove complete-fibre elimination. The full image, fixed-base splitting, complete overlaps, and global D3 atlas remain open.

3.4.2 The successor D3-1.3 programme

The successor target includes the retained process-tensor tester

$$S_{D3}(Y) = C_{ee}^{\text{op},1.3}(Y) = W_Y * T_{e,\text{act}}^{U,\text{ret}}$$

with threshold $c_e^{\min} = 0.987$. Retained-base factorization makes this quantity constant on each complete lift fibre. The finite-unitary deficit continuum with $\delta \in [0.02, 0.03]$ has $S_{D3} = 1 - \delta < 0.987$ and therefore empty fibres. A prompt-HNL continuum remains accepted. Consequently

$$\emptyset \subsetneq \text{im } \rho_{D3}^{1.3} \subsetneq Q_{\text{osc}}^{\text{SCIC}}, \quad |\text{im } \rho_{D3}^{1.3}| \geq \mathfrak{c}, \quad |Q_{\text{osc}}^{\text{SCIC}} \setminus \text{im } \rho_{D3}^{1.3}| \geq \mathfrak{c}.$$

On the independently defined RCF-finite semantic subcategory, source normalization, exact coefficient descent, chartification, and groupoid codescent yield closure. The finite-operational complement remains

$$\text{ROOS}_{\text{fin-op} \setminus \text{RCF-fin}}^{1.3}.$$

3.5 D4: public-summary and complete cosmological targets

The positive D4 target is the public-95 summary object. It contains a standard-thermal source compiler, three light population slots, a mass-indexed flat- Λ CDM background family, retained-oscillation transparency, and membership of the scalar mass sum in the released band

$$B_{95}^{\text{pub}} = (-\infty, \Sigma_{95}^{\text{pub}}].$$

The normal-ordering pullback is the nondegenerate interval $[0, m_\star]$, whereas the standard inverted-ordering floor lies outside the band. This is a set-theoretic summary-target disposition, not a transported posterior or realized ordering selector.

The marginal result is

$$\text{im } \rho_{D4}^{\text{pub95}} = Q_{\text{osc}}^{\text{SCIC}}, \quad |\pi_0 \mathcal{L}_{D4}^{\text{pub95}}(q)| \geq \mathfrak{c} \quad \forall q \in Q_{\text{osc}}^{\text{SCIC}}.$$

The minimal proved separator on the positive branch is Σ_ν . The broader ledger is `CosmologicalScaleHistory`. The comparison tags are

$$\text{Rel}_{D1}(D4) = \text{ScaleHistoryRefinement}, \quad \text{Rel}_{D3}(D4) = \text{DistinctSeparator}.$$

The complete cosmological target remains

$$\text{Status}(D4; T_{D4}^{\text{complete}}) = \text{NotReady}_{\text{JLR-D4-1}}.$$

3.6 D5: seven symmetry and charge targets

D5 supplies seven exact target objects and target-indexed base images:

$$\begin{aligned} \text{im } \rho_{\text{acc}}^{(4)} &= Q_{\text{Acc} \leq 4}, & \text{im } \rho_{\text{bg,min}} &= Q_{\text{MinBG}}, \\ \text{im } \rho_{\text{loc,3N}} &= Q_{\text{Loc3N}}, & \text{im } \rho_{\text{glob}} &= Q_{\text{GlobProt}}, \\ \text{im } \rho_{\text{exp}[O]} &= Q_{\text{ExpLift}(O)}, & \text{im } \rho_{\text{sp}[q_\Phi]} &= Q_{\text{Cov}_\Phi(q_\Phi)}, \\ \text{im } \rho_{\text{rem}[\Gamma, q_\Phi]} &= Q_{\text{RemFaith}(\Gamma, q_\Phi)}. \end{aligned}$$

The first four are nonempty proper-continuum subatlases. Local and exact-global accepted continua are completed by a source-formed finite-order RG/EFT right inverse. Explicit, spontaneous, and remnant images are exact source-semantic subatlases, but their general cardinality remains `NotReady` where no separate bound is proved.

D5 also proves connected spectral charge uniformity on targets carrying exact continuous or exact residual covariance of the complete mass-response and spectral-incidence structure. That theorem concerns symmetry-bearing connected components and does not establish terminal-carrier identity.

At D3-1.3 scope,

$$D3D5Status(T_{acc}^{(4)}) = D3D5Status(T_{glob}) = \text{IndependentCoordinate}.$$

The other five rows remain NotReady, and no marginal common-generator theorem is available.

3.7 Marginal endpoint matrix

Leg	Base image	Fibre effect	Joint significance
D1	Full base	Continuum split	Absolute support must share one mass record with D4; not an independent free product.
D2	Full base	Continuum split, including defined zero	Exact-global/local $B - L$ can force the zero subfibre; accidental $d \leq 4$ permits dimension-five LNV.
D3-1.1	At least continuum	Explicit surviving continuum; global image open	Parent joint closure inherits D3 global NotReady.
D3-1.3	Proper continuum	Empty deficit fibres and accepted continuum	Supplies a continuum complement for every joint image; RCF-finite atlas closed, parent remainder ROOS.
D4 public	Full base	Continuum split on standard-thermal branch	Refines D1 absolute scale through Σ_v and one common population history.
D4 complete	Unavailable	Unavailable	Every branch containing it is NotReady JLR-D4-1.
D5 first four	Proper continuum	Target-indexed exact atlases	Joint compatibility remains a new PCR-5E theorem.
D5 break-ing/remnant	Exact source-semantic images; cardinality partly NotReady	Target-dependent	Requires one common violation/breaking/stabilizer/matching object.

4 Target-indexed simultaneous target family

4.1 The target category and slice structure

Let Tar_v denote the category whose objects are fully typed neutrino target packets and whose morphisms are standing-preserving target embeddings. Every morphism retains the old target records, witness family, lawful-equivalence data, failure semantics, and continuation operations. A change of background model, detector interface, process comparison class, uncertainty semantics, or source comparison class is part of the object and cannot be hidden inside a morphism.

PCR-5E works in the slice category $\text{Tar}_v/T_{\text{osc}}$. Every D target comes with its retained embedding of the oscillation target. This slice remembers the common base and prevents a target join from identifying records merely because they have similar names.

Definition 4.1 (Marginal target diagram). For a branch τ , the marginal target diagram $\mathcal{D}_T^\tau$ has objects

$$T_{D1}, T_{D2}, T_{D3}^{\tau_3}, T_{D4}^{\tau_4}, T_{D5}^{\tau_5}$$

and the five retained embeddings from T_{osc} . Arrows between marginal targets are included only when supplied by a frozen target comparison theorem; otherwise the diagram contains no inferred relation.

4.2 Shared and target-specific records

The target join requires an explicit normalization of shared fields. The following are identified through their common retained object:

- the charged-current preparation/detection boundary;
- the event-complete retained oscillation channel;
- the PCR-5B seven-channel standing packet;
- the base source frame, boundary attachment, and matching orientation;
- the common vacuum/deformation branch where the marginal targets use the same branch;
- lawful gauge, basis, chart, and source isomorphisms that act diagonally on all retained copies.

The following remain target-specific unless a theorem supplies a comparison map:

- D1 endpoint spectral measure, campaign response, and nuisance geometry;
- D2 transition amplitude, nuclear/detector/statistical frame, and null-search acceptance;
- D3 direct-state, active-closure, process-tensor, and CLFV records;
- D4 population history, background solution, perturbation hierarchy, and mass-sum evaluator;
- D5 action, charge lattice, anomaly, protector, violation, breaking carrier, stabilizer, remnant, and faithful quotient.

4.3 Cross-phenomenon target grammar

The target join is not complete until every physically shared quantity has a consistency relation. The cross grammar is generated before source or survivor inspection from the types of the marginal records. Its primitive classes are:

- X1. **Mass/spectral coherence:** the D1 endpoint spectral measure and D4 population mass records must descend from one light-sector spectral object; D3 heavy thresholds and D5 mass/matching records must use the same source masses where they overlap.
- X2. **Boundary coherence:** D1 beta-decay, D2 nuclear transition, D3 direct/CLFV, and oscillation boundary records share the same charged-current and charged-lepton representation data where their source paths meet.
- X3. **Charge and conjugation coherence:** D2 LNV amplitudes and D5 $B - L$ action/breaking records must satisfy the same charge-violation equations on the actual neutrino-bearing path.
- X4. **Population/activity coherence:** every D3 or D5 heavy sector receives the D4 activity value and kinetic landing fixed by the declared SOE-D4-1 source point; no nonzero old coupling is converted into D4 population by target interpretation and no active D4 relation is suppressed by numerical smallness.
- X5. **Anomaly and representation coherence:** all D3/D5 local or global structures use one representation/charge ledger and one anomaly calculation.
- X6. **Vacuum and breaking coherence:** D4 background, D5 spontaneous/remnant structure, and any scalar or threshold extension must refer to one compatible physical vacuum orbit or declare a target update.
- X7. **Matching and continuation coherence:** D1/D2 low-energy responses, D3 heavy matching, D4 transport, and D5 RG/EFT transport must commute on shared scales and retain every target-visible threshold.
- X8. **Common-cone coherence:** every retained marginal witness diagram must admit one physical apex and one compatible system of projection arrows.
- X9. **Complete finite higher incidence:** for every finite arity $n \geq 3$, every target-active typed hyperedge among marginal source or physical sorts is retained with its orientation, applicability guard, continuation action, failure status, and lawful automorphism action. X9 is the complete family of such hyperedges, not a label standing in for an unexamined remainder.

Definition 4.2 (Cross-witness signature). Let W_{cross}^{τ} be the many-sorted signature generated by X1–X9, with applicability guards inherited from the branch certificate. Its equations and failure types are denoted R_{cross}^{τ} .

4.4 Semantic joint uses before the finite cross presentation

The completeness of the cross grammar is not inferred from the displayed list. It is established from a semantic trace formed directly from the frozen target and source types.

Definition 4.3 (Joint target-facing semantic trace). For a raw same-target physical presentation W , define

$$\tau_E^\tau(W) = (\tau_{\text{osc}}(W), \tau_{D1}(W), \dots, \tau_{D5}(W), [\mathcal{D}_{\text{inc}}(W)]_{\text{phys}}).$$

The first six entries are the independently frozen marginal target-facing traces. The final entry is the physical isomorphism class of the complete finite target-active incidence hypergraph: repeated bearer identifications, all typed cross arrows and finite hyperedges, their applicability and source-activity guards, the common vacuum/background/matching objects, continuation actions, failure incidences, and common-cone data.

Definition 4.4 (Semantic joint target use). A semantic joint target use is a same-target, lawful-redescription-invariant, fail-closed physical use whose defined value is extensional with respect to the target-facing data fixed by the branch certificate. The class is denoted $\text{JointUse}_{\text{sem}}(T_E^\tau)$. A use may act on one marginal trace, on a shared bearer, or on an irreducible finite cross incidence; it may not import an undeclared selector, source operation, target record, or continuation.

Theorem 4.5 (E-S1 – joint semantic-use and cross-relation saturation). For every branch τ :

1. every $u \in \text{JointUse}_{\text{sem}}(T_E^\tau)$ factors uniquely through τ_E^τ on its domain of definition;
2. the marginal factors of τ_E^τ are exactly the marginal trace families already proved complete in PCR-2 and the D-series handoffs;
3. the nonmarginal factor $[\mathcal{D}_{\text{inc}}]_{\text{phys}}$ is represented without loss by X1–X9, where X1–X8 are typed projections and X9 retains every remaining finite target-active hyperedge at its actual arity;
4. lawful source, gauge, basis, chart, and parameter transport acts functorially on the trace and cannot create or erase a target-active incidence.

Consequently $(W_{\text{cross}}^\tau, R_{\text{cross}}^\tau)$ is saturated relative to the declared finite source comparison class. A relation requiring a new sort, operation, infinite operational category, or target record is an explicit source or target update rather than a hidden current relation.

Proof. The proof is by reduction to the two upstream semantic completeness theorems rather than by inspection of the displayed X-list.

Marginal part. PCR-2 defines target-facing semantic uses before choosing a finite evaluator profile and proves that every such use factors through the complete physical interface trace. The D-series handoffs repeat that discipline for each enriched target. Therefore any dependence of a joint use on one marginal leg factors through the corresponding marginal trace $\tau_{D\alpha}$.

Joint-incidence part. Suppose the marginal traces are fixed. PCR-3 identifies a finite same-target source presentation with its complete activity-reduced carrier trace, including every retained bearer, constructor, typed arrow, finite hyperedge, support relation, orientation, parameter domain, and constitutive equation. PCR-4 adds the physical overlap diagram, residual/failure incidences, and common-cone class. Hence a same-target use that can still vary must depend on how two or more already-fixed marginal records are co-instantiated in that complete source/physical incidence object. Since the declared comparison class is finitely presented, the dependence is carried by a finite typed incidence with its applicability, activity, continuation, failure, and automorphism data. That is exactly a component of $[\mathcal{D}_{\text{inc}}]_{\text{phys}}$.

Binary projections of the complete incidence object are sorted into X1–X8 by their bearer types. Every irreducible incidence of arity at least three is retained literally, at its actual arity and with its full data, in the indexed family X9; no pairwise decomposition is imposed. Thus equality of τ_E^τ is equality of every marginal semantic trace and of the complete physical incidence diagram. Extensionality of semantic uses then gives equal outputs. The factor is unique on the image of the trace.

Conversely every X1–X9 evaluator is a typed functional of τ_E^τ . Lawful source, gauge, basis, chart, and parameter arrows preserve activity, typing, incidence arity, orientation, continuation, failure status, and the common-cone isomorphism class, so the factorization is natural. A proposed relation not represented by the complete trace must add a sort, operation, target record, continuation, or operational category; by the fixed-domain typing rule it is a source or target update, not an omitted current relation. $\square$

4.5 E01: joint-target formation

Construction 4.6 (Typed target colimit). The joint target is the colimit in the slice over T_{osc} , augmented by the independently formed saturated cross signature:

$$T_E^\tau = \text{colim}_{\text{Tar}_\tau/T_{\text{osc}}} \mathcal{D}_T^\tau \cup (W_{\text{cross}}^\tau, R_{\text{cross}}^\tau).$$

The union symbol here denotes disjoint typed extension after normalization of the common retained records, not an untyped set union.

Theorem 4.7 (E01 – joint-target formation). *For every branch certificate TauCert_E^τ , the object T_E^τ of Theorem 4.6 exists and is unique up to unique isomorphism in the target slice. It retains every marginal target record and continuation, contains no duplicate ill-typed witness slot, and contains the complete pre-evaluation cross-witness grammar certified by Theorem 4.5. A marginal field conflict has exactly one of three dispositions:*

1. lawful normalization through a common retained object;
2. a typed branch obstruction recorded in T_E^τ ;
3. an explicit source or target update producing a different certificate.

Proof. The five marginal objects share the base target through specified embeddings, so their coproduct in the slice identifies only the images of those embeddings. Every other field remains tagged by its marginal target. Shared records with proved comparison maps are normalized by the corresponding coequalizer. Because target fingerprints, codomains, applicability predicates, uncertainty objects, and continuation operations are part of the typing data, incompatible duplicates cannot be silently identified.

The cross signature is generated from the semantic trace before survivor inspection and is saturated by Theorem 4.5. Adjoining it is a free typed extension followed by quotienting by the branch-fixed equations. The universal properties of the slice colimit, free extension, and quotient give existence and uniqueness up to unique isomorphism.

If two marginal fields are comparison-equivalent, the coequalizer gives lawful normalization. If their requirements are jointly meaningful but inconsistent, the target retains a branch obstruction. If they change the domain, evaluator, source inventory, or continuation structure, the branch certificate changes and the result is a source or target update. These cases are exhaustive by target typing. $\square$

Corollary 4.8 (No survivor-dependent target completion). *No cross relation may be added solely because a selected marginal tuple needs it to survive. Any such addition is a target update and must be evaluated under a new branch certificate.*

4.6 Target fingerprints

The target fingerprint $\text{FP}(T_E^\tau)$ is the ordered tuple of:

(Domain, Base, Records, WitnessSorts, Applicability, Acceptance, Uncertainty,
SourceScope, Background, OperatorCategory, LawfulEquivalence, Continuation, Failure, Update).

The fingerprint is included verbatim in the supplementary formal branch catalogue. Equality of fingerprints is necessary for path equivalence and same-scope comparison.

4.7 Vectorlike-joint source comparison class

The two positive branches are first formed as raw precursor target-and-comparison certificates from the exact-response vectorlike Dirac source grammar exported by D5. Their complete source domains are fixed here before any D3 acceptance, D3/D5 marginal landing, joint obstruction, base-image, fibre-cardinality, standing-coordinate diagnosis, or PCR-6 endpoint is inspected. A later source-qualified comparison certificate may restrict this raw class only by the independently proved D3/D5 marginal landing theorem and must itself be frozen before the genuinely joint D1/D2/D4 calculation.

Choose a compact positive intrinsic light-response interval

$$I_m^\circ = [m_a, m_b] \in (0, m_\star)$$

inside the frozen D4 public-summary normal-ordering interval. Its strict upper separation from $m_\star$ leaves a nondegenerate carrier-indexed absolute-scale direction. Define

$$\tau_\star := m_\star^2 - m_b^2 > 0, \quad I_\tau^\circ = [0, \tau_b] \in [0, \tau_\star), \quad \tau_b > 0,$$

and for every $(m_0, \tau) \in I_m^\circ \times I_\tau^\circ$ set

$$m_i^{(\tau)}(m_0) := \sqrt{m_i(m_0)^2 + \tau}.$$

This is the pre-frozen common D1/D4 metric-source domain. It preserves every oscillation mass-squared difference and satisfies the public-summary bound uniformly because $\sqrt{m_0^2 + \tau} < m_\star$. The parameter τ is not inserted into the D3/D5 vectorlike source coefficients; it is carried only by the explicitly declared D1 hyperprocess/D4 population extension and has an exact reduct to the intrinsic vectorlike source. Choose a compact source threshold interval

$$I_\lambda^{\text{src}} = [\lambda_a, \lambda_b] \in (1, 2), \quad H_\lambda = \lambda \Lambda,$$

inside the D5 exact-response matching interval, and define

$$s_{\text{src}}^{\text{max}} = \min \left\{ \frac{1}{4}, \frac{v}{8\sqrt{2}\lambda_b\Lambda} \right\}, \quad I_s^{\text{src}} = (0, s_{\text{src}}^{\text{max}}).$$

For $m_0 \in I_m^\circ$, $\lambda \in I_\lambda^{\text{src}}$, and $s \in I_s^{\text{src}}$, let $x(m_0, \lambda, s)$ be the finite PCR-3/B15 source with the D5 coefficients

$$Y_\nu = \frac{\sqrt{2}}{v} c U \Sigma(m_0) V^\dagger, \quad Y_{LN} = -\frac{\sqrt{2}}{v} s H_\lambda U,$$

$$\mu = s \Sigma(m_0) V^\dagger, \quad M_N = c H_\lambda \mathbf{1}, \quad c = (1 - s^2)^{1/2}.$$

The field, charge, source-incidence, anomaly, vacuum, and finite-order matching data are those of the source-realized D5 vectorlike Dirac threshold family. This is a source grammar, not a D3 pass assertion.

Definition 4.9 (Vectorlike-joint raw source class). The class $\mathbf{S}_{\text{VJ}}^{\text{raw}}$ is the lawful source groupoid generated by the objects $x(m_0, \lambda, s)$ on

$$\Theta_{\text{VJ}}^{\text{raw}} = I_m^\circ \times I_\lambda^{\text{src}} \times I_s^{\text{src}},$$

together with their source, gauge, basis, and framed matching arrows. The class is activity-reduced before any target path is formed. Nonzero active–vectorlike incidences retain the vectorlike block; disconnected or source-null additions do not create extra objects.

Let I_{76} be a compact nondegenerate interval contained in the already certified D2 accepted amplitude locus, and let $\mathbf{a}_{\text{D4}}^{\text{seed}}$ be the SOE-D4-1 seed-active/off-seed-null activity vector certified upstream by SAR-D4-1. These are fixed from marginal downstream interface data, not from the PCR-5E joint calculation.

For $\tau_5 \in \{T_{\text{acc}}^{(4)}, T_{\text{glob}}\}$, let $\mathbf{S}_{\text{VJ,raw}}^{\tau_5}$ be the full lawful source groupoid formed from $\mathbf{S}_{\text{VJ}}^{\text{raw}}$ by adjoining:

1. the D1 endpoint-neutral hyperprocess/metric source extension on I_τ° over each intrinsic point $m_0 \in I_m^\circ$, together with its exact base-reduct functor;
2. the D2 correction domain $Z_{T_{\text{acc}}^{(4)}} = I_{76}$ or the exact-global defined-zero domain $Z_{T_{\text{glob}}} = \{0\}$;
3. the fixed D4 activity vector $\mathbf{a}_{\text{D4}}^{\text{seed}}$, its three light seed populations carrying the common shifted mass record $\mathbf{m}^{(\tau)}(m_0)$, and the declared null/terminal objects on every off-seed D4 sort;
4. the D5 target tag τ_5 and its source action, anomaly, protector, or bounded-operator predicate.

No D3 acceptance, common-apex verdict, base-image result, or fibre-cardinality result is used in this source formation.

Definition 4.10 (Vectorlike-joint raw precursor target-comparison certificate). Define

$$T_{E, \text{VJ,raw}}^{\tau_5} := T_E^{(1.3, \text{pub95}, \tau_5)} \Big|_{\mathbf{S}_{\text{VJ,raw}}^{\tau_5}}.$$

The restriction changes only the declared source comparison class. Every parent target record, evaluator, applicability rule, uncertainty object, continuation, failure status, and lawful-equivalence datum is retained.

Theorem 4.11 (E-S3 – vectorlike-joint raw precursor freeze). *The fingerprints $\text{FP}(T_{E, \text{VJ}, \text{raw}}^{T_{\text{acc}}^{(4)}})$ and $\text{FP}(T_{E, \text{VJ}, \text{raw}}^{T_{\text{glob}}})$ are fixed before D3 acceptance, simultaneous source survival, common-apex evaluation, base-image calculation, fibre counting, standing-coordinate diagnosis, or PCR-6 handoff. Their source domains are the complete raw domains*

$$I_m^\circ \times I_\lambda^{\text{src}} \times I_s^{\text{src}} \times I_\tau^\circ \times Z_{\tau_5},$$

not an after-the-fact accepted locus. The positive D3/D5 stratum (I_E, s_E) constructed by the later landing theorem is a solved sublocus of this already frozen raw precursor and does not redefine its fingerprint. The source-qualified certificate formed from that marginal theorem is a distinct, explicitly comparison-qualified child certificate; it does not replace or complete the raw precursor or the parent target.

Proof. Every source field in the two precursor fingerprints is imported from a marginal source theorem or from the displayed source grammar. The vectorlike coefficients and parameter domains are fixed before D3 response evaluation. The D1/D4 common absolute-scale interval and its exact base-reduct data are frozen before D3/D5 evaluation. The D2 interval or exact-zero cell comes from the frozen D2 handoff. The D4 activity vector is primitive source data in SOE-D4-1 and its coherence theorem is upstream of PCR-5E. The D5 tag is one of the seven previously separated targets. Hence certificate formation contains no marginal or joint survivor, base image, fibre count, common-generator diagnosis, or preferred PCR-6 endpoint. Solving the D3/D5 landing problem inside this pre-existing source comparison class is evaluation of a frozen certificate, not target backfitting. $\square$

4.8 Completeness status at target formation

Target formation itself can be complete while a target evaluator remains unavailable. The status of T_E^τ is therefore a pair

$$\text{TargetStatus}(\tau) = (\text{Formed}, \text{Evaluable}).$$

All twenty-eight parent branches are formed. A branch with $\tau_4 = \text{complete}$ is formed but not fully evaluable because JLR-D4-1 is absent. This distinction prevents the missing evaluator from being mistaken for a malformed target or an empty physical fibre. The two E-S3 vectorlike-joint raw precursor certificates are formed and evaluable before the D3/D5 landing calculation. Their later Pass, Fail, N/A, ROOS, source-update, and target-update strata are outputs of evaluation inside those frozen raw certificates, not new target definitions. The E-S5 source-qualified child certificates are formed only after the independently proved E-S4 marginal bridge and are then frozen before every genuinely joint D1/D2/D4 calculation.

5 Common-base pullback and joint source theory

5.1 The concrete base groupoid

PCR-5B represents the oscillation quotient by the concrete pointed standing-core groupoid $\mathcal{N}_{\text{osc}}$. Its connected components are the SCIC classes, but the groupoid retains base arrows, isotropy, framed provenance, and the standing packet

$$\Xi_{\text{osc}} = (T_{\text{face}}, R_{\text{stand}}, \Delta_{\text{del}}, C_{\text{act}}, M_{\text{match}}, B_{\text{bdy}}, F_{\text{same}}).$$

For $q \in \mathcal{Q}_{\text{osc}}^{\text{SCIC}}$, write $\mathcal{N}_{\text{osc}}[q]$ for the full connected subgroupoid representing q .

Each marginal restriction functor

$$R_\alpha : \mathcal{N}_\alpha \longrightarrow \mathcal{N}_{\text{osc}}$$

is part of the marginal data object. The associated connected marginal fibre is

$$\mathcal{L}_\alpha(q) = \mathcal{N}_\alpha \times_{\mathcal{N}_{\text{osc}}}^h \mathcal{N}_{\text{osc}}[q].$$

5.2 E02: common-base homotopy pullback

Definition 5.1 (Raw common-base tuple groupoid). For a branch τ and base class q , define

$$\mathcal{P}_{\text{marg}}^\tau(q) = \mathcal{L}_{D1}(q) \times_{\mathcal{N}_{\text{osc}}[q]}^h \mathcal{L}_{D2}(q) \times_{\mathcal{N}_{\text{osc}}[q]}^h \mathcal{L}_{D3}^{\tau_3}(q) \times_{\mathcal{N}_{\text{osc}}[q]}^h \mathcal{L}_{D4}^{\tau_4}(q) \times_{\mathcal{N}_{\text{osc}}[q]}^h \mathcal{L}_{D5}^{\tau_5}(q).$$

An object is a marginal five-tuple together with explicit isomorphisms between all retained base restrictions. Morphisms contain the marginal arrows and the coherence two-cells induced by the homotopy pullback.

Theorem 5.2 (E02 – common-base pullback). *For every branch τ and connected base class q , the groupoid $\mathcal{P}_{\text{marg}}^\tau(q)$ exists and is invariant under replacement of any marginal presentation by a lawful isomorphic frame. A tuple whose marginal restrictions lie in different connected components of $\mathcal{N}_{\text{osc}}$ has no object in $\mathcal{P}_{\text{marg}}^\tau(q)$ for any q .*

Proof. The category of small groupoids admits homotopy pullbacks. Iterating them over the fixed connected subgroupoid $\mathcal{N}_{\text{osc}}[q]$ produces $\mathcal{P}_{\text{marg}}^\tau(q)$. A lawful change of marginal frame is an equivalence over $\mathcal{N}_{\text{osc}}$; homotopy pullback is invariant under equivalence, so the resulting common-base groupoid is unchanged up to equivalence.

If two marginal restrictions lie in distinct connected components, no base arrow identifies them. The isomorphism data required for an object of the homotopy pullback therefore do not exist. This excludes wrong-base tuples before source amalgamation and without appealing to coarse family labels. $\square$

Remark 5.3 (Common base is not common source). The object $\mathcal{P}_{\text{marg}}^\tau(q)$ records compatible retained cores. It does not assert that the five lifts arise from one source hypergraph, one vacuum, one background, one matching map, or one physical apex. Those are subsequent equalizer conditions.

5.3 The source base and marginal reducts

Let $\mathcal{S}_{\text{B15}}$ be the frozen source-frame category exported by PCR-5B. Each marginal source extension comes with a reduct

$$U_\alpha : \mathcal{S}_\alpha \longrightarrow \mathcal{S}_{\text{B15}}$$

and a source-facing trace. The joint source must recover all five marginal sources through exact reducts while representing each shared source node exactly once.

Definition 5.4 (Raw source amalgamation). The raw source amalgamation is the homotopy limit

$$\mathcal{S}_{\text{raw}}^\tau = \text{holim} \left(\mathcal{S}_{D1} \rightarrow \mathcal{S}_{\text{B15}} \leftarrow \mathcal{S}_{D2}, \dots, \mathcal{S}_{D5}^\tau \rightarrow \mathcal{S}_{\text{B15}} \right).$$

Its objects contain a common B15 source frame, the five marginal source extensions, and coherent source-reduct isomorphisms.

This raw limit still lacks cross relations. It can contain tuples that use the same base source but assign inconsistent mass records, charge lattices, vacua, thresholds, or response amplitudes.

5.4 Cross-source signature

The cross-source theory $\mathcal{S}_{\text{cross}}^\tau$ is generated from the target cross grammar. Its primitive sorts include:

- **LightSpec**: the common light spectral/mass-response object;
- **HeavyThresh**: heavy-state thresholds and oriented matching records;
- **Amp₇₆**: the complete D2 transition-amplitude object;
- **PopHist**: D4 population and transport histories;
- **ChargeLat**, **Action**, **Protector**, **Break**, and **Remnant**: D5 data;
- **Vac**, **BG**, **Match**, and **Cont**: common vacuum, background, matching, and continuation objects;
- **Inc_{higher}[e]**: for each actual active constitutive hyperedge e whose marginal-tag support has cardinality at least three, the complete ordered support, activity guard, constitutive equation, continuation action, and failure frontier.

The principal source equations are:

$$\begin{aligned} \mu_\beta &= \Pi_\beta(\text{LightSpec}), \\ \Sigma_\nu &= \Pi_\Sigma(\text{LightSpec}, \text{PopHist}), \\ A_{76}^{\text{tot}} &= \Pi_{76}(\text{LightSpec}, \text{HeavyThresh}, \text{ChargeLat}, \text{Break}), \\ C_{ee}^{\text{op}} &= \Pi_{\text{act}}(\text{LightSpec}, \text{HeavyThresh}, \text{Cont}), \\ \text{ChargeViolation}(A_{76}^{\text{tot}}) &= \Pi_{\Delta Q}(\text{ChargeLat}, \text{Action}, \text{Break}, \text{Remnant}), \\ \text{Stress}_\nu &= \Pi_T(\text{LightSpec}, \text{PopHist}, \text{BG}). \end{aligned}$$

Each equation has an applicability guard. For example, a heavy state contributes to D4 only when its production or transport incidence is source-active in the frozen D4 time domain. On the vectorlike-joint source scope, the D4 activity maps are frozen before target evaluation to the seed-active/off-seed-null assignment certified by SAR-D4-1; no inference from heavy mass, small mixing, or numerical decoupling is used.

5.5 Activity reduction before path closure

Source activity is pointwise and exact. A source incidence is removed only when its source equation gives literal null activity at the parameter point. No numerical smallness threshold is used.

Definition 5.5 (Joint activity reduct). For $H \in \mathcal{S}_{\text{raw}}^\tau \times \mathcal{S}_{\text{cross}}^\tau$, let $\text{ActRed}_E(H)$ be the source hypergraph obtained by:

1. evaluating every primitive and derived activity predicate;
2. deleting source-null incidences and disconnected latent spectators;
3. retaining every nonnull incidence, irrespective of magnitude;
4. recomputing target paths, support closure, and higher-arity incidence only after the deletion.

This order is essential. A disconnected $B - L$ -violating spectator cannot become an explicit-breaking D5 lift; a source-null correction cannot create D2 multiplicity; and a heavy sector can be off-support in D4 only when the frozen source-activity guard assigns its D4 production, collision, decay/injection, and stress-energy incidences to their exact null objects. Kinematic inaccessibility or small coupling alone never supplies this nullity.

5.6 Conservative joint source category

Construction 5.6 (Joint source category). The conservative joint source category $\mathcal{S}_E^\tau$ is the full subgroupoid of activity-reduced objects in

$$\mathcal{S}_{\text{raw}}^\tau \times \mathcal{S}_{\text{cross}}^\tau$$

that satisfy all source-constitutive equations and admit exact reducts to the five marginal source categories.

Proposition 5.7 (Exact marginal source reducts). *For every branch τ , there are functors*

$$U_{E \rightarrow \alpha}^\tau : \mathcal{S}_E^\tau \longrightarrow \mathcal{S}_\alpha$$

whose further reduct to $\mathcal{S}_{B15}$ is independent of α up to the specified coherent isomorphism. Irreducible higher-arity incidence is preserved by $\mathcal{S}_E^\tau$ and may be forgotten only by a marginal reduct whose target does not type that incidence.

Proof. The objects of $\mathcal{S}_{\text{raw}}^\tau$ already contain coherent marginal reducts. The cross source theory adds equations and incidences but does not duplicate the marginal source objects. Restriction to the appropriate marginal sorts therefore defines $U_{E \rightarrow \alpha}^\tau$. The common B15 reduct is part of the raw homotopy-limit data. Higher-arity relations are explicit source objects; a reduct forgets them only where its codomain lacks the corresponding sort, and this forgetting is recorded rather than interpreted as source nullity. $\square$

5.7 Source-normal-form completeness

Theorem 5.8 (E-S2 – joint source-normal-form completeness). *Within the declared finite PCR-3/B15 comparison class and the five versioned source extensions, every semantic source presentation of the complete joint target has, after pointwise activity reduction, a source-coherent normal form in $\mathcal{S}_E^\tau$. Conversely every object of $\mathcal{S}_E^\tau$ has the declared semantic source interpretation. Equality of the complete activity-reduced joint source trace is exactly source-presentation equivalence up to lawful source reparameterization.*

Proof. Each marginal source extension already carries its source-normal-form and exact reduct theorem. The raw homotopy limit identifies the repeated B15 source rather than copying it. By [Theorem 4.5](#), every additional standing-bearing joint dependence is a finite typed incidence in the complete physical/source overlap hypergraph and is represented by X1–X9 with its activity, applicability, continuation, and failure data. Pointwise activity

reduction removes exactly source-null incidences before path closure. The resulting finite hypergraph therefore contains every semantic source use and no target-inert spectator. Lawful source isomorphisms and parameter transport preserve this trace, while trace equality reconstructs the same normal form by the marginal extensionality theorems and the literal incidence data. This gives both directions. $\square$

5.8 Source-completeness boundary

By Theorem 4.5, the target-facing cross grammar is semantically saturated. Source-first reconstruction then gives the corresponding complete finite source presentation: every target-active use is represented either in one marginal source trace or by an actual typed incidence in X1–X9. In particular, higher incidence is indexed by the actual constitutive hyperedges rather than represented by one unspecified residual. The theorem does not claim unrestricted ultraviolet completeness. A proposed source architecture outside these formation rules is a source update, not a hidden member of the current category.

Status 5.9 (Joint source readiness). A branch receives `SourceReady` when every marginal source handoff is complete at that branch and every target-generated cross source relation is typed. A D3-1.1 branch may be source-ready while fibre-atlas finality remains `NotReady`. A D4-complete branch is target-formed but evaluator-not-ready; its source category is retained for future evaluation rather than declared empty.

6 Simultaneous physical realization and obstruction normal form

6.1 Semantic physical objects

A joint source object is not yet physical. The semantic physical comparison class consists of complete realizations whose target-facing records are all represented on one object and whose shared witness diagrams have a common cone. This is the direct PCR-5E analogue of PCR-4 simultaneous survival.

Definition 6.1 (Joint physical presentation). A joint physical presentation over $H \in \mathcal{S}_E^\tau$ is a tuple

$$W_E = (W_{\text{base}}, W_{D1}, W_{D2}, W_{D3}, W_{D4}, W_{D5}, \mathcal{D}_{\text{inc}}, \omega_E)$$

where:

- W_{base} is one physical realization of the retained oscillation core;
- $W_{D\alpha}$ are target-specific cells attached to the same base object;
- $\mathcal{D}_{\text{inc}}$ is the complete physical incidence diagram, including cross and higher-arity arrows;
- ω_E is a common-cone witness from one semantic apex to every object in $\mathcal{D}_{\text{inc}}$.

The tuple is not a product of independently realized worlds. Every repeated physical bearer is identified through the source and incidence data.

6.2 Complete residual family

The residual family is formed from the target and source grammars before evaluation. It has the following sectors.

- R1. **Inherited base residuals** $\mathcal{E}_{\text{osc}}$: the PCR-4 role, channel, response, gauge, anomaly, vacuum, matching, and common-cone equations.
- R2. **Pure marginal residuals** $\mathcal{E}_{D\alpha}$: each D paper’s own source, response, physicality, covariance, acceptance, and continuation equations.
- R3. **Shared-parameter residuals** $\mathcal{E}_{\text{par}}$: equality and lawful transport of common masses, mixings, charges, thresholds, background anchors, and nuisance data.
- R4. **Boundary residuals** $\mathcal{E}_{\text{bdy}}$: compatibility of preparation/detection, beta-decay, nuclear, direct-state, CLFV, and cosmological attachment data.
- R5. **Mass/population residuals** $\mathcal{E}_{\text{mass-pop}}$: D1 endpoint support and D4 mass/population records descend from one light spectral object.

- R6. **LNV/charge residuals** $\mathcal{E}_{\text{LNV-charge}}$: D2 amplitudes have the charge-violation status prescribed by the D5 action, breaking ideal, or remnant congruence.
- R7. **Threshold/kinetic residuals** $\mathcal{E}_{\text{th-kin}}$: D3/D5 heavy states are included in D4 kinetic closure exactly when their frozen D4 activity guards are true; on the vectorlike-joint source scope every off-seed D4 leg is absent only through the source-coherent null/terminal objects certified by SAR-D4-1 and STR-D4-1, never through numerical smallness or a merely large threshold.
- R8. **Anomaly/global residuals** $\mathcal{E}_{\text{anom}}$: all local/global charges and representations use one anomaly and global-form ledger.
- R9. **Vacuum/background residuals** $\mathcal{E}_{\text{vac-bg}}$: D4 background and D5 breaking/stabilizer data inhabit one compatible vacuum and deformation branch.
- R10. **Matching/continuation residuals** $\mathcal{E}_{\text{match-cont}}$: every low/high-energy comparison and continuation square commutes.
- R11. **Higher-arity residuals** $\mathcal{E}_{\geq 3}$: irreducible relations involving three or more marginal sectors.
- R12. **Common-cone residual** $\mathcal{E}_{\text{cone}}$: the physical incidence diagram has one apex.

Write

$$\mathcal{E}_E^\tau = \mathcal{E}_{\text{osc}} \sqcup \bigsqcup_{\alpha} \mathcal{E}_{D\alpha} \sqcup \mathcal{E}_{\text{par}} \sqcup \mathcal{E}_{\text{bdy}} \sqcup \mathcal{E}_{\text{mass-pop}} \sqcup \mathcal{E}_{\text{LNV-charge}} \sqcup \mathcal{E}_{\text{th-kin}} \sqcup \mathcal{E}_{\text{anom}} \sqcup \mathcal{E}_{\text{vac-bg}} \sqcup \mathcal{E}_{\text{match-cont}} \sqcup \mathcal{E}_{\geq 3} \sqcup \mathcal{E}_{\text{cone}}.$$

6.3 Applicability and total status

Every residual $E \in \mathcal{E}_E^\tau$ has an applicability predicate App_E formed before its value. If $\text{App}_E = 0$ by target/source typing, the status is certified N/A. If $\text{App}_E = 1$, the residual evaluates as pass, fail, undefined, response-out-of-scope, source update, target update, or named readiness gap.

Definition 6.2 (Joint residual status). For a source/physical presentation x ,

$$\text{Stat}_E(x) \in \{\text{N/A}, \text{Pass}, \text{Fail}, \text{Undef}, \text{ROOS}, \text{SU}, \text{TU}, \text{NotReady}[\kappa]\}.$$

The first-failure record includes the residual identifier, source path, target tag, required witness, and reopening condition.

Lemma 6.3 (Status disjointness). *The listed statuses are pairwise disjoint. In particular, ROOS, SU, TU, and NotReady are not forms of solved emptiness.*

Proof. The statuses have distinct typing conditions. A fail is the evaluated falsity of an applicable residual. Undefined means the required in-scope value is not defined. ROOS means the semantic source point remains meaningful but the declared response evaluator does not cover it. Source and target updates change the certificate. NotReady records a missing theorem, evaluator, or closure certificate. None asserts the contradiction required for a solved-empty fibre. $\square$

6.4 E03: one-point simultaneous survival

Definition 6.4 (Joint physical groupoid). The groupoid PhysMod_E^τ is the full subgroupoid of joint physical presentations over $\mathbf{S}_E^\tau$ for which every applicable residual in $\mathcal{E}_E^\tau$ passes and every N/A status is certified. Morphisms are lawful physical isomorphisms preserving the complete target fingerprint, source trace, incidence diagram, response records, and common-cone witness.

Theorem 6.5 (E03 – one-point simultaneous survival). *Every object of PhysMod_E^τ is one source-coherent physical realization of the complete branch target. It has exact marginal physical reducts, represents every target-active cross and higher-arity incidence, and evaluates the retained oscillation interface with all D1–D5 structures present. Conversely, any semantic physical realization of T_E^τ in the declared comparison class determines an object of PhysMod_E^τ up to lawful isomorphism.*

Proof. By construction, an object of PhysMod_E^τ lies over one joint source object, and the common-cone residual requires one apex. The marginal reduct functors of Theorem 5.7 and the physical restriction of the incidence

diagram produce the five marginal physical objects. Passing the pure marginal residuals proves that each reduct lies in the corresponding marginal physical category. Passing the cross, background, matching, continuation, higher-arity, and cone residuals proves co-instantiation rather than splicing.

The retained oscillation evaluator is part of $\mathcal{E}_{\text{osc}}$ and the in-situ transparency residual. It is applied before deleting any D cells; hence the base is recovered on the enriched object itself.

Conversely, let W be a semantic realization of the complete target. By the semantic-use and cross-relation saturation theorem [Theorem 4.5](#) and the joint source-normal-form theorem [Theorem 5.8](#), its target-facing and source-facing data factor through τ_E^τ , so every target-active pairwise and higher-arity incidence is represented in the joint source object rather than inferred from a marginal label. Its marginal restrictions determine the D objects, and its physical overlap diagram supplies the common cone. Every target residual passes by semantic adequacy. Therefore W defines an object of PhysMod_E^τ . Lawful presentation choices give isomorphic objects. $\square$

Corollary 6.6 (No marginal splicing). *A tuple of marginal physical points with different vacua, backgrounds, base sources, matching maps, or apex witnesses is not an object of PhysMod_E^τ , even when every marginal point passes independently.*

6.5 In-situ base conservativity

Let Φ_4 be the complete PCR-4 retained evaluator and Ξ_{osc} the SCIC packet. The joint extension cells generate a relative target ideal $\mathcal{I}_E^\tau$ when evaluated by old-target functionals.

Theorem 6.7 (Joint retained-oscillation transparency). *For every $W_E \in \text{PhysMod}_E^\tau$,*

$$\Phi_4^{\text{ret},E}(W_E) = \Phi_4(W_{\text{base}}), \quad \Xi_{\text{osc}}^{\text{ret},E}(W_E) = \Xi_{\text{osc}}(W_{\text{base}}).$$

These identities hold with the D1–D5 cells physically present and do not use marginal or joint correction deletion.

Proof. Each marginal handoff contains an in-situ retained-base theorem or an exact retained restriction. The joint source equations identify the repeated base cells and require every cross incidence to be old-target transparent. Thus the joint-only generators lie in the intersection of the kernels of the old-target evaluators. The old evaluator algebra is closed under composition and continuation, so the full relative ideal is annihilated. The equalities follow channel by channel. If a cross incidence changed an old-target channel, the corresponding residual would fail and the point would not lie in PhysMod_E^τ . $\square$

6.6 E04: marginal-to-joint obstruction

Definition 6.8 (Obstruction groupoid). For $q \in \mathcal{Q}_{\text{osc}}^{\text{SCIC}}$, let $\mathcal{O}_E^\tau(q)$ be the groupoid whose objects are common-base marginal tuples equipped with their complete residual-status vectors and first-failure records. Morphisms preserve target tags, applicability, and failure provenance.

Definition 6.9 (Joint compatibility equalizer). The joint compatibility groupoid is the full sub-groupoid

$$\mathcal{J}_E^\tau(q) \subseteq \mathcal{P}_{\text{marg}}^\tau(q)$$

consisting of tuples that admit a joint source object and a one-apex physical lift with all applicable residuals passing.

Theorem 6.10 (E04 – exact marginal-to-joint obstruction). *For every branch τ and base class q ,*

$$\mathcal{L}_J^\tau(q) \simeq \mathcal{J}_E^\tau(q) \simeq \text{Eq}\left(\mathcal{P}_{\text{marg}}^\tau(q) \rightrightarrows \mathcal{O}_E^\tau(q)\right).$$

The fibre is solved empty if and only if every common-base tuple has a typed applicable failure obstruction and no tuple has Undef, ROOS, SU, TU, or NotReady status. If a non-failure status remains, the fibre receives the corresponding readiness classification rather than emptiness.

Proof. An object of the equalizer is precisely a common-base tuple whose two residual maps agree with the all-pass/N/A certificate and whose source-amalgamation data exist. By [Theorem 6.5](#), this is equivalent to a joint one-apex physical object, hence to an object of the joint fibre.

If every tuple has a certified applicable failure, no joint object exists and the failure atlas is exhaustive; the fibre is solved empty. Conversely, solved emptiness at the declared source scope requires that every source presentation be

dispositioned by a contradiction or failed residual. An undefined, response-out-of-scope, update, or readiness point is not a contradiction and therefore blocks solved emptiness. $\square$

Corollary 6.11 (Intersection is only an upper bound).

$$\text{im } \rho_J^\tau \subseteq \bigcap_{\alpha} \text{im } \rho_{\alpha},$$

and the inclusion can be strict because $O_E^\tau(q)$ may contain a joint-only obstruction.

7 Joint restriction, no merger, fibres, and path equivalence

7.1 Joint standing equivalence

Let W_E^τ be the complete branch target-admitted witness family, including all marginal, cross, failure, and continuation witnesses. Let U_E^τ be the licensed continuation monoid. For $W \in \text{PhysMod}_E^\tau$, define the complete joint continuation signature

$$\Sigma_E^\tau(W) = (w(uW))_{(w,u) \in W_E^\tau \times U_E^\tau}.$$

The joint SCIC equivalence is

$$W \sim_E^\tau W' \iff \Sigma_E^\tau(W) = \Sigma_E^\tau(W').$$

The activation burden is not assumed from the displayed coordinate list. It is discharged by the target/source/residual construction: every same-scope standing-bearing difference changes a target field, a source-incidence consequence, a residual, a failure status, or a licensed continuation witness.

Theorem 7.1 (Joint SCIC activation). *At every declared completed or target-qualified branch scope, W_E^τ is target-complete. Consequently $\sim_E^\tau$ is the unique same-scope standing carrier identity relation compatible with stable reference, lawful redescription, continuation stability, and reuse consistency.*

Proof. The target colimit includes all marginal target witnesses and the independently generated cross grammar. The source category includes every target-active source incidence. The physical residual normal form includes compatibility, failure, background, matching, continuation, and common-cone data. Thus an anchor change leaves the branch scope; a tensor-level same-scope change alters at least one included witness; and a skin/provenance change alters none. Fixed-domain carrier descent and the no-second-classifier theorem then give target completeness and uniqueness of the induced quotient. $\square$

7.2 E05: restriction and no merger

Definition 7.2 (Joint restriction). The functor

$$R_J^\tau : \mathcal{N}_E^\tau \longrightarrow \mathcal{N}_{\text{osc}}$$

forgets the D1–D5-only cells and cross relations while retaining their effects on every base channel. The marginal restrictions

$$R_{J \rightarrow \alpha}^\tau : \mathcal{N}_E^\tau \longrightarrow \mathcal{N}_\alpha$$

forget all but the base and the α -leg cells.

Theorem 7.3 (E05 – joint restriction, descent, and no merger). *The restriction functors are well defined on lawful isomorphism classes and satisfy*

$$R_\alpha \circ R_{J \rightarrow \alpha}^\tau \cong R_J^\tau \quad \text{for every marginal leg } \alpha.$$

They descend to connected-component quotients

$$\rho_J^\tau : \mathcal{Q}_J^\tau \longrightarrow \mathcal{Q}_{\text{osc}}^{\text{SCIC}}, \quad \rho_{J \rightarrow \alpha}^\tau : \mathcal{Q}_J^\tau \longrightarrow \mathcal{Q}_\alpha.$$

If $q \neq q'$ in $\mathcal{Q}_{\text{osc}}^{\text{SCIC}}$, no joint class restricts to both. Hence enrichment cannot merge distinct base SCIC classes.

Proof. A lawful joint arrow preserves the complete source trace, target fingerprint, response frames, background, matching, and continuation data. Its base restriction is therefore a lawful arrow in $\mathcal{N}_{\text{osc}}$, so R_J^τ is functorial and representative independent. The marginal commutative squares follow from the exact source and physical reducts.

If $q \neq q'$, PCR-5B/SCIC completeness supplies a retained base witness and continuation separating them. By Theorem 6.7, the same witness has the same value on the enriched object as on its retained base. A joint object cannot therefore restrict to both classes. This proves no merger and the descent of restriction to connected components. $\square$

7.3 Connected joint lift fibres

Definition 7.4 (Connected joint fibre). For $q \in \mathcal{Q}_{\text{osc}}^{\text{SCIC}}$,

$$\mathcal{L}_J^\tau(q) = \mathcal{N}_E^\tau \times_{\mathcal{N}_{\text{osc}}}^h \mathcal{N}_{\text{osc}}[q].$$

Its connected components are the enriched joint SCIC classes above q .

The fibre retains automorphisms. A single connected component may contain many presentations, and a set of sampled representatives cannot determine its isotropy or overlap structure.

7.4 Source charts and solution objects

Let $a = (F, \sigma, \kappa)$ index a B15 source theorem plan, a joint extension theorem plan, and a target branch. The raw parameter object Θ_a^τ contains all source parameters, response frames, background data, matching variables, action/breaking data, and cross-incidence coordinates. The applicability-guarded residual system is

$$\mathcal{E}_a^\tau(\theta) = 0, \quad \text{App}_{a,j}^\tau(\theta) \in \{0, 1\}.$$

The exact solved object is

$$Z_a^\tau = \{\theta \in \Theta_a^\tau : \text{every applicable residual passes and every N/A is certified}\}.$$

The lawful presentation groupoid Γ_a^τ is generated by source hypergraph isomorphisms, source automorphisms, fixed-boundary gauge and basis changes, response-frame isomorphisms, background chart maps, charge-lattice normalization, lawful parameter charts, and the target-admitted matching/gluing transformations.

The chart groupoid is $[Z_a^\tau / \Gamma_a^\tau]$. Its map to the concrete joint standing-core groupoid records the complete solved joint packet and framed provenance.

7.5 E06: exact fibre atlas and codescent

Theorem 7.5 (E06 – exact joint fibre and atlas assembly). *Fix a branch scope at which every source chart and complement is either solved or carries an exact typed readiness status. For every $q \in \mathcal{Q}_{\text{osc}}^{\text{SCIC}}$, the joint fibre is equivalent to the codescent of its source-indexed Čech nerve:*

$$\mathcal{L}_J^\tau(q) \simeq \text{codesc } \mathcal{U}_{E,\bullet}^\tau(q).$$

The global joint groupoid and quotient are

$$\mathcal{N}_E^\tau \simeq \text{codesc } \mathcal{U}_{E,\bullet}^\tau, \quad \mathcal{Q}_J^\tau \cong \coprod_{q \in \mathcal{Q}_{\text{osc}}^{\text{SCIC}}} \pi_0 \mathcal{L}_J^\tau(q).$$

For a target-qualified subcategory, these formulas hold with the subcategory tag, while the parent complement remains explicit.

Proof. The source grammar provides a jointly surjective family of chart functors onto the solved semantic category at the announced scope. Pairwise overlaps are homotopy fibre products over the concrete joint standing core. Higher overlaps are iterated homotopy fibre products with the diagonal action of the core automorphism group. The resulting simplicial groupoid is the source-indexed Čech nerve.

Descent is effective because objects agreeing on overlaps carry compatible source, response, physical, and provenance gluing data; conversely every semantic object has a local source-normal form. Codescent reconstructs the joint groupoid. Taking connected components and using no merger gives the disjoint decomposition over the base. If a parent chart is outside the closed subcategory, it is retained in the complement with its ROOS or NotReady tag and is not included in the subcategory codescent. $\square$

Remark 7.6 (Exactness is symbolic). The theorem does not require a finite list of joint models. A chart may have continuous, functional, or categorical moduli. Exactness means that its source domain, residual system, lawful groupoid, overlaps, and complement are specified at the inherited symbolic source resolution.

7.6 E07: order and path independence

The marginal targets can be added in many orders. PCR-5C shows that staged target extension is coherent only when the final targets are genuinely the same object over the base.

Definition 7.7 (Lawful target-addition path). A path p is a finite sequence of target embeddings beginning at T_{osc} and ending at an object isomorphic to T_E^τ , with source reducts and physical restrictions at every step. Two paths are endpoint-equivalent when their final target fingerprints agree and the comparison isomorphism preserves all target data.

Theorem 7.8 (E07 – order and path independence). *If p and p' are endpoint-equivalent lawful target-addition paths for the same branch τ , then their joint source, physical, fibre, and quotient objects are naturally equivalent in the slice over T_{osc} . The equivalence preserves base arrows, isotropy, source activity, residual status, and continuation signatures.*

If the paths end in different D3 programmes, different D4 resolutions, or different D5 target records, no path-independence theorem is asserted; the relation is a target update.

Proof. For endpoint-equivalent paths, the staged source categories are iterated homotopy limits of the same finite target/source diagram. Homotopy limits are invariant under rebracketing and permutation up to coherent equivalence. Nested activity reduction commutes with staged forgetting because activity is evaluated pointwise from the final source equations, not from the order of construction. The physical category is the equalizer of the same complete residual family, and the connected fibres are homotopy pullbacks of equivalent functors. These equivalences are compatible with codescent and yield the quotient equivalence.

A path changing D3 programme, D4 resolution, or D5 target changes the endpoint fingerprint. The preceding argument no longer has one common diagram, so path independence is inapplicable rather than false. $\square$

7.7 Failure and complement decomposition

For every branch and base class the raw fibre decomposes as

$$\mathcal{P}_{\text{marg}}^\tau(q) = \mathcal{L}_J^\tau(q) \sqcup \mathcal{F}_E^\tau(q) \sqcup \mathcal{U}_E^\tau(q) \sqcup \mathcal{R}_E^\tau(q) \sqcup \mathcal{S}_E^\tau(q) \sqcup \mathcal{T}_E^\tau(q) \sqcup \mathcal{N}_E^\tau(q),$$

representing pass, fail, undefined, ROOS, source update, target update, and NotReady components. The union is disjoint at status level even when a source presentation admits several first-failure explanations; a canonical ordered failure rule records the earliest load-bearing obstruction.

8 Common vectorlike realization of D3-1.3 and D5

This section discharges the load-bearing marginal source interface required by the positive joint branches. ARC-U1 has already admitted D3-1.3 as an additional separately tagged PCR-5E successor branch. It does not begin from a D3 survivor label. It begins from the raw source class frozen before response evaluation by [Theorem 4.11](#), lands that same D5 exact-response vectorlike Dirac source in the frozen D3-1.3 source and process ontology, and evaluates the complete H/U/L target on one physical apex. No D1, D2, D4, joint-image, fibre-cardinality, or PCR-6 conclusion is used in the landing theorem.

8.1 Common positive light-sector source

Fix one lawful three-light-mode oscillation frame with boundary/mixing data

$$(U, V, \mathbf{P}, \widehat{\mathbf{w}}^{(e)}), \quad \widehat{w}_i^{(e)} = |U_{ei}|^2, \quad \sum_i \widehat{w}_i^{(e)} = 1,$$

and fixed nonzero mass-squared differences in the normal-ordering component. For

$$m_0 \in I_m^\circ = [m_a, m_b] \Subset (0, m_\star]$$

define the positive light spectrum

$$\mathbf{m}(m_0) = (m_1(m_0), m_2(m_0), m_3(m_0)), \quad \Sigma(m_0) = \text{diag}(m_1, m_2, m_3),$$

and the source-formed light object

$$\mathbf{M}_{\text{light}}(m_0) = (\mathbf{m}(m_0), \mathbf{P}, \widehat{\mathbf{w}}^{(e)}, \text{CC}, \text{Prop}, \text{Match}_{\text{light}}).$$

The active normalization factor introduced below is retained separately by D3. On the D1 endpoint chart it multiplies the common signal normalization and does not alter the normalized endpoint-shape weights $\widehat{\mathbf{w}}^{(e)}$. Hence the D1 shape/support record and the D4 mode-indexed mass record descend from the same light object without conflating the D3 closure coordinate with a mass coordinate.

8.2 Exact-response vectorlike source normal form

For $(m_0, \lambda, s) \in \Theta_{\text{VJ}}^{\text{raw}}$, put

$$H_\lambda = \lambda \Lambda, \quad c = (1 - s^2)^{1/2}, \quad M_D^\star(m_0) = U \Sigma(m_0) V^\dagger,$$

and define

$$Y_V(m_0, s) = \frac{\sqrt{2}}{v} c U \Sigma(m_0) V^\dagger, \quad Y_{LN}(\lambda, s) = -\frac{\sqrt{2}}{v} s H_\lambda U,$$

$$\mu(m_0, s) = s \Sigma(m_0) V^\dagger, \quad M_{N,\lambda,s} = c H_\lambda \mathbf{1}.$$

With three active right-handed neutrinos and equal-rank singlet multiplets N_L, N_R , all of conventional $B - L = -1$, the source operators are

$$-\bar{L} \tilde{H} Y_{\nu_R} - \bar{L} \tilde{H} Y_{LN} N_R - \bar{N}_L \mu \nu_R - \bar{N}_L M_N N_R + \text{h.c.}$$

The complete Dirac mass map has the exact factorization

$$\mathbb{M}_{m_0, \lambda, s} = \mathcal{U}_L(s) \begin{pmatrix} \Sigma(m_0) & 0 \\ 0 & H_\lambda \mathbf{1} \end{pmatrix} \mathcal{U}_R(s)^\dagger.$$

The light and heavy charged-current blocks are cU and $-sU$. The light pole spectrum is $\mathbf{m}(m_0)$, the heavy pole spectrum is the exactly degenerate triple H_λ , and the kinetic terms remain canonical.

Lemma 8.1 (Source coherence, activity, and D5 physicality). *Every object $x(m_0, \lambda, s) \in \mathbb{S}_{\text{VJ}}^{\text{raw}}$ has a finite source-coherent PCR-3 normal form in the LOCAL/HID/HYB source grammar. For $s > 0$, the active-vectorlike Yukawa incidence and the heavy charged-current projection are nonzero, so activity reduction does not remove the vectorlike block. On one physical apex the source is gauge invariant, exactly $B - L$ covariant, vectorlike-anomaly neutral, vacuum compatible, and equipped with the D5 finite-order matching right inverse that fixes the complete target-retained light continuation while retaining H_λ and its oriented matching arrow.*

Proof. All fields and arrows are among the finite local, hidden, matching, and hybrid constructors admitted by the PCR-3 source theory. The displayed coefficients satisfy the constitutive mass-map equations exactly, and the unitary block factorization supplies the physical light/heavy decomposition. Since $s > 0$, both the source coupling to N_R and the charged-current heavy projection are active, so the heavy block lies on a target path and is not a spectator. Each displayed operator has net $B - L = 0$; the equal-rank N_L, N_R pair is vectorlike and contributes no net local anomaly. The scalar vacuum and kinetic data are shared with the base source. The D5 recursive matching theorem applies to the same source coefficients and preserves all retained light outputs while leaving the physical threshold visible. These statements concern source formation and physical coherence, not D3 target acceptance. $\square$

8.3 Frozen D3-1.3 response system

Let $\mathcal{J}_H$, $\mathcal{J}_U$, and $\mathcal{J}_L$ be the finite H, U, and L response registries of the D3-1.3 target on the vectorlike source topology. Each entry contains its source/scale/topology applicability guard, exact response transport, accepted set, and Pass/N/A/ROOS/update semantics. Let $\mathcal{J}_{\text{proc}}$ be the finite process-scope and retained-tester record. For a fixed applicability cell, write

$$R_{H,j}(m_0, \lambda, s), \quad R_{U,k}(m_0, \lambda, s), \quad R_{L,\ell}(m_0, \lambda, s)$$

for the corresponding transported responses.

The D3 direct-state theorem supplies continuity in the source coupling and an accepted neighbourhood of the exact zero response on every applicable null-search chart. The exact-unitary point lies in the interior of every applicable null-centred U acceptance region, and the D3 target provides a positive low-mass universality margin. These facts are properties of the frozen marginal target; they are not inferred from PCR-5E survival.

Theorem 8.2 (E-S4A – uniform vectorlike D3/D5 strict-margin product-box theorem). *There exist a compact nondegenerate rational threshold interval*

$$I_E = [\lambda_-, \lambda_+] \Subset I_\lambda^{\text{src}}$$

and a compact positive interval

$$I_s^E = [s_-, s_+] \Subset (0, s_{\text{src}}^{\text{max}})$$

such that the full Cartesian box

$$I_m^\circ \times I_E \times I_s^E$$

is contained in one source-topology, applicability, routing, matching-orientation, continuation, and failure cell of the frozen D3-1.3/D5 marginal evaluator. For every point in that box:

1. *every H chart is in its exact Pass, N/A, or ROOS class and every applicable H response lies in its frozen accepted set;*
2. *every applicable U response lies in its frozen accepted set and the retained tester satisfies*

$$C_{ee}^{\text{op},1.3} = 1 - s^2 > 0.987;$$

3. *the complete L response lies strictly inside the frozen acceptance set;*
4. *process scope and quantitative coverage are complete;*
5. *no source-activity, target, topology, applicability, matching, continuation, or failure frontier is crossed;*
6. *the exact-global and bounded-accidental D5 predicates hold on the same source and physical apex.*

The theorem is uniform over the entire pre-frozen interval I_m° ; it is not obtained by choosing an interval inside the projection of a thinner routing cell.

Proof. Guard factorization. The D3-1.3 H, U, and process-scope target records contain heavy-state production/propagation/decay data, active-sector closure, weak-universality data, the retained process tester, and the fixed relative oscillation packet. They contain no D1 absolute-support or D4 mass-sum evaluator. On the displayed vectorlike source, every H/U/process applicability guard therefore factors through

$$(\lambda, s; \mathcal{G}_{\text{src}}, U, V, \Delta m_{21}^2, \Delta m_{31}^2)$$

and is independent of the common light absolute offset m_0 . For every $s > 0$, the same active-vectorlike incidences are nonzero, so the source topology and activity vector are constant throughout the punctured source domain. The D5 action, anomaly, and protector equations are exact polynomial identities in the displayed charges and source maps and hold throughout that domain.

Uniform H and U margins at a positive source point. Choose λ_0 away from the finite set of H/U topology, scale, lifetime, and matching guard frontiers. On a sufficiently small one-sided neighbourhood of $s = 0$, every target-active direct-state response is of the form

$$R_{H,j}(m_0, \lambda, s) = s^{r_j} \bar{R}_{H,j}(\lambda, s), \quad r_j \geq 1,$$

with $\bar{R}_{H,j}$ continuous and bounded on the compact source box. Hence all applicable null-search responses converge uniformly to the accepted zero response as $s \downarrow 0$. Likewise every nontrivial U deviation is a continuous function of

s^2 , and the mandatory retained tester is exactly $1 - s^2$. Since the exact-unitary point has strict U margin, there is a positive s_0 for which every applicable H and U inequality is strict at (λ_0, s_0) , uniformly for all $m_0 \in I_m^\circ$, and with $s_0^2 < 1 - 0.987$.

Uniform L margin. The heavy contribution vanishes identically for every (m_0, λ, s) because the heavy poles are degenerate and

$$\sum_k U_{ek} U_{\mu k}^* = 0.$$

For the light contribution, unitarity permits subtraction of any common loop value:

$$\mathcal{A}_{\mu e}^L(m_0, s) = c^2 \sum_i U_{ei} U_{\mu i}^* \left[F_\gamma \left(\frac{m_i(m_0)^2}{m_W^2} \right) - F_\gamma \left(\frac{m_1(m_0)^2}{m_W^2} \right) \right].$$

On the compact light interval, the mean-value theorem gives

$$|\mathcal{A}_{\mu e}^L(m_0, s)| \leq \frac{K_\gamma}{m_W^2} \sum_i |U_{ei} U_{\mu i}^*| |\Delta m_{i1}^2|,$$

where K_γ is the supremum of $|F'_\gamma|$ on the compact argument interval. The right-hand side is independent of m_0 and is strictly below the frozen L acceptance bound by the inherited light-GIM margin. Thus the L chart is uniformly accepted on the full I_m° .

Open product cell. At the positive point (λ_0, s_0) , every applicable numerical inequality is strict, every active incidence is nonzero, and no guard equality or matching-orientation frontier is met. The finite routing vector and all response maps are continuous on the fixed positive-source topology. Therefore the complete Pass/routing locus contains an open rectangle

$$J_\lambda \times J_s \ni (\lambda_0, s_0)$$

in the (λ, s) -plane. The preceding factorization and uniform L bound make this rectangle valid for every $m_0 \in I_m^\circ$, not merely for a projected subset. Choose compact rational subintervals

$$I_E \subseteq J_\lambda, \quad I_s^E \subseteq J_s$$

with positive length and with $s_+^2 < 1 - 0.987$. Process-scope completeness is constant on this source topology because every target-active finite slot is present in the deterministic extension. Exact D5 covariance and anomaly cancellation are source identities. This proves the stated full Cartesian product inclusion. $\square$

Corollary 8.3 (Uniform positive D3 response radius). *Let*

$$s_\star := s_+.$$

Then every point of

$$I_m^\circ \times I_E \times I_s^E$$

has the complete H/U/L/process and D5 status stated in Theorem 8.2; in particular $s_\star^2 < 1 - 0.987$ and $s_\star < s_{\text{src}}^{\max}$.

Definition 8.4 (Canonical marked response choice). Let s_E be the first positive dyadic rational 2^{-n} , in the fixed natural-number order, lying in the interior of the pre-proved interval I_s^E . Let I_E be the first compact rational interval selected by the fixed lexicographic rectangle rule in the proof of Theorem 8.2. This deterministic solved-chart choice is made inside the pre-frozen source comparison certificate and depends only on the frozen D3/D5 source and target records. It contains no joint survivor, image, cardinality, or preferred PCR-6 endpoint.

Put $c_E = (1 - s_E^2)^{1/2}$, $\delta_E = s_E^2$, and

$$x_{m_0, \lambda} = x(m_0, \lambda, s_E).$$

Corollary 8.5 (E-S4 – D3-1.3/D5 common vectorlike source landing and physical survival). *For every $(m_0, \lambda) \in I_m^\circ \times I_E$, under the ARC-U1-authorized D3-1.3 branch, the same source $x_{m_0, \lambda}$ is simultaneously:*

1. *a coherent finite PCR-3/B15 source and a finite-operational D3-1.3 process law;*
2. *a one-apex physical realization with every inherited and D3 common-cone residual equal to zero;*
3. *H-admissible with every direct-state instrument in its exact Pass, N/A, or ROOS class;*

4. *U-admissible on every applicable chart, with retained tester*

$$C_{ee}^{\text{op},1.3}(x_{m_0,\lambda}) = c_E^2 = 1 - \delta_E > 0.987;$$

5. *L-admissible for the complete $\mu \rightarrow e\gamma$ response;*

6. *process-scope complete, quantitatively covered, and retained-oscillation transparent with the complete B15 matching and continuation packet;*

7. *exact-global and bounded-accidental B – L admissible on the D5 side.*

Unequal values of λ remain distinguished by the physical threshold H_λ and its oriented B15 matching record. The theorem closes this explicit finite-operational analytic source chart directly; it does not claim closure of the remaining D3 finite-operational source universe.

Proof. The full Cartesian solved region, positive-source activity stability, and uniform H/U/L/process margins are supplied by E-S4A, [Theorem 8.2](#). The remaining work is the explicit same-source physical landing.

Source law and process scope. By [Theorem 8.1](#), the source is a finite object of the PCR-3 source theory, not a process label attached after evaluation. Its finite-dimensional propagation law and finitely many detector/intervention slots admit the positive process-comb representation required by D3-1.3. Every source-active U primitive is represented in the deterministic process extension, so the quantitative-coverage condition is satisfied.

One physical apex. Gauge invariance, exact B – L covariance, vectorlike anomaly cancellation, canonical kinetic terms, the common scalar vacuum, the exact block mass factorization, charged-current attachment, and finite-order matching coexist in one Lagrangian/matching/propagation object. No H, U, L, D5, or retained-base datum is taken from a different source point.

H response. For every applicable direct-state chart, source formation gives zero response at $s = 0$, and the exact response is continuous in s on the selected compact routing cell. The dyadic value $s_E < s_\star$ therefore lies in the common accepted neighbourhood established by [Theorem 8.2](#). Nonapplicable instruments retain N/A or ROOS according to their actual mass, lifetime, topology, and scale guards. No scalar limit is transported between topologies.

U response. The active block is $N = c_E U$, so

$$NN^\dagger = c_E^2 \mathbf{1}.$$

The retained process tester is exactly $c_E^2 > 0.987$. Every other applicable U response passes by the uniform strict-margin product box of [Theorem 8.2](#); every inapplicable or out-of-response chart retains its typed status. Thus a favourable tester coordinate is not being substituted for the complete U packet.

L response. No independent CLFV source is present. The heavy poles are exactly degenerate and their charged-current matrix is $-s_E U$, hence the heavy loop contribution is

$$\mathcal{A}_{\mu e}^H \propto s_E^2 F_\gamma(H_\lambda^2/m_W^2) \sum_{k=1}^3 U_{ek} U_{\mu k}^* = 0$$

by row orthogonality. The light contribution is the inherited GIM-suppressed term, multiplied by the common factor c_E^2 , and remains below the frozen MEG-II acceptance bound. The L pass is therefore a source-defined physical result, not a fitted null.

Retained base and matching. The D5 finite-order recursive right inverse preserves every target-retained light coefficient, boundary value, discrete continuation status, and failure record over I_E . Deterministic D3 restriction recovers the complete oscillation standing core in situ. The heavy threshold and oriented matching arrow remain standing-visible, so unequal λ are not related by lawful source, gauge, basis, chart, or matching transport.

D5 targets. Every displayed operator is exactly B – L covariant, the vectorlike pair is anomaly neutral, and the three active ν_R fields supply the required chiral packet. The faithful global action and continuation-stable protector therefore hold. Exact-global standing implies the bounded accidental predicate through active dimension four.

All conclusions are proved on the displayed common source and apex. The proof uses the D3 target response maps and the D5 source equations but no joint-image or cardinality assumption. $\square$

Corollary 8.6 (Complete marginal bridge export). *For every $(m_0, \lambda) \in I_m^\circ \times I_E$, the complete marginal status packet is*

$$\text{Status}_{D3/D5}^{1.3, \tau_5}(x_{m_0, \lambda}) = (\text{SourcePass}, \text{PhysicalPass}, \text{H/U/LPass}, \text{ProcessPass}, \text{RetainedBasePass}, \text{D5Pass})$$

for $\tau_5 \in \{T_{\text{acc}}^{(4)}, T_{\text{glob}}\}$. *The export contains the exact source groupoid, parameter interval, response margins, Pass/N/A/ROOS routing, matching and continuation packet, and failure frontier. It identifies an explicit solved Pass stratum inside the raw precursor source-comparison certificates already frozen by Theorem 4.11.*

Theorem 8.7 (Total vectorlike-joint marginal status decomposition). *For each $\tau_5 \in \{T_{\text{acc}}^{(4)}, T_{\text{glob}}\}$, the pre-frozen raw precursor source groupoid $S_{\text{VJ}, \text{raw}}^{\tau_5}$ admits the disjoint total decomposition*

$$S_{\text{VJ}, \text{raw}}^{\tau_5} = S_{\text{VJ}, \text{Pass}}^{\tau_5} \sqcup S_{\text{VJ}, \text{Fail}}^{\tau_5} \sqcup S_{\text{VJ}, \text{N/A}}^{\tau_5} \sqcup S_{\text{VJ}, \text{ROOS}}^{\tau_5} \sqcup S_{\text{VJ}, \text{SU}}^{\tau_5} \sqcup S_{\text{VJ}, \text{TU}}^{\tau_5}.$$

The status of an object is the complete product of its source, physical, H, U, L, process-scope, retained-base, matching/continuation, and D5 target statuses. The Pass stratum contains

$$I_m^\circ \times I_E \times \{s_E\} \times I_\tau^\circ \times Z_{\tau_5}, \quad Z_{T_{\text{acc}}^{(4)}} = I_{76}, \quad Z_{T_{\text{glob}}} = \{0\}.$$

Every point outside that displayed solved stratum retains its actual total status and typed failure or update certificate; no complement is converted into an empty fibre. The decomposition is invariant under the lawful source and parameter groupoid.

Proof. The D3-1.3 evaluator is total and fail closed on the finite-operational comparison class: every H, U, L, process, applicability, retained-base, matching, and continuation entry has Pass, Fail, N/A, ROOS, source-update, or target-update status. The D5 accidental and exact-global evaluators are likewise total on the displayed source grammar. Their finite product, together with source and one-apex physicality, gives the disjoint decomposition. Congruence under lawful arrows follows from the marginal transport theorems. The explicit Pass inclusion is Theorem 8.5 together with the frozen D2 source domains and the source-guarded D4 extension. Since the raw precursor certificate was fixed in Theorem 4.11 before any of these values were evaluated, the decomposition is an endpoint of the frozen target rather than a survivor-defined target. $\square$

8.4 Source-qualified vectorlike-joint comparison class

Definition 8.8 (Source-qualified vectorlike-joint comparison category). For $\tau_5 \in \{T_{\text{acc}}^{(4)}, T_{\text{glob}}\}$, let

$$S_{\text{VJ}}^{\tau_5} := S_{\text{VJ}, \text{raw}}^{\tau_5} \big|_{I_m^\circ \times I_E \times \{s_E\} \times I_\tau^\circ \times Z_{\tau_5}}, \quad Z_{T_{\text{acc}}^{(4)}} = I_{76}, \quad Z_{T_{\text{glob}}} = \{0\}.$$

This full lawful subgroupoid is formed from the E-S4 marginal bridge export: the same D5 source-formed vectorlike family has complete D3-1.3 H/U/L, process-scope, retained-base, matching/continuation, and D5 Pass status on $I_m^\circ \times I_E \times \{s_E\}$. The independent coordinate $\tau \in I_\tau^\circ$ belongs only to the pre-frozen D1/D4 conservative source extension and is erased by the D3/D5 reduct, so the E-S4 marginal status is uniform in τ . The D1, D2, and D4 source domains and target records were already present in the E-S3 raw precursor; no joint cross residual, common-apex verdict with all five legs, image, fibre, standing-coordinate, cardinality, ROC, or PCR-6 result is used in this restriction.

Define the source-qualified target-and-comparison certificate

$$T_{E, \text{VJ}}^{\tau_5} := T_{E, \text{VJ}, \text{raw}}^{\tau_5} \big|_{S_{\text{VJ}}^{\tau_5}}.$$

It is a comparison-qualified child certificate of the raw precursor and parent branch, not a claim that either larger scope has been completed.

Theorem 8.9 (E-S5 – source-qualified vectorlike-joint comparison freeze). *The fingerprints*

$$\text{FP}(T_{E, \text{VJ}}^{T_{\text{acc}}^{(4)}}), \quad \text{FP}(T_{E, \text{VJ}}^{T_{\text{glob}}})$$

are fixed by E-S3 and the independently proved E-S4 marginal bridge before any genuinely joint D1/D2/D4 cross-relation calculation, five-leg one-apex amalgamation, joint base-image determination, lift-fibre reconstruction,

standing-coordinate diagnosis, cardinality theorem, Role-Occupancy Closure, or PCR-6 handoff. Their source domains are exactly

$$I_m^\circ \times I_E \times \{s_E\} \times I_\tau^\circ \times Z_{\tau_5},$$

and their complements inside the E-S3 raw precursor retain the total Pass/Fail/N/A/ROOS/source-update/target-update decomposition of Theorem 8.7. Therefore the later proper-continuum theorem is target-and-comparison-qualified but is not survivor-defined or endpoint-backfitted.

Proof. E-S3 fixes the complete raw source grammar, parameter box, D1/D2/D4 source domains, D5 target tag, target evaluators, continuations, failure codomains, and lawful arrows before D3/D5 evaluation. E-S4 then proves, solely from the frozen D3-1.3 and D5 marginal targets, that the canonical interval I_E and dyadic point s_E carry complete marginal Pass data on the same source and physical apex. Restricting a previously fixed comparison category by an independently established marginal theorem produces an explicit comparison-qualified child certificate. At this stage no joint D1–D5 cross system has been solved and no joint image, fibre, cardinality, source-sufficiency, standing-coordinate, or physical-nature conclusion exists. Hence the source-qualified fingerprint is prior to every result for which it serves as domain. The total complement remains recorded by Theorem 8.7, so no unqualified parent claim or hidden emptiness is introduced. $\square$

9 Cross-phenomenon compatibility equations

The semantic saturation theorem Theorem 4.5 fixes the complete cross-relation domain before the branch calculations. This section solves the relations used by the vectorlike-joint targets and records the relations that remain readiness-gated at parent scope.

9.1 D1–D4 common light spectral object

Let a conventional light-sector joint source carry a positive spectral resolution

$$M_{\text{light}} = (\{m_i\}_{i=1}^r, \{P_i\}_{i=1}^r, \{w_i^{(e)}\}_{i=1}^r), \quad m_i \geq 0, \quad \sum_i w_i^{(e)} = 1, \quad w_i^{(e)} \geq 0.$$

The D1 electron-boundary effective mass and the D4 mass sum are

$$m_\beta^2 = \sum_i w_i^{(e)} m_i^2, \quad \Sigma_\nu = \sum_i m_i$$

on the standard light branch.

Lemma 9.1 (Mass-sum dominance of the beta endpoint). *For every positive light spectral object,*

$$m_\beta \leq \max_i m_i \leq \Sigma_\nu.$$

Consequently, every D4 public-summary point with $\Sigma_\nu \leq \Sigma_{95}^{\text{pub}} < 0.45 \text{ eV}$ automatically satisfies the scalar KATRIN bound $m_\beta < 0.45 \text{ eV}$.

Proof. Since the $w_i^{(e)}$ are nonnegative and sum to one,

$$m_\beta^2 = \sum_i w_i^{(e)} m_i^2 \leq \max_i m_i^2.$$

Taking square roots gives the first inequality. The second follows from nonnegativity of all masses. The numerical comparison is part of the frozen target bands. $\square$

Theorem 9.2 (D1–D4 mass-response factorization). *On the standard-thermal light-sector branch, the D1 and D4 minimal response maps factor through one source-formed light spectral object:*

$$S_{D1} = f_\beta(M_{\text{light}}), \quad S_{D4} = f_\Sigma(M_{\text{light}}, \text{PopHist}_{\text{th}}).$$

The D4 public-summary condition is a scale-history refinement of D1, not an independent Cartesian coordinate. The joint D1–D4 source space is therefore the equalizer of their mass records rather than the product of arbitrary marginal fibres.

Proof. The D1 endpoint spectral measure is the pushforward of the same light mass projectors through the electron boundary weights. The D4 standard-thermal source uses those light mass records in the three compiled population slots and evaluates their scalar sum inside a fixed population/background history. Hence both responses factor through M_{light} . A product of arbitrary marginal fibres would allow inconsistent mass lists or electron weights and therefore violate the shared-parameter residual. The equalizer imposes identity of the underlying source object. $\square$

Corollary 9.3 (No D1 elimination on the D4 positive branch). *Within the D4 public-summary positive interval, D1 contributes no additional scalar-band elimination. It may still refine the joint fibre through endpoint spectral shape, support, nuisance, or nonstandard-response coordinates not fixed by Σ_v .*

9.2 D2–D5 charge and LNV compatibility

Let A_{76}^{tot} be the complete source-physical D2 transition amplitude. Its D5 charge violation is computed on the same active neutrino-bearing source path. A disconnected violating spectator is removed by activity reduction and cannot satisfy the cross equation.

Definition 9.4 (Joint LNV/charge residual). For a D5 target τ_5 , define

$$E_{2,5}^{\tau_5}(W) = \text{ChargeViolation}_{\tau_5}(A_{76}^{\text{tot}}(W)).$$

The pass condition is determined by the D5 target:

- exact global or unbroken local: $E_{2,5} = 0$;
- accidental through $d \leq 4$: no active $d \leq 4$ violation, while higher-dimensional violation is permitted;
- explicit breaking: a source-active target-bearing violation incidence is required;
- spontaneous/remnant: the violation must lie in the breaking ideal and satisfy the faithful quotient congruence.

Theorem 9.5 (Exact-global D2 zero-subfibre theorem). *On a joint branch with exact-global $B - L$ protection and no target update,*

$$A_{76}^{\text{tot}} = 0.$$

The D2 leg survives precisely on its defined zero-amplitude subfibre. Non-observation is not used in the proof.

Proof. The exact-global target requires a faithful action and continuation-stable protector on every active neutrino-bearing source, matching, and response path. The D2 transition changes lepton number by two conventional units. A nonzero target-bearing amplitude would therefore violate the protector. Since no breaking carrier or explicit violation is part of the exact-global target, the only passing physical amplitude is zero.

D2 independently constructs the defined zero-amplitude survivor $z = 0$ in its marked affine family. That survivor is a source-physical equality on one apex and lies in the empirical accepted region. Thus the joint exact-global branch retains D2 only through this subfibre. $\square$

Corollary 9.6 (LNV null search does not prove exact-global symmetry). *The implication used above is*

$$\text{exact-global protector} \implies A_{76}^{\text{tot}} = 0,$$

not its converse. A D2 zero response does not by itself construct a global action, anomaly packet, or protector.

Theorem 9.7 (Accidental-target compatibility with higher-dimensional LNV). *On the bounded accidental target $T_{\text{acc}}^{(4)}$, an active target-bearing LNV source of dimension greater than four may coexist with accidental $B - L$ conservation through $d \leq 4$. Hence the nondegenerate D2 accepted amplitude interval is not collapsed by the accidental target, provided all renormalizable $B - L$ -violating source incidences are source-null.*

Proof. The accidental target quantifies only the complete active operator content through dimension four. A higher-dimensional source is outside that bounded operator envelope and therefore does not falsify the accidental predicate. The D2 response may be generated by that source while the retained active process and all $d \leq 4$ operators preserve the accidental target. Source activity reduction removes any null renormalizable violation. The remaining D2 amplitude interval is accepted by the D2 target and compatible with the D5 accidental target. $\square$

Remark 9.8 (Breaking and remnant branches). For explicit, spontaneous, and remnant targets, the cross equation is source-semantic and exact, but the general joint fibre remains readiness-gated. A nonzero D2 amplitude must be carried by an actual target-bearing violation incidence; for spontaneous or remnant targets its primitive charge must lie in the breaking ideal and satisfy the faithful quotient congruence. The existence of representative examples does not close all source families or cardinalities.

9.3 D2–D3 support typing

The accidental marked D2 correction is an exact target-scale transition leg supported on the typed ^{76}Ge source/nuclear/detector path. It is target-bearing for D2 but contains no H, U, or L source slot of D3.

Theorem 9.9 (D2–D3 typed support transparency). *On the vectorlike-joint accidental comparison class, the D2 correction reduct to the D3 source theory is the null D3 extension of the common base source. Consequently it changes no D3 direct-state, active-closure, CLFV, process-scope, boundary, or matching evaluator. On the exact-global branch the correction source is absent and the statement is immediate.*

Proof. The D2 source-first colligation is attached along the D2 nuclear transition bearer and its detector/statistical interface. The marked source formation rule supplies no source incidence from that correction to a D3 H, U, or L operation. Activity reduction is performed before target-path closure, so the D3 reduct forgets the D2-only cells while retaining the common light source and physical apex. The D2 source remains non-spectator for D2 because its ^{76}Ge transition block is target-bearing. Typed support, rather than numerical smallness, therefore separates the two response legs. $\square$

9.4 D3–D4 source-guarded population compatibility

The D4 standard-thermal positive branch contains three light seed populations. SOE-D4-1 treats the new D4 activity maps as source parameters and guards; it contains no clause deriving an off-seed D4 activity value from an old nonzero coupling. SAR-D4-1 therefore authorizes a coherent seed-active/off-seed-null relative extension of every B15 source, and STR-D4-1 evaluates kinetic saturation at that actual source point.

Theorem 9.10 (Source-guarded D3–D4 relative extension). *Let H be any finite source-coherent B15 realization, including a D3/D5 vectorlike heavy sector with nonzero old-source couplings. Apply the frozen SOE-D4-1 relative extension that:*

1. *activates the three standard-thermal light population slots;*
2. *assigns their mass records, Fermi–Dirac data, weak-decoupling initial conditions, stable free streaming, background, and finite hierarchy;*
3. *assigns the declared null or terminal D4 objects to every off-seed kinetic/stress-energy sort, with the corresponding D4 activity values false.*

Then the extended source is coherent. Every off-seed production, collision, decay/injection, hidden/environmental attachment, and stress-energy kernel is exactly zero by its own frozen activity guard. Kinetic saturation is exactly the three light seed modes, while the complete old D3/D5 source, threshold, matching, and higher incidence remain present in the source trace.

Proof. SAR-D4-1 is a finite schema audit of SOE-D4-1: activity maps occur as primitive source data and guards, no frozen constructor or constitutive equation has an activity value as a derived consequent of another, and every inactive D4 relation admits the declared null or terminal object. Thus the stated partially active assignment satisfies the full D4 source equations while retaining H literally.

At that coherent point, every off-seed enabling predicate has its own D4 activity factor equal to zero. KSD-D4-1 therefore closes the kinetic support at the retained light seed, and STR-D4-1 supplies the exact zero kernels and stress-energy descent. No inference from a large threshold or small coupling is used. A future same-scope theorem forcing an off-seed activity value true would be a SourceUpdate and would reopen this result rather than retroactively changing the present source point. $\square$

Warning 9.11 (No small-coupling or threshold laundering). The theorem does not infer D4 nullity from weak mixing, Boltzmann suppression, or a heavy mass. The old vectorlike source and its nonzero couplings remain in the trace. Only the independently frozen D4 activity values and guarded null objects make the new D4 kinetic legs exactly null.

9.5 D3–D5 comparison

D5 proves two-way target-faithful deformations for the accidental and exact-global rows at D3-1.3 scope. One family varies D3 active closure while preserving D5 standing; another varies the D5 separator while preserving D3

active closure. This is stronger than image nonalignment and weaker than full joint factorization.

Theorem 9.12 (Joint preservation of D3–D5 marginal independence). *On the common physical comparison subcategories for $T_{\text{acc}}^{(4)}$ and T_{glob} , the D3-1.3 active-closure separator and the respective D5 separator remain functionally independent after adjoining the D1 and D4 mass-compatible cells. For the exact-global branch, the D2 coordinate is restricted to zero but does not make either D3 or D5 a function of the other.*

Proof. The D5 comparison theorem supplies two target-faithful physical deformation directions on its common D3-1.3/D5 category, not merely an alignment of base images. Apply the D1/D4 attachment functor that uses the common light mass object and the source-guarded relative extension of Theorem 9.10. This functor retains the old D3/D5 source literally and holds the light, background, and endpoint records fixed during both deformation directions. On the accidental branch choose one fixed accepted D2 amplitude; on the exact-global branch the D2 amplitude is identically zero. These constant added coordinates cannot turn either nonconstant D3/D5 separator into a function of the other. The two-way deformation therefore persists on the enlarged target-and-comparison-qualified joint comparison category containing the vectorlike-joint source. This theorem concerns the coordinate diagnostic; it is not used to infer membership or cardinality of the vectorlike-joint child quotient. $\square$

Status 9.13 (Unresolved D3–D5 rows). The minimal-background, local-3N, explicit-breaking, spontaneous, and remnant comparisons remain NotReady. PCR-5E does not infer dependence, independence, or common generation from that absence.

9.6 D4–D5 vacuum and action compatibility

On the unbroken accidental or exact-global vectorlike-joint child branches, D5 does not introduce a breaking vacuum orbit. The D4 flat- Λ CDM background and the D5 action therefore share the base vacuum branch through covariance of the light stress-energy object. For spontaneous/remnant targets, one must additionally prove that the D5 breaking vacuum, its stabilizer, and the D4 background/history occupy one compatible branch; this is among the unresolved joint residuals.

9.7 Higher-arity relation: mass, charge, population, and response support

A nonzero LNV source can affect the light mass response and the D4 population history while also changing the D5 symmetry status. This is a genuinely three-way source relation:

$$\text{LNVSource} \longrightarrow (\text{LightSpec}, \text{ChargeViolation}, \text{PopHist}).$$

On the accidental vectorlike-joint target the source is allowed, its D3 support is typed null by Theorem 9.9, and its D4 kinetic activity is fixed by the SOE-D4-1 extension. On the exact-global target the source is absent and the relation collapses to the zero-amplitude subfibre. On spontaneous/remnant branches the same relation is controlled by the breaking ideal. It may not be replaced by pairwise consistency alone.

Proposition 9.14 (Cross-relation completeness on vectorlike-joint targets). *For the accidental and exact-global source-qualified targets $T_{E,VJ}^{\tau_s}$, the residuals solved in this section exhaust every target-active cross relation. In addition to the displayed pairwise relations, all finite higher incidences are retained through the complete physical incidence trace of Theorem 4.5.*

Proof. The source comparison certificate Theorem 8.9 contains only the common light Dirac package, the exact-response vectorlike threshold source, D1 endpoint cells, the D2 zero or higher-dimensional amplitude cell, the D4 seed-active/off-seed-null relative extension, and the accidental or exact-global D5 packet. The complete trace records all shared mass, boundary, charge, activity, anomaly, vacuum, matching, continuation, and common-cone bearers.

By Theorem 4.5, any additional standing-bearing cross use must be a functional of this trace. Binary projections are X1–X8 and are solved by the D1–D4, D2–D5, D2–D3, D3–D4, D3–D5, and D4–D5 relations above. Any irreducible relation of arity at least three is retained literally in X9 and evaluated on the same source/apex; the displayed mass/charge/population relation is one irreducible example. Any other actual higher incidence generated by the marked source sorts—including charge/matching/closure, boundary/process/matching, or source/activity/stress-energy incidence—is retained at its actual arity and evaluated on the same apex. No breaking carrier, remnant quotient, extra cosmological species, second vacuum, or undeclared source operation belongs to this certificate. A relation requiring one is a source or target update, not an omitted current residual. $\square$

10 Vectorlike-joint simultaneous families and proper-continuum closure

The source-qualified comparison certificates are frozen by Theorem 8.9 inside the ARC-U1-authorized D3-1.3 branch. This section performs the genuinely joint work: it attaches the D1, D2, and D4 source/response cells to the common D3-1.3/D5 vectorlike source, solves the complete cross system on one physical apex, and computes the vectorlike-joint child image and fibre lower bounds. The proper-continuum endpoint is retained without qualification loss inside the exact target-and-comparison fingerprint.

10.1 D1 endpoint and source-guarded D4 cells

For every $(m_0, \tau) \in I_m^\circ \times I_\tau^\circ$, define the carrier-indexed common absolute-scale record

$$\mathbf{m}^{(\tau)}(m_0) = (m_1^{(\tau)}(m_0), m_2^{(\tau)}(m_0), m_3^{(\tau)}(m_0)), \quad m_i^{(\tau)}(m_0) = \sqrt{m_i(m_0)^2 + \tau},$$

with

$$t_\beta(m_0, \tau) = \sum_i \widehat{w}_i^{(e)} (m_i^{(\tau)}(m_0))^2, \quad \Sigma_v(m_0, \tau) = \sum_i m_i^{(\tau)}(m_0).$$

Attach the D1 endpoint apparatus through the declared SOE-D1-1 hyperprocess envelope of the intrinsic vectorlike source and the one-support metric law $\mu_{\beta, m_0, \tau} = \delta_{t_\beta(m_0, \tau)}$. The D1 base-reduct functor removes the metric law and wrapper and returns the original PCR-3/B15 vectorlike source exactly. The physical active normalization c_E^2 is shared with the D3 source and is fixed throughout the source-qualified comparison class; the normalized endpoint-shape weights remain $\widehat{w}_i^{(e)}$.

Attach the D4 public-summary source by the SOE-D4-1 seed-active relative extension of Theorem 9.10. The three standard-thermal light population slots are active and carry the same common mass record $\mathbf{m}^{(\tau)}(m_0)$. The choice of I_τ° in E-S3 guarantees $\sqrt{m_0^2 + \tau} < m_\star$, so the full interval remains in the public-summary band. Every vectorlike-heavy D4 production, collision, decay/injection, hidden/environmental, and stress-energy activity value is source-false with its declared null or terminal object. The old vectorlike source, its nonzero active incidence, and the physical threshold H_λ remain present in the common source trace. Hence the D4 heavy kernels are exactly zero by their own source guards, not by weak mixing, Boltzmann suppression, or an informal “heavy enough” argument. The D1 and D4 cells are tied by the same source-formed parameter τ ; no second mass record is admitted.

10.2 D2 amplitude cells

Let $A_{76}^\star \neq 0$ and let I_{76} be the source-formed nondegenerate accepted interval exported by D2.

On the accidental branch attach the target-bearing higher-dimensional LNV source

$$A_{76}^{\text{tot}}(z) = z A_{76}^\star, \quad z \in I_{76},$$

on the same physical apex through the D2 exact-amplitude colligation and split attachment. Its D3 reduct is typed null by Theorem 9.9; its D4 kinetic status is fixed by the declared SOE-D4-1 activity point; and its operator order is outside the active $d \leq 4$ envelope of the accidental target.

On the exact-global branch no active LNV source is present. The D2 cell is the defined physical zero point

$$z = 0, \quad A_{76}^{\text{tot}} = 0,$$

which passes the D2 search target and the continuation-stable exact-global protector simultaneously.

10.3 D5 action cells

For the exact-global branch retain the faithful global $B - L$ action already carried by $x_{m_0, \lambda}$, together with its common charge lattice, anomaly/global record, protector, matching packet, and vacuum. Every source and physical arrow is covariant.

For the accidental branch retain the same exact-response vectorlike matter content but impose only the bounded active-operator predicate through dimension four. The higher-dimensional D2 source is permitted and does not enter that bounded active operator set.

10.4 Solved vectorlike-joint source strata

Inside the pre-frozen source comparison classes of Theorem 8.9, define the full lawful solved subgroupoids

$$\begin{aligned} S_{\text{VJ},\text{sol}}^{T_{\text{glob}}} &= S_{\text{VJ}}^{T_{\text{glob}}} \big|_{I_m^\circ \times I_E \times \{s_E\} \times I_\tau^\circ \times \{0\}}, \\ S_{\text{VJ},\text{sol}}^{T_{\text{acc}}^{(4)}} &= S_{\text{VJ}}^{T_{\text{acc}}^{(4)}} \big|_{I_m^\circ \times I_E \times \{s_E\} \times I_\tau^\circ \times I_{76}}. \end{aligned}$$

These subgroupoids are the connected strict-margin source strata proved by Theorem 8.5; they are not inserted into the target fingerprint after seeing the joint endpoint. The complement inside the frozen vectorlike-joint target retains its actual D3 response, applicability, failure, or readiness status and is not called empty.

10.5 Source objects

Definition 10.1 (Exact-global vectorlike-joint source). For $(m_0, \lambda, \tau) \in I_m^\circ \times I_E \times I_\tau^\circ$, define

$$H_{\text{VJ},\text{glob}}(m_0, \lambda, \tau) = \text{ActRed}_E \left(x_{m_0,\lambda} \cup E_\beta^{\text{BB}}(m_0, \tau) \cup A_{76}(0) \cup K_{\text{th}}^{\text{guarded}}(m_0, \tau) \cup B_{L,\text{glob}} \right).$$

Definition 10.2 (Accidental vectorlike-joint source). For $(m_0, \lambda, \tau, z) \in I_m^\circ \times I_E \times I_\tau^\circ \times I_{76}$, define

$$H_{\text{VJ},\text{acc}}(m_0, \lambda, \tau, z) = \text{ActRed}_E \left(x_{m_0,\lambda} \cup E_\beta^{\text{BB}}(m_0, \tau) \cup A_{76}(z) \cup K_{\text{th}}^{\text{guarded}}(m_0, \tau) \cup B_{L,\text{acc}}^{(4)} \right).$$

The union symbols denote typed source amalgamation over the shared light source, boundary, mass records, vacuum, charge ledger, matching arrows, and continuation objects. They do not duplicate a carrier, source bearer, or physical apex.

10.6 Constraint-preserving one-apex amalgamation

Theorem 10.3 (Vectorlike-joint one-apex amalgamation). *Every source in $S_{\text{VJ},\text{sol}}^{T_{\text{glob}}}$ or $S_{\text{VJ},\text{sol}}^{T_{\text{acc}}^{(4)}}$ has a source-faithful one-apex physical realization*

$$W_{\text{VJ},\text{glob}}(m_0, \lambda, \tau) \in \text{PhysMod}_E^{(1.3, \text{pub95}, T_{\text{glob}}; \text{VJ})},$$

$$W_{\text{VJ},\text{acc}}(m_0, \lambda, \tau, z) \in \text{PhysMod}_E^{(1.3, \text{pub95}, T_{\text{acc}}^{(4)}; \text{VJ})}.$$

All inherited, marginal, pairwise, finite higher-arity, background, matching, continuation, and common-cone residuals vanish on the corresponding object. The complete oscillation standing packet is recovered in situ while every D1–D5 cell remains physically present.

Proof. Inherited base and D3/D5 source. Theorem 8.5 supplies one finite source-coherent physical object with exact H/U/L dispositions, process coverage, retained base, matching, action, anomaly, protector, vacuum, and common cone.

D1. The endpoint measure is the one-support law $\delta_{I_\beta(m_0, \tau)}$ on the source-authorized hyperprocess envelope. Its exact base reduct returns $x_{m_0,\lambda}$, while its complete endpoint apparatus and nuisance attachment are the accepted D1 retained-transparent construction. The fixed active normalization belongs to the shared boundary/source packet, and the D4 public band implies the scalar D1 bound by Theorem 9.1.

D2. On the accidental target the exact amplitude cell realizes zA_{76}^* in the accepted interval. It is target-bearing for D2, lies above the bounded $d \leq 4$ D5 envelope, and has the typed D3-null reduct proved in Theorem 9.9. On the exact-global target the source and amplitude are exactly zero, preserving the protector and passing the defined D2 zero-response target.

D4. The seed-active relative extension of Theorem 9.10 supplies the three light populations carrying $\mathbf{m}^{(\tau)}(m_0)$, the mass-indexed background, conservation law, stress-energy descent, and finite perturbation hierarchy. Its retained-oscillation restriction erases the D4-only population and absolute-scale cells and returns the same intrinsic vectorlike base source as the D1 base reduct. Every vectorlike-heavy D4 activity guard is false at the frozen source point, so every off-seed D4 kernel is exactly zero while the old vectorlike source remains active for D3/D5.

Cross and higher incidence. D1 and D4 share the same light mass record. D2 and D5 satisfy Theorems 9.5 and 9.7. D2 and D3 satisfy Theorem 9.9. D3 and D4 satisfy the source-guarded activity theorem. D3 and D5 use the same source, charge, anomaly, threshold, and matching object. D4 and D5 share a covariant vacuum and light stress-energy object. Every finite higher incidence is retained and exhausted by Theorem 9.14. The complete attachments therefore form one physical incidence diagram with the displayed apex. Joint retained transparency follows from Theorem 6.7. $\square$

10.7 Base image and fibre geometry

For fixed intrinsic vectorlike parameters $(m_0, \lambda) \in I_m^\circ \times I_E$, define the source-first common absolute-scale leaf

$$\mathcal{F}_{m_0, \lambda}^{\beta\Sigma} = \{H_{\text{VJ, glob}}(m_0, \lambda, \tau) : \tau \in I_\tau^\circ\}$$

on the exact-global branch, and the analogous leaf at any fixed $z \in I_{76}$ on the accidental branch. The intrinsic PCR-3/B15 vectorlike source $x_{m_0, \lambda}$, its heavy threshold, source graph, matching orientation, and D3/D5 packet are fixed on the leaf. Only the explicitly pre-frozen D1 hyperprocess metric law and the D4 seed-population mass record vary with τ .

Let

$$U_{\beta\Sigma}^{\text{osc}} : \mathcal{F}_{m_0, \lambda}^{\beta\Sigma} \longrightarrow \mathcal{N}_{\text{osc}}$$

be the composite of the exact D1 hyperprocess base reduct, the D4 retained-oscillation restriction, and the ordinary D2/D3/D5 forgetful restriction. By construction it returns the concrete standing core of $x_{m_0, \lambda}$. Denote its base SCIC class by

$$q_{m_0, \lambda} := [x_{m_0, \lambda}]_{\text{SCIC}}.$$

Definition 10.4 (Frozen oscillation matching-generator family and relative reduct). Let $\mathcal{G}_M^{\text{osc}}$ be the complete primitive matching-generator family entering the frozen PCR-5A/PCR-5B channel M^{match} . Its generators are the target-authorized evaluations of:

1. source-path and matching-incidence data;
2. retained resolution and effective-response validity;
3. matching orientation;
4. rank and source responsibility;
5. physical heavy threshold and operator-order filtration;
6. matching applicability, continuation, and fail-closed boundary;
7. SCIC-visible matching provenance, covariance, existence, and no-backflow status.

The family contains no D1 endpoint-support or D4 population/mass-sum primitive, because those are added only in the enriched target. For fixed (m_0, λ) , define the frozen relative matching reduct

$$\mathcal{R}_{m_0, \lambda}^{\text{rel}} := U_{\beta\Sigma}^{\text{osc}}(H_{\text{VJ, glob}}(m_0, \lambda, \tau)).$$

The right-hand side is independent of τ by the exact source reduct; the generator audit below proves that this independence extends to the complete matching channel rather than assuming it.

Theorem 10.5 (E-S7A – frozen oscillation matching-generator factorization). *Fix $(m_0, \lambda) \in I_m^\circ \times I_E$. For every generator $g \in \mathcal{G}_M^{\text{osc}}$ there is a unique function $\bar{g}$ on the relative reduct such that*

$$g(H_{\text{VJ, glob}}(m_0, \lambda, \tau)) = \bar{g}(\mathcal{R}_{m_0, \lambda}^{\text{rel}}) = g(x_{m_0, \lambda})$$

for all $\tau \in I_\tau^\circ$; the same equality holds on the accidental branch at fixed z . Consequently

$$M^{\text{match}}(m_0, \lambda, \tau) = M^{\text{match}}(m_0, \lambda, \tau') \quad \forall \tau, \tau' \in I_\tau^\circ.$$

No intrinsic UV source parameter is varied in this theorem: the absolute-scale direction is carried by the declared conservative D1/D4 extension.

Proof. We inspect the complete frozen generator family.

Source path and matching incidence. The inner vectorlike source, its finite source graph, light relative-response map, heavy block, and oriented matching arrow are literally the same object for every τ . The SOE-D1-1 hyperprocess constructor preserves every primitive source node and incidence and has an exact reduct to that object. The SOE-D4-1 retained-oscillation restriction removes only D4 population, background, and absolute-scale cells. Hence the old oriented matching arrow is unchanged.

Retained resolution and effective-response validity. The D1 base reduct and D4 retained restriction are identity maps on the frozen oscillation response, its resolution class, and its validity certificate. The common shift

$$(m_i^{(\tau)})^2 - (m_j^{(\tau)})^2 = m_i^2 - m_j^2$$

therefore appears only in enriched absolute-support/population records and changes no base resolution generator.

Heavy threshold and operator order. The physical heavy spectrum is the fixed degenerate triple H_λ , and the source operator category and order filtration are independent of τ . The D1 metric law and D4 population record introduce no old-target heavy threshold or operator-order arrow.

Orientation, rank, and source responsibility. The matrices $\mathcal{U}_L(s_E)$, $\mathcal{U}_R(s_E)$, the active and heavy charged-current blocks, all ranks, and the carrier owner of every matching arrow are fixed. The D1 hyperprocess wrapper is attached to the same electron-boundary carrier owner and current-source deletion is natural; the D4 seed populations are attached to the same retained light modes. No source responsibility is reassigned.

SCIC-visible provenance, covariance, existence, and no backflow. The same vectorlike matching diagram exists and is covariant for every τ . The new D1/D4 cells are typed enriched-target objects with declared forgetful maps. They neither invert the old heavy-to-light arrow nor supply an old-target selector. Thus the no-backflow generator has the same value.

Applicability, continuation, and failure. The matching applicability guards, orientation cell, source topology, heavy threshold, rank, and base continuation/failure frontier are fixed by E-S4A and E-S4. D1/D4 acceptance and failure data are retained in the enriched packet and are not recoded as base matching data.

Every primitive matching generator is therefore constant on the leaf and factors through the literal base reduct. PCR-5A generator saturation then gives the complete equality of M^{match} . Uniqueness of $\bar{g}$ follows because the reduct is surjective onto the displayed base core. $\square$

Theorem 10.6 (E-S7 – carrier-indexed common absolute-scale base conservativity). *Fix $(m_0, \lambda) \in I_m^\circ \times I_E$, $s = s_E$, and $\tau, \tau' \in I_\tau^\circ$. The corresponding vectorlike-joint sources satisfy*

$$\Phi_4^{\text{ret}}(H_{\text{VJ, glob}}(m_0, \lambda, \tau)) = \Phi_4^{\text{ret}}(H_{\text{VJ, glob}}(m_0, \lambda, \tau')),$$

and

$$\Xi_{\text{osc}}(H_{\text{VJ, glob}}(m_0, \lambda, \tau)) = \Xi_{\text{osc}}(H_{\text{VJ, glob}}(m_0, \lambda, \tau')).$$

More explicitly,

$$\begin{aligned} T_\tau^{\text{face}} &= T_{\tau'}^{\text{face}}, & R_\tau^{\text{stand}} &= R_{\tau'}^{\text{stand}}, & \Delta_\tau^{\text{del}} &= \Delta_{\tau'}^{\text{del}}, \\ C_\tau^{\text{act}} &= C_{\tau'}^{\text{act}}, & M_\tau^{\text{match}} &= M_{\tau'}^{\text{match}}, & B_\tau^{\text{bdy}} &= B_{\tau'}^{\text{bdy}}, & F_\tau^{\text{same}} &= F_{\tau'}^{\text{same}}. \end{aligned}$$

The same statement holds on the accidental branch at fixed z . Hence every τ -leaf lies over the single base class $q_{m_0, \lambda}$, while unequal τ values remain standing-distinct in the enriched vectorlike-joint target.

Proof. The composite reduct $U_{\beta\Sigma}^{\text{osc}}$ is exact on the complete old source and standing core. It removes the D1 metric law and hyperprocess wrapper, the D4 population/background cells, and the other enriched-only records, while returning $x_{m_0, \lambda}$ with its source graph, boundary, activity, deletion, matching, continuation, and failure data unchanged.

The target-face, boundary, activity, and deletion channels are therefore literally constant. The standing response is constant because every shifted mass vector has the same mass-squared differences and the same boundary/mixing data. The matching channel is constant by E-S7A, [Theorem 10.5](#). The base continuation and failure channel is constant because the exact reduct preserves the old continuation monoid and old failure frontier; the interval I_τ° was pre-frozen inside both D1 and D4 acceptance domains, so no enriched failure is laundered into an old-target status. Thus all seven packet coordinates agree.

The enriched responses are nonconstant. Since $\sum_i \widehat{w}_i^{(e)} = 1$,

$$t_\beta(m_0, \tau) = t_\beta(m_0, 0) + \tau$$

is strictly increasing in τ . Also

$$\frac{\partial}{\partial \tau} \Sigma_\nu(m_0, \tau) = \frac{1}{2} \sum_i \frac{1}{m_i^{(\tau)}(m_0)} > 0.$$

Therefore unequal τ values have unequal complete D1/D4 continuation signatures. The fixed-base conclusion follows from packet completeness and SCIC descent, while enriched inequality follows from the standing D1/D4 witnesses. No intrinsic vectorlike mass matrix, heavy threshold, or old matching object is returned. $\square$

Corollary 10.7 (Fixed-base common absolute-scale foliation). *For every $(m_0, \lambda) \in I_m^\circ \times I_E$, the map*

$$I_\tau^\circ \longrightarrow \pi_0 \mathcal{L}_{J, \text{glob}}^{\text{VJ}}(q_{m_0, \lambda}), \quad \tau \longmapsto [W_{\text{VJ}, \text{glob}}(m_0, \lambda, \tau)]$$

is well-defined and injective. The analogous map is injective on the accidental branch at fixed z , with z available as an additional independent splitting direction.

Definition 10.8 (Canonical image-calibration light point). Let $m_\diamond$ be the first rational point in the interior of I_m° under the fixed enumeration of $\mathbb{Q}_{>0}$. This choice is made before the image calculation and has no standing authority beyond selecting one explicit threshold-calibration slice.

Lemma 10.9 (Threshold injection into the joint base image). *For each solved vectorlike-joint stratum, the map*

$$I_E \longrightarrow \text{im } \rho_{J, \text{VJ}}, \quad \lambda \longmapsto q_{m_\diamond, \lambda}$$

is injective. Hence $|\text{im } \rho_{J, \text{VJ}}| \geq \mathfrak{c}$.

Proof. Every $\lambda \in I_E$ has a one-apex lift at $(m_\diamond, \tau = 0)$ by Theorem 10.3. Unequal values of H_λ and their oriented matching arrows are not related by lawful source, gauge, basis, chart, process, or matching transport. Packet completeness therefore separates the base classes. $\square$

Lemma 10.10 (Continuum fibres above every vectorlike-joint base). *For every $(m_0, \lambda) \in I_m^\circ \times I_E$,*

$$|\pi_0 \mathcal{L}_{J, \text{glob}}^{\text{VJ}}(q_{m_0, \lambda})| \geq \mathfrak{c}, \quad |\pi_0 \mathcal{L}_{J, \text{acc}}^{\text{VJ}}(q_{m_0, \lambda})| \geq \mathfrak{c}.$$

On both targets the injection is supplied by $\tau \in I_\tau^\circ$; on the accidental target $z \in I_{76}$ supplies a second splitting direction.

Proof. E-S7 and Theorem 10.7 place the full nondegenerate interval I_τ° over the single base class $q_{m_0, \lambda}$. Strict monotonicity of both $t_\beta(m_0, \tau)$ and $\Sigma_\nu(m_0, \tau)$ separates the complete enriched D1/D4 continuation signatures. The same injection exists on the accidental branch; additionally, distinct accepted amplitudes $z \in I_{76}$ change the D2 standing response. All points have one-apex realizations by Theorem 10.3. $\square$

Lemma 10.11 (Continuum base complement). *For every D3-1.3 joint branch,*

$$|Q_{\text{osc}}^{\text{SCIC}} \setminus \text{im } \rho_J^\tau| \geq \mathfrak{c}.$$

In particular, this holds for both vectorlike-joint certificates.

Proof. The joint image is contained in the D3-1.3 marginal image. The finite-unitary deficit continuum has a retained process-tester value below the mandatory 1.3 threshold on every complete lift, so it lies in the complement of the D3 marginal image and therefore in the complement of every joint image carrying that target. $\square$

Theorem 10.12 (Vectorlike-joint proper-continuum joint closure). *For $\tau_5 \in \{T_{\text{acc}}^{(4)}, T_{\text{glob}}\}$, the independently frozen target-comparison certificate satisfies*

$$\begin{aligned} \emptyset \subsetneq \text{im } \rho_{J, \text{VJ}}^{(1.3, \text{pub95}, \tau_5)} \subsetneq Q_{\text{osc}}^{\text{SCIC}}, \\ |\text{im } \rho_{J, \text{VJ}}^{(1.3, \text{pub95}, \tau_5)}| \geq \mathfrak{c}, \quad |Q_{\text{osc}}^{\text{SCIC}} \setminus \text{im } \rho_{J, \text{VJ}}^{(1.3, \text{pub95}, \tau_5)}| \geq \mathfrak{c}, \quad |Q_{J, \text{VJ}}^{(1.3, \text{pub95}, \tau_5)}| \geq \mathfrak{c}. \end{aligned}$$

Moreover every displayed base class $q_{m_0, \lambda}$ carries at least continuum many enriched joint lifts. Thus both vectorlike-joint child quotients are non-singleton and the strong proper-continuum endpoint is retained unchanged.

Proof. The target and source comparison scope is frozen by Theorem 8.9. Nonemptiness and the image lower bound follow from Theorems 10.3 and 10.9; properness and the complement lower bound follow from Theorem 10.11; and the enriched quotient and fixed-base fibre lower bounds follow from Theorem 10.10. No conclusion is transported to the larger parent source comparison class. $\square$

10.8 Scope of the positive result

Status 10.13 (Vectorlike-joint branch status). For $\tau_5 \in \{T_{\text{acc}}^{(4)}, T_{\text{glob}}\}$, Writing TCQ for target-and-comparison-qualified,

$$\begin{aligned} \text{Status}_E(1.3, \text{pub95}, \tau_5; \text{VJ}) = & \text{ProperContinuum}_{\text{VJ}}^{\text{TCQ}} \\ & \oplus \text{ROOS}_{\text{fin-op}\backslash\text{closed parent}} \\ & \oplus \text{NotReady}_{\text{parent charts}}. \end{aligned}$$

The first summand is the exact theorem over the frozen target-and-comparison certificate. The remaining summands are explicit parent-scope boundaries; they are neither weakened nor silently eliminated by the vectorlike-joint calculation.

11 Target-branch analysis and completeness

11.1 Branch support as the support of the joint lift pseudofunctor

For a branch τ , define

$$\text{Supp}_J^\tau = \{q \in Q_{\text{osc}}^{\text{SCIC}} : \pi_0 \mathcal{L}_J^\tau(q) \neq \emptyset\}.$$

When the complete source atlas is closed, this is exactly $\text{im } \rho_J^\tau$. When an unresolved complement remains, PCR-5E records three disjoint base subsets:

$$Q_{\text{osc}}^{\text{SCIC}} = \text{Supp}_{J, \text{solved}}^\tau \sqcup \text{Excl}_{J, \text{solved}}^\tau \sqcup \text{Unres}_J^\tau.$$

The first contains proved nonempty fibres, the second proved empty fibres, and the third Undef/ROOS/update/NotReady fibres. Exact base selection requires $\text{Unres}_J^\tau = \emptyset$.

11.2 Branches with the complete D4 target

Theorem 11.1 (Complete-cosmology branch obstruction). *If $\tau_4 = \text{complete}$, then*

$$\text{Status}_E(\tau) = \text{NotReady}_{\text{JLR-D4-1}}$$

independently of τ_3 and τ_5 . No base fibre is thereby proved empty.

Proof. The complete joint target contains the complete D4 evaluator as an active target field. JLR-D4-1 is required to map every admissible joint source/background point to the fresh correlated likelihood. The evaluator is absent. Therefore the joint residual vector contains an unevaluable applicable component on every source chart. This is a named readiness gap, not a physical contradiction. The result is independent of the other target legs because adding them cannot construct the missing evaluator. $\square$

This theorem disposition covers fourteen of the twenty-eight branches.

11.3 Public-summary branches with D3-1.1

Theorem 11.2 (D3-1.1 global-completion obstruction). *For every $\tau = (1.1, \text{pub95}, \tau_5)$, the full parent branch satisfies*

$$\text{Status}_E(\tau) = \text{NotReady}_{\text{GFC-D3-1.1}}.$$

The explicit D3 prompt-HNL continuum may be used only to certify target-qualified nonvacuity on source subloci whose D5 and D4 cross residuals are separately solved; it cannot determine the full joint image or quotient.

Proof. The D3-1.1 marginal handoff lacks the complete global fibre atlas, exact base image, fixed-base splitting, and total overlap closure. PCR-5E joint fibres are subobjects of common-base pullbacks containing the D3 fibre. Without the D3 parent atlas, neither all joint source charts nor all joint complements can be classified. An explicit surviving continuum establishes nonemptiness where a common apex is constructed, but it does not solve every base fibre or overlap. Hence parent finality is NotReady. $\square$

This theorem disposition covers seven additional branches.

11.4 Public-summary branches with D3-1.3

The remaining seven branches use the successor D3 target and the D4 public-summary target.

Theorem 11.3 (Accidental and exact-global vectorlike-joint closure). *For $\tau_5 \in \{T_{\text{acc}}^{(4)}, T_{\text{glob}}\}$, Writing TCQ for target-and-comparison-qualified,*

$$\begin{aligned} \text{Status}_E(1.3, \text{pub95}, \tau_5) &= \text{ProperContinuum}_{\text{VJ}}^{\text{TCQ}} \\ &\oplus \text{ROOS}_{\text{fin-op} \setminus \text{RCF-fin}} \\ &\oplus \text{NotReady}_{\text{unclosed parent charts}}. \end{aligned}$$

Each solved vectorlike-joint subgroupoid has at least continuum many classes and a proper-continuum base image inside its independently frozen source-qualified target.

Proof. The positive theorem is [Theorem 10.12](#). The source comparison class was frozen before evaluation by [Theorem 8.9](#); the solved strict-margin subgroupoids were then derived by [Theorem 8.5](#). D3-1.3 retains a finite-operational non-RCF remainder as ROOS, and neither the solved subgroupoid nor its codescent exhausts the larger parent source universe. Therefore the parent branch is not identified with the vectorlike-joint quotient, while the positive proper-continuum theorem remains exact at its announced target-and-comparison fingerprint. $\square$

Theorem 11.4 (Five unresolved D5 joint rows). *For*

$$\tau_5 \in \{T_{\text{bg,min}}, T_{\text{loc,3N}}, T_{\text{exp}}[O], T_{\text{sp}}[q\Phi], T_{\text{rem}}[\Gamma, q\Phi]\},$$

the D3-1.3/public-summary parent branch is NotReady. More precisely:

$$\begin{aligned} T_{\text{bg,min}} &: \text{NotReady}_{\text{D3D5-MinBG deformation}}, \\ T_{\text{loc,3N}} &: \text{NotReady}_{\text{D3D5-Local action/anomaly preservation}}, \\ T_{\text{exp}}[O] &: \text{NotReady}_{\text{universal neutrino-matched transparent family}}, \\ T_{\text{sp}}[q\Phi] &: \text{NotReady}_{\text{common action/vacuum/stabilizer/matching}}, \\ T_{\text{rem}}[\Gamma, q\Phi] &: \text{NotReady}_{\text{faithful quotient/congruence/continuation}}. \end{aligned}$$

No listed status implies an empty joint branch.

Proof. D5 proves that at least one target-faithful deformation direction is absent in each row, so marginal dependence or independence is not closed. PCR-5E additionally requires one-apex compatibility with D1, D2, and D4. The named missing data are target-active components of that compatibility problem. Without them the joint source/physical atlas cannot be exhausted. Since representative source points may still exist, the correct disposition is NotReady rather than solved empty. $\square$

11.5 E10: base-class selection theorem

Theorem 11.5 (E10 – base support and selection strength). *For every branch τ ,*

$$\text{im } \rho_J^\tau = \text{Supp}_J^\tau \subseteq \bigcap_{\alpha} \text{im } \rho_{\alpha}$$

whenever the complete branch atlas is closed. At current authority:

1. *every complete-D4 branch has unresolved support rather than an image theorem;*
2. *every D3-1.1/public-summary branch has unresolved support outside any explicitly solved sublocus;*
3. *the solved vectorlike-joint accidental and exact-global D3-1.3/public-summary subgroupoids have proper-continuum base image and continuum complement;*
4. *the other D3-1.3/public-summary rows have unresolved support at the named D5 comparison boundary.*

No branch has a proved singleton base image.

Proof. The equality with fibre support is the definition of the restriction image. Inclusion in the intersection follows because a joint object has each marginal reduct. The four branch conclusions are [Theorems 11.1 to 11.4](#). The vectorlike-joint solved image contains continuum many base classes by [Theorem 10.9](#); therefore it is not singleton. The remaining branches have no complete image theorem, so singleton support cannot be asserted. $\square$

11.6 Classification of the twenty-eight branches

D3	D4	D5 row	PCR-5E parent status
1.1	complete	all seven	$\text{NotReady}_{\text{JLR-D4-1}}$ and independently $\text{NotReady}_{\text{GFC-D3-1.1}}$.
1.3	complete	all seven	$\text{NotReady}_{\text{JLR-D4-1}}$; D3/D5 substatus retained but cannot complete the target.
1.1	pub95	accidental	$\text{NotReady}_{\text{GFC-D3-1.1}}$; explicit common-source subloci do not close the parent atlas.
1.1	pub95	minimal background	$\text{NotReady}_{\text{GFC-D3-1.1}}$ plus unresolved D3–D5 background-inventory comparison.
1.1	pub95	local-3N	$\text{NotReady}_{\text{GFC-D3-1.1}}$ plus local action/anomaly compatibility gap.
1.1	pub95	exact global	$\text{NotReady}_{\text{GFC-D3-1.1}}$; exact-global zero-LNV subfibre remains a possible explicit sublocus.
1.1	pub95	explicit breaking	$\text{NotReady}_{\text{GFC-D3-1.1}}$ and universal target-bearing matching gap.
1.1	pub95	spontaneous	$\text{NotReady}_{\text{GFC-D3-1.1}}$ and common breaking-vacuum/stabilizer gap.
1.1	pub95	remnant	$\text{NotReady}_{\text{GFC-D3-1.1}}$ and faithful-remnant quotient/continuation gap.
1.3	pub95	accidental	Target-and-comparison-qualified vectorlike-joint proper-continuum closure; parent retains non-RCF ROOS and unclosed charts.
1.3	pub95	minimal background	$\text{NotReady}_{\text{D3D5-MinBG}}$.
1.3	pub95	local-3N	$\text{NotReady}_{\text{D3D5-Local}}$.
1.3	pub95	exact global	Target-and-comparison-qualified vectorlike-joint proper-continuum closure; D2 restricted to defined zero; parent retains non-RCF ROOS and unclosed charts.
1.3	pub95	explicit breaking	$\text{NotReady}_{\text{universal neutrino-matched transparent explicit family}}$.
1.3	pub95	spontaneous	$\text{NotReady}_{\text{common breaking apex}}$.
1.3	pub95	remnant	$\text{NotReady}_{\text{faithful remnant common apex}}$.

The table has sixteen displayed rows because each of the first two rows represents seven D5 branches. It covers all twenty-eight certificates without collapsing their individual D5 tags in the supplementary formal ledger.

11.7 No parent branch finality

Corollary 11.6 (Parent PCR-5E NotReady theorem). *At the complete parent scope of every branch $\tau \in \mathfrak{B}_E$, at least one mandatory E-H field remains incomplete. Therefore*

$$\text{Cert}_{E,\text{parent}}^\tau = \text{NotReady}_{\text{PCR5E}}.$$

This does not erase the target-and-comparison-qualified vectorlike-joint theorems.

Proof. The branch matrix exhausts $\mathfrak{B}_E$. Complete-D4 branches lack E-H4 and E-H9 at JLR-D4-1. D3-1.1 branches lack the complete fibre atlas and finality. D3-1.3/public-summary accidental and exact-global branches retain a parent ROOS/unclosed-chart complement. The other five rows retain named cross-target gaps. Thus no parent branch satisfies all release gates. The vectorlike-joint solved subgroupoids have their own exact target-and-comparison scope and handoff object. $\square$

12 Standing-coordinate incidence and common-generator diagnostics

The joint quotient and the standing-coordinate diagnostic are different objects. The quotient determines carrier identity at the joint target. The incidence object records which already-computed standing separators are activated by which phenomenon targets. It has no independent authority to change membership or equivalence.

12.1 Minimal separators

A separator is target-admitted and standing-bearing when changing its value while holding the target anchor and every lawful-equivalence datum fixed changes the complete target continuation signature. A minimal separator is the canonical quotient of the relevant standing response: any other admitted object determining the same response must map onto it. This universal-property formulation prevents a convenient microscopic presentation from being promoted to a common generator merely because two responses can be calculated from it.

For the present branches, the relevant separator classes are

- S_{osc} : relative propagation/phase/continuation structure,
- S_{D1} : electron-boundary absolute spectral support,
- S_{D2} : complete ^{76}Ge transition-amplitude/rate class,
- S_{D3} : $C_{ee}^{\text{op},1.3}$ together with the target-retained heavy matching class,
- S_{D4} : Σ_ν on the public-summary branch,
- S_{D5,τ_5} : target-specific action/anomaly/protector/violation/stabilizer/remnant data.

A source family name, fitted parameter label, or ultraviolet ancestry is not a separator unless it changes one of these admitted standing consequences.

12.2 E-R1: incidence object

Definition 12.1 (Phenomenon-to-coordinate incidence). Let

$$\mathcal{P}_\nu = \{\text{osc}, D1, D2, D3, D4, D5\}$$

and let C_ν^τ be the target-tagged set of exact marginal and joint separators. Define

$$I_\nu^\tau \subseteq \mathcal{P}_\nu \times C_\nu^\tau$$

by $(\alpha, s) \in I_\nu^\tau$ exactly when s is minimal for an α -target distinction at branch τ .

Theorem 12.2 (E-R1 – standing-coordinate incidence). *At every completed or source-qualified branch scope, I_ν^τ is invariant under lawful source isomorphism, gauge and basis transformation, response-frame change, parameter-chart change, and presentation replacement. It does not refine $\sim_E^\tau$ and therefore is not a second carrier classifier.*

Proof. Minimal separator identity is defined by the universal factorization property of target-admitted witnesses and by target-faithful deformation. Lawful transformations preserve those witnesses, their codomains, their continuation action, and their failure frontiers. They therefore preserve the canonical response quotient and the incidence relation. The incidence object is formed after the joint carrier quotient and never enters the definition of $\sim_E^\tau$; it cannot refine the joint partition. $\square$

12.3 Incidence on the vectorlike-joint accidental target

On the source-qualified vectorlike-joint accidental target of Theorem 8.9, the exact incidence is

Phenomenon	Minimal standing coordinate
osc	S_{osc}
$D1$	$G_{\beta\Sigma} \xrightarrow{\tilde{f}_\beta} \mu_\beta$
$D2$	$A_{76}^{\text{tot}} = zA_{76}^\star$
$D3$	$C_{ee}^{\text{op},1.3} = 1 - \delta_E$ and $H_\lambda \in M_{\text{match}}$
$D4$	$G_{\beta\Sigma} \xrightarrow{\tilde{f}_\Sigma} \Sigma_\nu$
$D5$	$T_{\text{acc}}^{(4)}$ bounded operator-support predicate.

Here $\delta_E = s_E^2$ is fixed on the marked comparison certificate, whereas the physical threshold H_λ varies on the source-formed matching interval. The D2 correction is target-bearing for D2 but is a typed null under the D3 source reduct by [Theorem 9.9](#); it therefore cannot manufacture D3 acceptance. D3 and D5 are independent in the target-faithful sense. D1 and D4 are distinct response faces of the canonical common mass-response quotient proved below.

12.4 Incidence on the vectorlike-joint exact-global target

On the exact-global source-qualified target,

$$A_{76}^{\text{tot}} = 0$$

is a source-protector-imposed subfibre condition. D2 remains a target leg, but its response coordinate is frozen by the exact-global protector. The D5 minimal separator is the faithful global action/protector packet, not the numerical value zero. The D3 threshold/matching separator remains independently variable.

12.5 Canonical minimal D1–D4 common generator

Let $M_{\text{abs}}^{\text{VJ}}$ be the source-formed carrier-indexed common absolute-scale records occurring in the two vectorlike-joint targets. An object is represented by $(m_0, \tau) \in I_m^\circ \times I_\tau^\circ$, with shifted light masses $\mathbf{m}^{(\tau)}(m_0)$, while the intrinsic vectorlike base source, oscillation differences, charged-current boundary, standard-thermal history class, and source-comparison certificate are fixed on each leaf. Define the combined response

$$S_{\beta\Sigma}(M) = (S_{D1}(M), S_{D4}(M)) = (f_\beta(M), f_\Sigma(M)).$$

Definition 12.3 (Canonical D1–D4 response quotient). For $M, M' \in M_{\text{abs}}^{\text{VJ}}$, write

$$M \sim_{\beta\Sigma} M' \iff w(u(S_{\beta\Sigma}(M))) = w(u(S_{\beta\Sigma}(M')))$$

for every D1- or D4-admitted witness w and every licensed continuation u in the source-qualified joint target. Set

$$\mathbf{G}_{\beta\Sigma} := M_{\text{abs}}^{\text{VJ}} / \sim_{\beta\Sigma} \cong \text{im}(S_{\beta\Sigma}).$$

The quotient map is denoted $q_{\beta\Sigma}$.

Theorem 12.4 (E-S6 – D1–D4 canonical complete-trace minimal common-generator theorem). *On each vectorlike-joint standard-thermal target:*

1. *there are unique target-preserving maps $\bar{f}_\beta$ and $\bar{f}_\Sigma$ such that*

$$S_{D1} = \bar{f}_\beta \circ q_{\beta\Sigma}, \quad S_{D4} = \bar{f}_\Sigma \circ q_{\beta\Sigma};$$

2. *for every admitted standing object G with*

$$S_{D1} = g_\beta \circ G, \quad S_{D4} = g_\Sigma \circ G,$$

there is a unique map h on $\text{im } G$ such that

$$q_{\beta\Sigma} = h \circ G;$$

3. *globally, the quotient is faithfully represented by the absolute lightest-mass coordinate*

$$a = \sqrt{m_0^2 + \tau} \in I_{\text{abs}} := \left\{ \sqrt{m_0^2 + \tau} : (m_0, \tau) \in I_m^\circ \times I_\tau^\circ \right\},$$

so $\mathbf{G}_{\beta\Sigma} \cong I_{\text{abs}}$; on every fixed intrinsic base leaf $m_0 = \text{const}$, its restriction is faithfully represented by $\tau \in I_\tau^\circ$.

Thus $\mathbf{G}_{\beta\Sigma}$, represented faithfully by the retained light mass-response object on this locus, is the canonical minimal common generator. D1 and D4 remain non-isomorphic targets because their witness families, response environments, continuation maps, and failure frontiers are different.

Proof. The factorization in item 1 follows from the quotient definition. For item 2, suppose $G(M) = G(M')$. The two displayed factorizations imply equality of the complete D1 and D4 response outputs under every licensed

continuation, hence $M \sim_{\beta\Sigma} M'$. Therefore $q_{\beta\Sigma}$ is constant on each fibre of G and induces a unique map h on $\text{im } G$. This is the universal coarsest-factor property; no smaller admitted standing quotient can still determine both responses.

For item 3, write $a = \sqrt{m_0^2 + \tau}$. The fixed oscillation splittings give

$$m_i^{(\tau)}(m_0) = \sqrt{a^2 + \Delta_i},$$

and the standard-thermal mass sum

$$\Sigma_\nu(a) = \sum_i \sqrt{a^2 + \Delta_i}$$

is strictly increasing on the compact positive interval I_{abs} . Equality of the D4 component therefore implies equality of a , and the fixed splittings and boundary data then imply equality of the complete common absolute-scale record. Conversely equal a gives equal D1 effective endpoint support and equal D4 mass sum. Thus the quotient is represented faithfully by I_{abs} . At fixed m_0 , the map $\tau \mapsto a$ is strictly increasing, so the restriction is faithfully represented by I_τ° . D1 adds the endpoint spectral response and D4 adds population/background acceptance; neither target is collapsed into the other. $\square$

12.6 D2–D5 exact-global obstruction, not common generation

Proposition 12.5 (Protector obstruction). *On the exact-global vectorlike-joint target, the relation*

$$S_{D5, \text{glob}} = \text{ExactGlobalPass} \implies A_{76}^{\text{tot}} = 0$$

is a compatibility obstruction. It is not a common-generator factorization of the D2 and D5 separators.

Proof. The D5 separator is the action/protector packet. The D2 separator is the transition-amplitude class. The protector constrains the permitted value of the amplitude but does not factor through it: distinct exact-global actions can yield the same zero amplitude, and a zero amplitude can arise without any exact-global action. Conversely the D2 response contains nuclear, detector, and statistical data absent from the protector. No two-way factorization exists. $\square$

12.7 D3–D5 independent coordinates

Proposition 12.6 (Independent-coordinate status). *At D3-1.3 scope, the D3 threshold/active-closure separator and the D5 accidental or exact-global separator are independent coordinates on the target-and-comparison-qualified common physical comparison category obtained by adjoining the fixed D1, D2, and D4 cells to the D5 D3-comparison object. This category contains the vectorlike-joint physical family. Neither separator factors through the other, and each has a nonconstant target-faithful deformation with the other fixed.*

Proof. The common source-landing theorem [Theorem 8.5](#) supplies an accepted vectorlike object in the comparison category. Varying λ changes the D3 physical threshold and oriented matching record while preserving the exact D5 action, anomaly, and protector. Conversely, the D5 target-faithful deformations imported in [Theorem 9.12](#) alter the bounded or exact-global symmetry packet while preserving the D3 separator. The source-guarded D1/D4 attachment functor and a fixed accidental D2 point or exact-global D2 zero point preserve both deformation directions. Hence independence is established on the enlarged target-and-comparison-qualified comparison category. The statement is a standing-coordinate diagnostic and is not used as a membership or cardinality premise for the vectorlike-joint quotient. $\square$

12.8 E-R2: independent-coordinate factorization

Marginal functional independence is not sufficient for a product theorem. A joint fibre factorizes only if the source support, residual system, continuation, shared parameters, background, matching, and isotropy also factor.

Theorem 12.7 (E-R2 – exact factorization criterion). *Let A be a set of marginal legs on a branch τ . Suppose:*

1. *their source extensions share only the retained base and declared diagonal automorphism action;*
2. *every cross residual among them is a consequence of the base equalizer;*

3. *their witness algebras tensor-factor over the base;*
4. *their continuation operations commute and introduce no shared threshold, vacuum, background, or matching data;*
5. *each leg admits target-faithful deformations with the others fixed.*

Then the comparison functor from the joint fibre to the compatible homotopy product of marginal fibres is an equivalence.

On the vectorlike-joint targets, the hypotheses fail for the full five-leg family: D1/D4 share $\mathbb{G}_{\beta\Sigma}$, D2/D5 exact-global share a nontrivial obstruction, and D3/D4 share the source-activity and matching equalizers of [Theorem 9.10](#). Hence no full product factorization is licensed. The D3/D5 accidental and exact-global pair factor conditionally after the common mass, D2, and D4 coordinates are fixed.

Proof. Under the five hypotheses, the joint source category is the homotopy product of the marginal source categories over the base, and the joint residual equalizer decomposes into the product of marginal residual equalizers. The diagonal automorphism action accounts for shared base isotropy. The comparison functor is fully faithful because a joint arrow is exactly a compatible tuple of marginal arrows, and essentially surjective because every compatible tuple has a unique source/physical amalgamation under source-support separation and absence of cross residuals.

For the vectorlike-joint full family, [Theorems 9.10](#), [12.4](#) and [12.5](#) exhibit non-base cross relations. After fixing those common coordinates, [Theorem 12.6](#) supplies the two-way target-faithful deformation and no remaining D3–D5 cross residual, giving the conditional pairwise factorization. $\square$

12.9 E-R3: common-generator and emergent synergy

Theorem 12.8 (E-R3 – branch diagnostic). *The vectorlike-joint diagnostics are:*

1. *D1/D4: CommonGenerator $_{\beta\Sigma}$ at the source-qualified standard-thermal scope;*
2. *D2/D5 exact-global: ObstructionCoupled, not common generation;*
3. *D3/D5 accidental and exact-global: IndependentCoordinate;*
4. *all-phenomenon common generator: NotEstablished;*
5. *emergent joint-only standing generator: NotEstablished on the vectorlike-joint families and NotReady on the parent branches.*

Proof. Items 1–3 are [Theorems 12.4](#) to [12.6](#). A common all-phenomenon generator would have to factor simultaneously to mass response, LNV amplitude, active closure, population transport, and target-specific symmetry data. The D3/D5 independence theorem and the D2/D5 obstruction defeat such a factorization on the vectorlike-joint categories. No new target-admitted invariant appears only after these targets are combined; the new content is compatibility equations among already standing objects. For parent branches, unresolved source charts and D5 comparison rows prevent exhaustion, so the correct status is NotReady rather than nonexistence. $\square$

12.10 Incidence summary

Leg	Coordinate class	Vectorlike-joint relation	Authority
Oscillation	Relative phase/continuation	Frozen base	Does not determine absolute scale or LNV character.
D1	Absolute spectral support	Canonical common generator with D4	Endpoint response remains distinct from cosmological response.
D2 accidental	Conjugation/LNV response	Independent target-bearing correction; typed-null under D3 reduct	Nonzero accepted interval survives.

Leg	Coordinate class	Vectorlike-joint relation	Authority
D2 global	Conjugation/LNV response	Forced zero subfibre by protector	Zero does not imply the protector.
D3	Active closure/heavy matching	Independent of D5 acc/global; exact D4 off-seed source nullity	Parent non-RCF remainder retained.
D4	Population/transport mass sum	Common generator with D1 plus source-activity equalizer with D3	Summary target only; no complete likelihood or ordering trace.
D5 accidental	Bounded operator support	Independent of D3	Does not imply exact global symmetry.
D5 global	Exact action/protector	Independent of D3; obstructs D2 nonzero	Connected spectral charge uniformity is not terminal identity.
Other D5	Target-specific	NotReady at joint comparison	No inferred dependence, independence, or common generator.

13 Joint source composition and no-repair sufficiency

13.1 No-repair deletion discipline

Let $W \in \text{PhysMod}_E^\tau$ have a finite active source-kind set $K_E(W)$. For a constitutively well-typed subset $S \subseteq K_E(W)$, deletion means:

- retain every parameter, coupling, vacuum, background, matching arrow, boundary record, and target record at its parent value;
- delete the source kinds in $K_E(W) \setminus S$ and all incidences whose bearer is deleted;
- do not retune, replace a source, change branch, re-glue, or select a new representative;
- recompute activity, target paths, physical residuals, and the complete joint standing packet.

Definition 13.1 (Joint source-sufficiency function).

$$N_{E,W} : 2^{K_E(W)} \longrightarrow \{0, 1, \perp\}$$

where $N_{E,W}(S) = 1$ if the fixed no-repair deletion to S still realizes the complete joint packet, 0 if an applicable residual fails, and $\perp$ if the deletion is constitutively ill-typed or leaves a named readiness gap.

The minimal-support antichain is

$$\text{MinSupp}_E(W) = \{S : N_{E,W}(S) = 1 \text{ and } N_{E,W}(S') \neq 1 \text{ for every } S' \subsetneq S\}.$$

13.2 E08: exact source-sufficiency theorem

Theorem 13.2 (E08 – joint source-sufficiency and synergy). *At every completed or target-and-comparison-qualified source scope, $N_{E,W}$ is invariant under lawful source isomorphism and the antichain $\text{MinSupp}_E(W)$ is exact. A survivor is:*

- single-route if the unique minimal support has one source kind;
- redundant if two incomparable minimal supports exist;
- synergistic if the full active set is the unique minimal support;
- partially synergistic if every minimal support has more than one source kind and several incomparable supports exist;

- higher-arity if at least one indispensable retained relation has arity greater than two and cannot be reconstructed from its proper incidence faces.

No status may be established using retuning or target change.

Proof. The function is defined from the fixed source hypergraph and complete joint evaluator, both invariant under lawful source isomorphism. Finite support makes the minimal elements well defined. The listed dispositions exhaust the possible antichain forms. A higher-arity tag records an additional structural property of the minimal supports and their incidence, not a new source label. Retuning and target change create a different source point or certificate and therefore do not evaluate $N_{E,W}$. $\square$

13.3 Vectorlike-joint source blocks

At the declared macroblock resolution, write

$$K_{\text{mark}} = \{L, H, P, G, V\},$$

where:

- L is the light carrier, mass, charged-current, and oscillation source block;
- H is the source-realized exact-response vectorlike D3/D5 threshold and matching block;
- P is the D4 population/background/transport block;
- G is the D5 action/protector or bounded accidental block;
- V is the optional target-bearing higher-dimensional LNV block on the accidental branch.

The D1 endpoint cells are response attachments to L , not a second matter source. The exact-global D2 zero cell has no active source kind. Common matching, vacuum, and higher incidence are relations among the blocks and are retained in every deletion in which their bearers remain.

13.4 Exact-global source composition

Proposition 13.3 (Exact-global macroblock synergy). *For an exact-global marked survivor,*

$$K_{\text{glob}} = \{L, H, P, G\}, \quad \text{MinSupp}_E(W_{\text{glob}}) = \{K_{\text{glob}}\}.$$

At macroblock resolution the source is synergistic. The synergy is target-compositional and does not by itself imply four terminal carriers.

Proof. Deleting L destroys the oscillation role, D1 endpoint measure, D4 light populations, and the carrier on which D5 acts. Deleting H removes the D3 heavy/active-closure target record and its matching threshold. Deleting P removes the D4 population/background target. Deleting G removes exact-global protection and invalidates the D2 zero-as-protected joint relation. Thus every proper subset fails at least one target residual. The full set passes by [Theorem 10.3](#).

The source blocks are target-role supports, not automatically terminal carrier nodes. Terminal identity is determined only after SCIC and Role-Occupancy Closure. $\square$

13.5 Accidental source composition

Proposition 13.4 (Accidental optional LNV route). *For an accidental marked survivor with nonzero D2 amplitude,*

$$K_{\text{acc}} = \{L, H, P, G, V\}, \quad \text{MinSupp}_E(W_{\text{acc}}) = \{\{L, H, P, G\}\}.$$

The block V creates a standing D2 split but is not necessary for target survival because deleting it yields the defined zero-amplitude D2 survivor while preserving the bounded accidental target.

Proof. The four blocks L, H, P, G are necessary by the same arguments as in [Theorem 13.3](#), replacing exact-global protection by the bounded accidental target. The D2 empirical target accepts the source-physical zero response.

Deleting V without retuning makes the high-dimensional LNV incidence source-null and leaves the other source equations unchanged, so the D2 leg moves to its zero subfibre and still passes. Hence V is not in the minimal support. Its presence nevertheless changes the D2 standing response and creates an enriched class. $\square$

Remark 13.5 (Redundancy versus optional enrichment). At a finer source resolution, several distinct high-dimensional D2 operators may furnish the same accepted response or different response classes. Their exact redundancy/synergy structure is inherited from the D2 source-composition hypergraph. The marked macroblock theorem does not collapse those intrinsic distinctions.

13.6 Higher-arity incidence

The marked exact-global source contains the relation

$$\text{LightSpec} \times \text{HeavyThresh} \times \text{GlobalAction} \longrightarrow \text{ProtectedMatching},$$

and the accidental source contains

$$\text{LightSpec} \times \text{LNVSOURCE} \times \text{PopHist} \longrightarrow (\text{MassResponse}, \text{LNVSOURCE}, \text{StressHistory}).$$

These relations are retained as higher-arity incidence. Their presence does not automatically make the carrier higher-arity at terminal resolution; it makes the joint source/standing packet higher-arity.

13.7 E09: emergent mixed selection

Theorem 13.6 (E09 – emergent mixed architecture disposition). *No emergent mixed-carrier selection theorem is established by the vectorlike-joint accidental or exact-global child branches. The vectorlike-joint child target requires several source blocks, but every indispensable block is already target-active in at least one marginal leg, and no joint-only separator excludes all pure carrier architectures beyond the common-source compatibility equations.*

For the parent branches, emergent mixed selection remains NotReady because the unresolved source charts and D5 comparison rows have not been exhausted.

Proof. An emergent mixed theorem would require a joint-only standing separator and a proof that every proper source architecture fails while a source-first mixed architecture survives. On the vectorlike-joint child branches, the new content consists of equalizers and obstructions among marginally standing data: the canonical common mass quotient, zero-LNV protection, source-guarded D4 activity, vectorlike threshold/matching, and one-apex coherence. They require joint support but do not define a new carrier separator absent from every marginal target. The explicit survivors are source-faithful compatible amalgamations of already admitted source blocks; the D3/D5 vectorlike block is certified by Theorem 8.5, not inserted by joint reinterpretation. Thus the proper status is NotEstablished, not a positive mixed selection theorem.

At parent scope, omitted source charts might contain a joint-only mixed survivor or a pure survivor; the readiness gaps prevent either conclusion. $\square$

13.8 Source-composition consequences

The E-H8 payload contains, for each completed object:

$$(K_E(W), N_{E,W}, \text{MinSupp}_E(W), \text{Disposition}(W), \text{HigherInc}(W), \partial\text{Fail}_{\text{del}}(W)).$$

For unresolved parent charts the same schema carries NotReady fields rather than inferred supports.

14 Selection, singleton criteria, and quotient geometry

14.1 Base selection and full selection

PCR-5C distinguishes selection of an oscillation base class from selection of one complete enriched carrier class. The distinction remains essential at PCR-5E:

$$|\operatorname{im} \rho_J^\tau| = 1 \quad \not\Rightarrow \quad |Q_J^\tau| = 1,$$

because the unique surviving base class may have a nontrivial joint fibre.

Definition 14.1 (Base-selection strength). A branch has:

- *no base selection* if $\operatorname{im} \rho_J^\tau = Q_{\text{osc}}^{\text{SCIC}}$;
- *proper base selection* if $\emptyset \subsetneq \operatorname{im} \rho_J^\tau \subsetneq Q_{\text{osc}}^{\text{SCIC}}$;
- *singleton base selection* if $|\operatorname{im} \rho_J^\tau| = 1$;
- *inconsistency* if $\operatorname{im} \rho_J^\tau = \emptyset$ and every base fibre is solved empty;
- *readiness-gated support* if an unresolved base subset remains.

The vectorlike-joint accidental and exact-global child branches have proper-continuum base selection. No parent branch has an exact completed image theorem.

14.2 E11: enriched singleton criterion

Theorem 14.2 (E11 – enriched singleton theorem). *Assume a branch τ has a complete joint atlas and SCIC activation certificate. Then $|Q_J^\tau| = 1$ if and only if there exists a unique base class $q_\star \in Q_{\text{osc}}^{\text{SCIC}}$ such that:*

1. $\pi_0 \mathcal{L}_J^\tau(q_\star)$ is singleton;
2. every other base fibre is solved empty;
3. the unique connected joint fibre contains no hidden standing split, unresolved complement, or second component introduced by overlap codescent.

A chosen representative, a singleton chart, or a singleton base image is insufficient.

Proof. By Theorem 7.5,

$$Q_J^\tau \cong \coprod_{q \in Q_{\text{osc}}^{\text{SCIC}}} \pi_0 \mathcal{L}_J^\tau(q).$$

If the quotient is singleton, exactly one summand is singleton and every other summand is empty. Completeness of the atlas and status partition ensures those empty summands are solved empty rather than unresolved. Codescent completeness excludes an unseen overlap component.

Conversely, if one fibre has exactly one connected component and all other fibres are solved empty, the coproduct contains exactly one element. SCIC activation ensures that no finer same-scope carrier distinction exists inside that component. $\square$

Corollary 14.3 (Vectorlike-joint child branches are non-singleton). *The target-and-comparison-qualified accidental and exact-global vectorlike-joint quotients are not singleton because each contains at least continuum many enriched joint classes.*

Corollary 14.4 (No singleton inference from readiness gaps). *If any base stratum is ROOS or NotReady, E11 cannot certify singletonhood even when every solved stratum but one is empty.*

14.3 Moduli rather than census

A joint fibre may be:

- finite discrete;
- countably stratified;

- a continuous semialgebraic or analytic moduli space;
- functional, with histories or kernels as parameters;
- categorical, with nontrivial isotropy and families of objects;
- mixed across these types.

The quotient is therefore not reduced to a scalar cardinal unless an exact cardinality theorem applies.

Definition 14.5 (Exact quotient classification). The classification $\text{Geom}(Q_J^\tau)$ is the codescent-preserved geometry of the connected-component object, including discrete strata, parameter spaces, functional coordinates, isotropy, overlap identifications, and readiness-tagged complements.

14.4 E12: exact cardinality and moduli theorem

Theorem 14.6 (E12 – exact cardinality/moduli theorem). *For every completed branch or target-qualified subcategory, the joint atlas determines exactly one of the following statuses:*

SolvedEmpty, Singleton, Finite _{$m>1$} , Countable, Continuous,
Functional, Categorical, Mixed, ReadinessGated.

A lower bound is not an exact cardinality, and a target-qualified classification does not classify the parent complement.

For the two vectorlike-joint child branches,

$$\text{Geom}(Q_{J,\text{mark}}^\tau) = \text{MixedContinuousCategorical}, \quad |Q_{J,\text{mark}}^\tau| \geq \mathfrak{c}.$$

At parent scope all twenty-eight branches are ReadinessGated.

Proof. The status list is exhaustive for the groupoid-valued atlas: its connected components are either absent, singleton, finite, countable, continuous, functional, categorical, or a disjoint/codescent mixture. A nonempty unresolved complement adds the readiness-gated status.

The marked parameter families contain real intervals in heavy threshold and light absolute scale, so their connected-component quotient has a continuous subspace. The source charts retain nontrivial presentation groupoids and overlap data inherited from B15 and the marginal extensions, so the exact object is categorical as well as continuous. Hence the mixed continuous/categorical classification and continuum lower bound.

Parent readiness follows from Theorem 11.6. □

14.5 Cardinality firewalls

The following cardinalities remain distinct:

1. the cardinality of the target role set;
2. the cardinality of the joint carrier quotient Q_J^τ ;
3. the number of internal spectral sectors in one realization;
4. the number of source blocks in one minimal support;
5. the dimension or cardinality of a parameter chart.

No equality among these is inferred without a dedicated theorem.

Proposition 14.7 (No spectral or parameter promotion). *Three light spectral modes, a three-block cosmological compiler, or a three-dimensional parameter chart does not imply three terminal joint carriers. Conversely, one terminal target role does not imply a singleton joint quotient.*

Proof. The objects live at different levels and are quotiented by different equivalence relations. Terminal carrier identity is defined by complete continuation signatures; spectral sectors and parameter coordinates may vary inside one signature-bearing object. The PCR-5 base already exhibits one terminal role with a non-singleton carrier quotient. □

14.6 Selection summary

Scope	Base-image status	Enriched quotient status
D3-1.3/public/accidental marked	Proper continuum; continuum complement	Mixed continuous/categorical; at least continuum; non-singleton.
D3-1.3/public/exact-global marked	Proper continuum; continuum complement	Mixed continuous/categorical; at least continuum; D2 zero subfibre; non-singleton.
D3-1.3/public parent acc/global	Parent support not fully closed because of ROOS/unclosed charts	Readiness-gated; marked lower bound preserved.
D3-1.3/public other D5 rows	Unresolved at target-specific D3–D5/cross boundary	Readiness-gated.
D3-1.1/public	Global image unresolved	Readiness-gated.
Any complete-D4 branch	Evaluator unavailable	Readiness-gated at JLR-D4-1.

15 Realized traces and Role-Occupancy Closure

15.1 Structural quotient and realized trace

The PCR-5E quotient is target-relative structural identity. It classifies source-coherent physical realizations under the complete joint continuation signature. A realized trace is a distinct object: a stable target-admitted record that identifies which structural class is instantiated, or an impossibility theorem excluding all but one class.

Definition 15.1 (Realized-carrier trace). A realized-carrier trace for a branch τ is a target-admitted map

$$\text{Tr}_E^\tau : \text{PhysMod}_E^\tau \longrightarrow \mathcal{R}_E^\tau$$

whose image contains a stable singleton realized class or whose certified complement excludes every competing joint class. The trace must be fixed in the target certificate before it is used as a selector.

15.2 E13: trace-versus-structure theorem

Theorem 15.2 (E13 – trace/structure separation). *Let Q_J^τ be the completed structural quotient of a branch.*

1. *If $|Q_J^\tau| = 1$, target-relative standing carrier identity is structurally closed, but no external empirical proposition follows unless the corresponding trace is already part of T_E^τ .*
2. *If $|Q_J^\tau| > 1$, a same-scope structural singleton can arise only through an independently formed realized trace already licensed by the target or through impossibility exclusion of every class but one.*
3. *A later likelihood preference, posterior ranking, naturalness criterion, canonical representative, or observation not contained in T_E^τ defines a target update and cannot retroactively change Q_J^τ .*

Proof. SCIC fixes identity through the complete target continuation signature. A singleton signature image closes identity at that target and nothing beyond it. For a non-singleton image, any further same-scope collapse must be witnessed by a target-authorized standing difference. The realized-trace and impossibility routes exhaust the ways such a difference can enter without choosing a representative. A later empirical comparison or ranking changes the witness family or acceptance rule; by fixed-domain discipline it is a new certificate. $\square$

Corollary 15.3 (No D4 ordering selector). *The D4 public-summary normal-ordering pullback is a target-specific scalar-band survivor family. It does not instantiate a pre-lock realized mass-ordering trace and cannot select an ordering branch in PCR-5E.*

Corollary 15.4 (No exact character from D2 null response). *Neither a D2 accepted zero amplitude nor a null search converts the joint carrier into an exact Dirac or exact Majorana realization without the corresponding target-fixed protector or violation theorem.*

15.3 Target-derived joint roles

Role-Occupancy Closure requires the role architecture to be formed before residue inspection. The joint target retains the one terminal complete-carrier role of the oscillation programme and adds internal target roles that describe work performed by the same or related physical architecture.

Definition 15.5 (Joint role architecture). For a branch τ , let

$$\mathcal{R}_E^\tau = \{r_{\text{car}}^{\text{term}}\} \sqcup \mathcal{R}_{\text{int}}^\tau,$$

where $r_{\text{car}}^{\text{term}}$ is the terminal complete-carrier role and $\mathcal{R}_{\text{int}}^\tau$ contains the target-typed internal roles:

$$\begin{aligned} & r_{\text{abs spec}}, \quad r_{\text{LNV resp}}, \quad r_{\text{active closure}}, \quad r_{\text{heavy match}}, \\ & r_{\text{population}}, \quad r_{\text{background}}, \quad r_{\text{symmetry/charge}}, \quad r_{\text{cross}}, \quad r_{\text{continuation}}. \end{aligned}$$

Some roles may be N/A on a target branch; none is promoted automatically into an additional terminal carrier role.

The terminal role set remains one-element because the complete joint target asks for one target-relative carrier architecture realizing the retained neutrino sector. PCR-6 may later determine whether the architecture contains one node, several nodes, or a higher-arity bearer. That physical-nature question does not alter the terminal target role fixed here.

15.4 Eligibility, live residue, and compatibility

Let $\text{Residue}_E^\tau = \mathcal{Q}_J^\tau$ at completed scope. Define the terminal eligibility relation

$$\mathbf{E}_E^\tau \subseteq \mathcal{Q}_J^\tau \times \{r_{\text{car}}^{\text{term}}\}$$

by complete target adequacy. A joint class is live when it occurs in at least one complete role-covering assignment satisfying the global compatibility predicate $\mathbf{J}_E^\tau$. The lawful automorphism group preserves the full assignment invariant tuple: target, role typing, residue, eligibility, compatibility, standing packet, source composition, boundary, continuation, failure, and provenance status.

15.5 E14: realization licensing

Theorem 15.6 (E14 – joint Role-Occupancy Closure). *For every branch with a complete joint quotient and target-derived role set,*

$$\text{RealizationLicensed}(C_E^\tau, T_E^\tau) \iff |\Sigma_E^\tau / \Gamma_{E,\tau}^{\text{lawful}}| = 1,$$

where Σ_E^τ is the groupoid of compatible role-covering assignments.

For each vectorlike-joint accidental or exact-global target-comparison certificate, the terminal role set has one element while the live joint residue contains at least continuum many SCIC classes. Hence

$$\neg \text{RealizationLicensed}(C_{E,\text{VJ}}^\tau, T_{E,\text{VJ}}^\tau),$$

and the diagnosis is terminal over-residue. For every parent branch, realization licensing is $\text{NotReady}_{\text{PCR5E}}$.

Proof. The biconditional is Role-Occupancy Closure applied to the branch-fixed assignment invariant. It removes only certificate-lawful automorphisms and requires pointwise eligibility plus global compatibility.

On a vectorlike-joint certificate, every class in the constructed quotient is eligible for the one terminal complete-carrier role because it is a complete target realization. The live residue therefore contains at least continuum many classes. No bijection exists from a one-element role set onto that residue, and no lawful automorphism identifies SCIC-distinct classes. The assignment orbit is not singleton; realization is unlicensed and the exact diagnosis is over-residue.

At parent scope the residue or its complement is incomplete, so the assignment groupoid cannot be exhausted. The proper status is NotReady rather than over-residue, under-residue, or incompatibility. $\square$

Remark 15.7 (Internal roles are not terminal carriers). The several internal phenomenon roles may be occupied by several source blocks or internal sectors. Their multiplicity is not the cardinality of the terminal carrier quotient. PCR-6 must construct the within-realization terminal bearer test explicitly.

15.6 Realization-status summary

Scope	ROC status	Reason
Vectorlike-joint accidental certificate	Over-residue; not licensed	One terminal role, at least continuum many complete joint classes.
Vectorlike-joint exact-global certificate	Over-residue; not licensed	D2 zero subfibre does not collapse D1/D4/D3/D5 enriched multiplicity.
D3-1.3/public parent acc/global	NotReady	Parent ROOS and unclosed source charts block complete live-residue calculation.
D3-1.3/public other D5 rows	NotReady	Target-specific common-source/cross residuals unresolved.
D3-1.1/public	NotReady	Global D3 fibre completion absent.
Any complete-D4 branch	NotReady	JLR-D4-1 evaluator absent.

16 Global endpoint theorem

16.1 Endpoint vocabulary

PCR-5E uses the following mutually exclusive terminal report classes at a fixed branch and source-comparison scope:

JointInconsistent	every source chart is solved or obstructed and no common physical object exists;
JointNotReady _{κ}	a named target, source, evaluator, overlap, continuation, or finality burden remains incomplete;
TargetComparisonQualifiedClosure	a complete theorem holds on a target whose source comparison class is an explicit fingerprint field, with the larger parent comparison class retained as a separate certificate;
SingletonEnrichedCarrier	the complete joint quotient is singleton and ROC licenses the target-derived role architecture;
FiniteMultiplicity _{m}	exactly $m > 1$ joint classes survive;
Countable/Continuous/Functional/Categorical/Mixed	the exact non-singleton moduli geometry is retained;
OnlyMixedOrSynergisticSurvivors	every surviving class has a mixed or higher-arity minimal source support;
AlignedSelectionDistinct	several phenomena select the same base support through different standing coordinates;
CommonGeneratorClosure	several separators factor through one proved canonical minimal joint invariant;
EmergentJointSynergy	a new standing invariant or source relation is activated only by the simultaneous target.

A scope can carry one geometric report and auxiliary coordinate/source reports. For example, the vectorlike-joint exact-global certificate is target-and-comparison-qualified, mixed continuous/categorical, over-residue, D1–D4 common-generator, D3–D5 independent, and D2–D5 obstruction-coupled.

16.2 Joint closure theorem

Theorem 16.1 (PCR-5E target-indexed joint closure with dependency repair). *Under the frozen PCR-1 through PCR-5D5 dependency package and at the finite source-presentation/symbolic B15 scope, the following statements hold.*

1. *ARC-U1 authorizes THUCL-D3-1.3 as an additional, separately tagged PCR-5E successor branch while preserving the original THUCL-D3-1.1 programme, theorem labels, and completion burden unchanged.*
2. *E00 fixes the twenty-eight parent target/version certificates before joint formation.*
3. *E-S1 derives and saturates the complete cross-relation grammar from the semantic joint target trace.*
4. *E-S2 reconstructs every semantic joint source presentation in the complete activity-reduced source grammar.*
5. *E-S3 fixes the complete raw vectorlike-joint precursor domains before D3/D5 response evaluation or any joint endpoint calculation.*
6. *E01 forms the exact typed target family in the slice over the oscillation target.*
7. *E02 forms the common-base homotopy pullback and excludes wrong-base tuples.*
8. *E-S4 lands the same source-realized D5 vectorlike Dirac family in the complete D3-1.3 H/U/L and finite-process target on one physical apex and gives the total typed status decomposition of the raw precursor.*
9. *E-S5 freezes the source-qualified vectorlike-joint child certificates before any genuinely joint D1/D2/D4 cross calculation or endpoint computation.*
10. *E03 constructs the conservative joint source theory and one-apex physical groupoid; joint membership is one-object physical survival rather than marginal intersection.*
11. *E04 gives a total obstruction normal form and separates solved emptiness from Undef, ROOS, source update, target update, and NotReady.*
12. *E05 proves representative-independent restriction, exact marginal reducts, and no merger of PCR-5B base classes.*
13. *E06 reconstructs every completed parent branch or target-and-comparison-qualified solved category by source-indexed groupoid codescent.*
14. *E07 proves order/path independence only for genuinely identical final target fingerprints.*
15. *E08 computes exact no-repair source-sufficiency antichains at completed scope.*
16. *E09 finds no emergent mixed-carrier selection theorem on the vectorlike-joint solved categories and retains parent NotReady status.*
17. *E10 gives the branch-specific base-support matrix.*
18. *E11 proves the enriched singleton criterion in both directions.*
19. *E12 classifies the vectorlike-joint solved quotients as mixed continuous/categorical with cardinality at least continuum and all parent branches as readiness-gated.*
20. *E13 separates structural closure from realized trace and exhausts same-scope selector routes.*
21. *E14 applies Role-Occupancy Closure; both vectorlike-joint solved quotients are terminal over-residue and no parent branch is realization-ready.*
22. *E15 exports exact E-H1–E-H10 handoff objects with parent and target-and-comparison-qualified finality tags.*
23. *E-R1 constructs the invariant standing-coordinate incidence relation.*
24. *E-R2 proves the exact factorization criterion and shows that the full five-leg vectorlike-joint fibre does not factor as a marginal product.*
25. *E-S6 proves the canonical complete-trace D1–D4 minimal common generator and its universal coarsest-factor property.*
26. *E-S7A proves matching-generator factorization and E-S7 proves the complete seven-channel PCR-5B packet is constant along each fixed- (m_0, λ) conservative absolute-scale leaf, establishing continuum enriched fibres without base drift.*

27. *E-R3 records D1–D4 common generation, D2–D5 exact-global obstruction coupling, and D3–D5 independence on the accidental and exact-global rows; no all-phenomenon common generator is established.*

Every parent branch has status `JointNotReady` with an exact tag. For $\tau_5 \in \{T_{\text{acc}}^{(4)}, T_{\text{glob}}\}$, the independently frozen vectorlike-joint target-comparison certificates satisfy

$$\emptyset \subsetneq \text{im } \rho_{J, \text{VJ}}^{(1.3, \text{pub95}, \tau_5)} \subsetneq Q_{\text{osc}}^{\text{SCIC}},$$

$$|\text{im } \rho_{J, \text{VJ}}^{(1.3, \text{pub95}, \tau_5)}| \geq \mathfrak{c}, \quad |Q_{\text{osc}}^{\text{SCIC}} \setminus \text{im } \rho_{J, \text{VJ}}^{(1.3, \text{pub95}, \tau_5)}| \geq \mathfrak{c}, \quad |Q_{J, \text{VJ}}^{(1.3, \text{pub95}, \tau_5)}| \geq \mathfrak{c}.$$

Hence neither vectorlike-joint quotient is singleton. No parent branch licenses a universal PCR-6 physical-nature theorem.

Proof. The architecture, source, and target repairs are Theorems 2.6, 4.5, 4.11, 5.8, 8.5, 8.9 and 10.6. The original E ladder is established in Theorems 2.3, 4.7, 5.2, 6.5, 6.10, 7.3, 7.5, 7.8, 11.5, 13.2, 13.6, 14.2, 14.6, 15.2, 15.6 and 17.2. The standing-coordinate refinements are Theorems 12.2, 12.4, 12.7 and 12.8. Parent readiness is Theorems 11.1 to 11.4; the strong vectorlike-joint endpoint is Theorem 10.12. The singleton and realization conclusions follow from E11 and E14. $\square$

16.3 Endpoint table by scientific question

Question	Current answer	Exact authority boundary
Does the complete D1–D5 target have a common source/physical formulation?	Yes, target-indexed	Structural source and one-apex categories are constructed for all branches; evaluability and finality vary.
Is the joint quotient the intersection of marginal images?	No	The joint fibre is an obstruction equalizer inside the common-base homotopy pullback.
Is the cross-relation grammar complete?	Yes at declared finite scope	E-S1 derives it from the complete semantic incidence trace; new sorts or operations are explicit updates.
Does the same D3/D5 source survive the full D3-1.3 target?	Yes on the VJ interval	E-S4A fixes the uniform product box and E-S4 evaluates H, U, L, process scope, retained base, matching, D5, and one-apex physicality on the same source and retains the total typed complement.
Does any completed target-comparison certificate have a nonempty joint residue?	Yes	Vectorlike-joint accidental and exact-global D3-1.3/public-summary certificates.
Does any current scope have a proved singleton base image?	No	VJ images contain continuum; other branches are unresolved.
Does any current scope have a singleton enriched quotient?	No	VJ quotients contain continuum; parent branches are NotReady.
Does exact-global $B - L$ select a D2 zero response?	Yes, one-way	The protector forces the defined zero D2 subfibre; zero does not imply the protector.

Question	Current answer	Exact authority boundary
Do D1 and D4 share a deeper coordinate?	Yes, VJ-qualified	Both factor through the canonical complete-trace quotient $G_{\beta\Sigma}$, with a proved universal minimality property.
Is D4 heavy nonparticipation inferred from mass or small coupling?	No	It follows exactly from the independently frozen off-seed source-activity guards.
Do D3 and D5 share a common generator?	No on two decided rows	Accidental and exact-global rows are independent; five rows remain NotReady.
Is a joint-only mixed carrier forced?	Not established	VJ blocks are jointly necessary, but no new terminal carrier separator is proved.
Is PCR-6 globally authorized?	No	Every parent E-H data set contains a readiness gap.
Is PCR-6 downstream used on a VJ solved stratum?	Yes, with strict tag	Only the two E-H-complete solved subcategories may be analyzed; no transport to the full VJ certificate or parent comparison class.

16.4 Strength and limitation

The strongest positive conclusion is not a unique carrier. It is an exact source/physical/fibre theorem and a proper-continuum non-singleton result on two independently frozen target-comparison certificates, together with an exhaustive parent readiness matrix. This is stronger than a list of compatible models and weaker than full joint closure of the finite-operational neutrino sector.

The result is fail-closed in the intended sense. A future positive JLR-D4-1 evaluator, D3-1.1 global atlas, D3-1.3 non-RCF closure, or D5 comparison theorem can be added through the declared update protocol. None is presumed absent in principle, and none is treated as currently solved.

17 Consequences for subsequent carrier-identity analysis

PCR-6 may begin only from a complete PCR-5E downstream interface at a declared scope. The downstream interface is reconstructive: downstream analysis may not infer omitted source, physical, continuation, or failure data from a headline endpoint.

17.1 The E-H1–E-H10 downstream data

Definition 17.1 (PCR-5E handoff package). For a completed branch or target-and-comparison-qualified solved subcategory C , define

$$\mathcal{H}_E^V(C) = (E-H1, \dots, E-H10)$$

with the fields below.

ID	Mandatory export
E-H1	Exact joint target certificate, including target records, applicability, uncertainty, continuation, target-version tags, and the E-S3 raw precursor fingerprint and the E-S5 source-qualified comparison fingerprint.
E-H2	Conservative joint source category, source reducts, activity guards, target-generated cross relations, semantic saturation, source-normal-form reconstruction, and the E-S4 same-source D3/D5 landing.

ID	Mandatory export
E-H3	One-apex simultaneous physical category, including inherited, marginal, pairwise, finite higher-arity, background, matching, continuation, and common-cone residuals.
E-H4	Final joint SCIC quotient and groupoid atlas for the completed solved subcategory, together with the total Pass/Fail/N/A/ROOS/source-update/target-update decomposition of its pre-frozen source comparison complement.
E-H5	Restriction maps to the oscillation base and every marginal target, with no-merger, representative-independence, and path-equivalence certificates.
E-H6	Standing-coordinate incidence object and every proved common-generator, independent-coordinate, obstruction-coupled, aligned-selection, or emergent-synergy relation, including the E-S6 universal-property certificate and the E-S7 fixed-base foliation certificate.
E-H7	Complete continuation monoid and every target-admitted terminal witness, compatibility evaluator, and failure functional at the exported scope.
E-H8	Joint no-repair source-sufficiency function, minimal-support antichain, S/R/P/Y or higher-arity disposition, and deletion failure frontier.
E-H9	Exact failure/update boundary, including D3 programme separation, D4 JLR-D4-1 status, all D5 target tags, every parent ROOS/NotReady remainder, and exact conditions for revision.
E-H10	Finality certificate declaring $\text{Ready}_{\text{target-and-comparison-qualified}}$, or $\text{NotReady}_{\text{PCR5E}}$ for within-realization physical-nature analysis.

17.2 E15: downstream finality theorem

Theorem 17.2 (E15 – PCR-6 handoff finality). *The E-H1–E-H10 package is complete for the solved subcategories*

$$\mathcal{S}_{\text{VJ},\text{sol}}^{T_{\text{acc}}^{(4)}} \quad \text{and} \quad \mathcal{S}_{\text{VJ},\text{sol}}^{T_{\text{glob}}}$$

inside the pre-frozen vectorlike-joint target-comparison certificates. Their finality status is

$$\text{Cert}_E^{\text{VJ}} = \text{Ready}_{\text{target-and-comparison-qualified}}.$$

PCR-6 may analyze the within-realization physical nature and minimal identity of objects in either exported solved subcategory only.

For every parent branch $\tau \in \mathfrak{B}_E$, at least one E-H field remains incomplete and

$$\text{Cert}_{E,\text{parent}}^\tau = \text{NotReady}_{\text{PCR5E}}.$$

PCR-6 may not fill a missing field by marginal intersection, representative choice, later characterization, or transport from a completed subcategory.

Proof. E-H1 is the exact pre-evaluation D3-1.3/public-summary/accidental or exact-global vectorlike-joint target-comparison certificate fixed by E-S3 and E-S5. E-H2 consists of Theorems 4.5, 5.8, 8.5 and 8.7; E-H3 is Theorem 10.3; E-H4 is the solved subgroupoid codescent quotient together with the total typed complement; E-H5 is Theorem 7.3; E-H6 is Theorems 12.2, 12.4, 12.7 and 12.8; E-H7 is the restricted complete continuation/witness package used in the joint SCIC activation theorem; E-H8 is Theorems 13.3 and 13.4; E-H9 is the exact source/target/failure frontier; and E-H10 records the target-and-comparison-qualified finality tag. Therefore every mandatory field is present at the solved-subcategory scope.

At parent scope, Theorem 11.6 identifies at least one missing mandatory field in every branch. A downstream inference of a missing joint atlas, source chart, evaluator, or failure frontier would add information not contained in PCR-5E and would create a new certificate. $\square$

17.3 Allowed and forbidden PCR-6 questions

On either exported solved subcategory, PCR-6 may ask:

- whether a joint realization has one terminal carrier node, several nodes, or an irreducible higher-arity bearer;

- whether its light spectral components independently carry complete terminal continuation signatures;
- whether source blocks are primitive carriers, internal factors, co-carriers, or response cells;
- whether a minimal physical-nature classification is constant across the exported quotient or varies by joint class.

It may not:

- identify the exported solved subcategory with the whole neutrino sector;
- discard the typed complement of the pre-frozen vectorlike-joint source comparison certificate;
- discard the parent ROOS or NotReady complement;
- change D2 zero-subfibre membership on the exact-global branch;
- replace the D4 public-summary target by the complete cosmological target;
- use D3-1.3 conclusions at D3-1.1 scope;
- merge SCIC-distinct joint classes because they share an interpreted physical nature.

17.4 Downstream scope table

Scope	E-H10 status	Permitted downstream use
D3-1.3/public/accidental VJ solved stratum	Ready, target-and-comparison-qualified	PCR-6 may analyze exported objects; nonzero D2 interval and bounded accidental D5 tag retained.
D3-1.3/public/exact-global VJ solved stratum	Ready, target-and-comparison-qualified	PCR-6 may analyze exported objects; D2 exact-zero subfibre and exact-global protector retained.
Full pre-frozen VJ comparison certificates	Typed decomposition; not universally solved	PCR-6 receives the solved stratum and exact complement, not a universal physical-nature theorem.
D3-1.3/public parent acc/global	NotReady PCR5E	Parent ROOS and unclosed charts remain.
D3-1.3/public other D5 rows	NotReady PCR5E	Target-specific joint gaps must close first.
D3-1.1/public	NotReady PCR5E	D3 Global Fibre Completion required.
Any complete-D4 branch	NotReady PCR5E	JLR-D4-1 required.

17.5 Supplementary formal record

The accompanying supplementary materials include a formal representation of the E-H1–E-H10 data for the two completed target-and-comparison-qualified sectors, together with the exact parent completeness boundaries. This representation is provided for navigation and reproducibility; it does not replace the E15 theorem or its proof.

18 Robustness and counterexample analysis

This section tests the principal same-scope counterexample routes to the joint closure theorem. An objection is relevant only if it preserves the exact target fingerprint, source comparison class, lawful-equivalence groupoid, continuation monoid, failure semantics, and announced scope while defeating a load-bearing implication or supplying a faithful same-scope counterexample. Changes of target, source ontology, witness family, or failure semantics are scope changes rather than counterexamples to the theorem proved here.

18.1 Resolution of previously identified proof obligations

Previously identified issue	Required result	The completed theorem chain closure
Survivor-defined source scope	Freeze the entire vectorlike-joint source comparison class before D3 acceptance or joint counting.	E-S3, Theorem 4.11 , fixes the complete raw domains $I_m^o \times I_\lambda^{\text{src}} \times I_s^{\text{src}} \times I_\tau^o \times Z_{\tau_5}$. E-S4 evaluates the complete marginal Pass stratum inside that raw domain, and E-S5 freezes the resulting source-qualified child certificate before the joint D1/D2/D4 calculation. Neither step redefines the parent fingerprint.
Unlicensed vectorlike substitution	Land the same D5 source-formed vectorlike Dirac family in the complete D3-1.3 source, H/U/L, process, retained-base, matching, continuation, and one-apex target.	E-S4, Theorems 8.5 and 8.7 , constructs the source normal form, proves complete H/U/L and process survival on a nondegenerate interval, and retains the exact typed complement on the pre-frozen source scope.
Hand-listed cross completeness	Define semantic joint uses before the finite cross grammar and prove every standing-bearing cross relation factors through the declared trace.	E-S1 and the joint source-normal-form theorem, Theorems 4.5 and 5.8 , derive X1–X9 from the complete semantic incidence trace and retain every finite higher-arity target-active hyperedge at its actual arity.
D1/D4 generator overreach	Prove a universal minimal factor rather than merely display one common source object.	E-S6, Theorem 12.4 , constructs the canonical complete-trace quotient $G_{\beta\Sigma}$, proves the universal coarsest-factor property, and identifies it faithfully with the positive absolute-offset interval.
D3 successor-branch architecture authority	Supply the explicit ARC update required by the permanent D3 programme lock before 1.3 is used inside PCR-5E.	ARC-U1 and the target-update memorandum admit 1.3 as an additional tagged branch while preserving every 1.1 target, label, and readiness burden.
Absolute-scale base drift	Prove the common D1/D4 τ -continuum stays over one complete PCR-5B packet while the intrinsic vectorlike source is fixed, including matching, deletion, continuation, and failure.	E-S7A and E-S7, Theorems 10.5 and 10.6 , prove generatorwise matching factorization, equality of all seven packet coordinates, and injective D1/D4 enriched splitting over the fixed class $q_{m_0, \lambda}$.
Uniform product-box quantifier gap	Prove a Cartesian solved region, not merely a cell with nondegenerate λ -projection; keep source topology fixed away from $s = 0$.	E-S4A, Theorem 8.2 , factors every guard, proves uniform H/U/L/process margins on $I_m^o \times I_E \times I_s^E$, and chooses s_E only after the positive box is frozen.
Frozen matching-generator gap	Audit every primitive PCR-5A/PCR-5B matching generator rather than infer M^{match} -constancy from target chronology.	E-S7A, Theorem 10.5 , factors path, resolution, orientation, rank, threshold, continuation/failure, and provenance generators through the exact composite base reduct.

ARC-U1 and the nine E-S auxiliary closure theorems E-S1–E-S7, E-S4A, and E-S7A strengthen the authority and premises beneath the positive endpoint. They do not delete a branch, shrink the announced image, alter either continuum lower bound, change a D3/D4/D5 target tag, or convert a parent readiness boundary into elimination.

18.2 Target and dependency attacks

Attack	Faithful attack form	Required defeat
D3 target laundering	Prove the 1.3 acceptance semantics using only the 1.1 target, or give an isomorphism of the complete target fingerprints.	Defeat Theorems 2.3 and 2.5. Shared upstream objects or vocabulary do not suffice.
D4 summary laundering	Supply the missing total theory-to-correlated-likelihood evaluator with the complete covariance, nuisance, and acceptance semantics.	Discharge JLR-D4-1 and defeat Theorems 2.8 and 11.1.
Generic $B - L$ collapse	Give target isomorphisms preserving action, anomaly, protector, breaking, stabilizer, remnant, matching, and continuation data.	Defeat Theorem 2.9; omitting a field is target weakening.
Endpoint backfitting	Identify a target, source, or cross field introduced only after the D3/D5 Pass locus, joint image, or cardinality was known.	Defeat E-S3, E-S5, and E-S1, Theorems 2.3, 4.5, 4.7, 4.11 and 8.9.
Premetric backflow	Show that a numerical mass, Yukawa value, metric ratio, or later physical-nature interpretation authorizes joint membership or SCIC identity.	Defeat the dependency lock, target formation, and joint SCIC activation theorems.

18.3 Source and one-apex attacks

Attack	Faithful attack form	Required defeat
Naive marginal intersection	Exhibit a tuple counted as joint although it has no common source or no physical apex.	Defeat Theorems 6.5, 6.10 and 10.3.
Wrong-base tuple	Produce a counted joint object whose marginal restrictions lie in different PCR-5B components.	Defeat Theorems 5.2 and 7.3.
Vectorlike source substitution	Show that D3 and D5 use different source objects, or that the proof evaluates only the retained tester and not the complete H/U/L/process packet.	Defeat Theorems 8.1, 8.5 and 8.6.
D3 H/U/L failure	Give an applicable frozen H, U, or L evaluator and a point of the displayed solved interval whose exact transported response fails its accepted set or whose status is misrouted.	Defeat Theorems 8.3 and 8.5.

Attack	Faithful attack form	Required defeat
Source splicing	Show that a positive joint object uses different light spectra, vacua, charge ledgers, matching maps, backgrounds, or continuation objects in different legs.	Defeat Theorem 10.3.
Missing higher incidence	Provide a same-scope finite target-active relation absent from the semantic incidence trace and changing physical survival.	Defeat Theorems 4.5, 5.8 and 9.14.
D4 numerical-null laundering	Show a nonzero heavy D4 production, transport, injection, hidden/environmental, or stress-energy kernel that was discarded because it is small or heavy.	Defeat Theorems 9.10 and 10.3. The proof requires source-false guards and exact null/terminal objects.
D2 spectator breaking	Show that the accidental D2 amplitude is disconnected from the neutrino target path or creates an active $d \leq 4$ violation.	Defeat Theorems 9.7 and 10.3.
Exact-global zero by observation	Show that the D2 zero subfibre is inferred from the experimental null rather than the source protector.	Defeat Theorem 9.5; the source implication is one-way and exact.
Common-cone failure	Exhibit incompatible marginal projection arrows or absence of a common physical cone for a counted object.	Defeat Theorem 10.3.

18.4 Quotient, cardinality, and coordinate attacks

Attack	Faithful attack form	Required defeat
Base merger	Find distinct PCR-5B classes whose retained separator is erased by the joint extension.	Defeat Theorems 6.7 and 7.3.
Presentation-dependent class	Find a lawful source, gauge, basis, frame, or parameter arrow that toggles joint SCIC status.	Defeat Theorems 7.1 and 7.3.
ROOS as empty	Identify a non-RCF or otherwise out-of-response point counted as eliminated.	Defeat Theorems 6.10, 8.7 and 11.3.
Lower bound reported as exact	Show that $ Q \geq \mathfrak{c}$ is presented as a complete scalar cardinality or that isotropy/moduli are omitted.	Defeat Theorem 14.6.
Threshold injection failure	Produce $\lambda \neq \lambda'$ in I_E with lawfully equivalent retained threshold and oriented matching records.	Defeat Theorem 10.9.
ARC authority failure	Show that the 1.3 branch is used without a prior target-update certificate, or that ARC-U1 alters or supersedes the 1.1 programme.	Defeat Theorem 2.6.
Base-drift failure	Exhibit $\tau, \tau' \in I_\tau^\circ$ at fixed intrinsic (m_0, λ, s_E) for which one of the seven frozen fields of Ξ_{osc} differs.	Defeat E-S7A/E-S7, Theorems 10.5 and 10.6.

Attack	Faithful attack form	Required defeat
Product-box failure	Exhibit $(m_0, \lambda, s) \in I_m^o \times I_E \times I_s^E$ with a changed routing guard, activity pattern, or failed applicable H/U/L/process evaluator.	Defeat E-S4A, Theorem 8.2.
Matching-generator failure	Exhibit a primitive frozen matching generator that is not preserved by the exact D1/D4 composite base reduct or changes with τ at fixed intrinsic (m_0, λ, s_E) .	Defeat E-S7A, Theorem 10.5.
Continuum-fibre failure	Produce distinct τ values whose complete D1/D4 enriched continuation signatures coincide, or show that their fixed-base premise fails.	Defeat Theorems 10.6, 10.10 and 12.4.
Mass-generator overreach	Give an admitted common factor G determining both complete D1 and D4 traces but admitting no map onto $G_{\beta\Sigma}$.	Defeat the universal property in Theorem 12.4.
D2/D5 false common generator	Construct a two-way target-preserving factorization between the exact-global protector and the complete D2 response.	Defeat Theorem 12.5.
D3/D5 false dependence	Break one of the two target-faithful deformation directions while retaining the common target and apex.	Defeat Theorem 12.6.
Synergy by retuning	Show that a minimal-support result changes parameters, replaces a source, or switches target after deletion.	Defeat Theorems 13.2 to 13.4.

18.5 Trace, realization, and downstream interface attacks

Attack	Faithful attack form	Required defeat
Empirical tie-break	Use posterior preference, fit quality, naturalness, or later observation to choose one current joint class without target update.	Defeat Theorem 15.2.
Role set chosen from residue	Define the joint terminal role architecture only after seeing the survivor count.	Defeat Theorem 15.6.
Pointwise eligibility as realization	Claim closure from individually eligible classes lacking a globally compatible assignment.	Defeat Role-Occupancy Closure in Theorem 15.6.
Illegal automorphism quotient	Merge assignments while changing a target witness, source-composition invariant, or continuation signature.	Defeat the certificate-lawful automorphism definition.
Qualified-to-parent transport	Use the vectorlike-joint E-H data to assert a parent-neutrino physical-nature result.	Defeat Theorem 17.2 and the E-H scope tags.
Carrier identity from spectral charge	Infer one terminal carrier directly from connected spectral charge uniformity.	Defeat the D5 authority boundary and the PCR-6 gate.

18.6 Counterexample burden for the proper-continuum theorem

A counterexample to Theorem 10.12 must preserve one of the pre-frozen certificates $T_{E,VJ}^{T_{acc}^{(4)}}$ or $T_{E,VJ}^{T_{glob}}$, including its complete source domain, target records, applicability rules, D4 source-activity guards, continuation monoid, failure semantics, and lawful equivalence. It must then prove at least one of the following:

1. the solved source interval of E-S4 is empty or contains an applicable H/U/L/process failure;
2. the one-apex joint residual family does not vanish on the displayed accidental or exact-global object;
3. unequal λ values become lawfully equivalent despite distinct physical thresholds and oriented matching records;
4. unequal m_0 values have equal complete D1/D4 continuation signatures despite strict mass-sum monotonicity;
5. the D3 finite-unitary deficit continuum admits a valid 1.3 joint lift;
6. the codescent quotient identifies classes separated by a target-admitted joint witness.

A richer source, different D3 target, complete-D4 update, different D5 target, or later physical interpretation is a new certificate, not a defeat of the stated endpoint.

18.7 Scope of the robustness analysis

The proper-continuum theorem rests on an independently fixed source comparison class, a complete same-source D3/D5 realization, exact source-guarded D4 nullity, semantic saturation of the joint relation system, one-apex physicality, and the universal D1/D4 common-generator theorem. The unresolved parent boundaries remain explicit. A contrary same-scope conclusion must therefore supply one of the counterexamples specified above or invalidate an upstream theorem on which the joint construction depends.

19 Scope extensions and conditions for revision

The strong vectorlike-joint endpoint and the parent readiness boundaries have different revision rules. A new result may enlarge the scope or defeat a theorem, but it may not retroactively reclassify a current point without an explicit target or source update.

19.1 Established interfaces

The following interfaces are closed in the completed theorem chain at their stated scope:

1. ARC-U1 supplies the explicit architecture authority for the additional target-tagged D3-1.3 PCR-5E branch without superseding D3-1.1;
2. E-S3 fixes the complete raw precursor vectorlike-joint source classes before D3 acceptance, joint survival, image, or cardinality;
3. E-S1 and the joint source-normal-form theorem saturate every finite target-active cross relation from the semantic joint trace;
4. E-S4 lands the same D5 vectorlike source in the complete D3-1.3 H/U/L and process target and supplies the total marginal status decomposition;
5. E-S5 freezes the source-qualified child certificates from the independently proved D3/D5 marginal bridge before the joint D1/D2/D4 calculation;
6. the D3/D4 bridge uses source-coherent seed-active/off-seed-null activity and exact null/terminal off-seed objects, not numerical suppression;
7. E-S6 identifies the D1/D4 common generator as the canonical complete-trace quotient with a universal coarsest-factor property.

8. E-S7A proves matching-generator factorization and E-S7 proves channel-by-channel constancy of the complete PCR-5B packet along every fixed- (m_0, λ, s_E) , τ -varying conservative absolute-scale leaf and hence the fixed-base continuum foliation.

A faithful revision of one of these interfaces requires a same-scope counterexample:

- **E-S3:** a source, target, or continuation field used by the positive theorem but absent from the pre-evaluation fingerprint;
- **E-S1:** a finite target-active joint relation changing standing while absent from the complete semantic incidence trace;
- **E-S4:** an applicable H, U, L, process, retained-base, matching, continuation, or D5 evaluator that fails on the displayed solved interval;
- **E-S5:** a joint-image, fibre, standing-coordinate, cardinality, ROC, or PCR-6 datum used to define the source-qualified child certificate;
- **D4 activity:** a frozen SOE-D4-1 constitutive law that forces an off-seed vectorlike-heavy kinetic or stress-energy activity value to be true on the marked source;
- **E-S6:** an admitted common standing factor determining both complete D1 and D4 traces from which the canonical quotient does not factor.
- **E-S7:** two points on one fixed- (m_0, λ, s_E) , τ -varying conservative absolute-scale leaf separated by any frozen PCR-5B packet coordinate, or unequal τ -points not separated by the enriched D1/D4 continuation signature.

Absent such an object, ARC-U1 and the auxiliary closure theorems remain available downstream and the proper-continuum endpoint is unchanged.

19.2 Closing the parent D3-1.1 and D3-1.3 gaps

The original D3-1.1 parent branches reopen only through completion of the frozen Global Fibre Completion programme at its own target semantics. The required export is a source-complete atlas with exact fibres, overlaps, support, failures, and finality. A theorem using the 1.3 quantitative coverage rule or process tester cannot discharge this burden.

The D3-1.3 parent branches reopen beyond the vectorlike-joint source scope only through an exact joint landing of the remaining finite-operational source charts. Each chart must retain its actual Pass, Fail, N/A, ROOS, source-update, or target-update status. Failure to find a non-RCF survivor is not an emptiness theorem.

19.3 Closing JLR-D4-1

The complete D4 branches reopen only with a total evaluator

$$\mathcal{E}_{\text{JLR}} : \text{Theory}_{D4}^{\text{complete}} \longrightarrow \text{Observable}_{D4}^{\text{complete}} \longrightarrow \text{Likelihood}_{D4}^{\text{corr}}$$

that preserves target identity, covariance, nuisance treatment, approximation status, provenance, ordering, and background semantics. The scalar public band is not a substitute.

19.4 Closing the five unresolved D5 joint rows

D5 row	Required new PCR-5E result
Minimal background	A D3-varying one-apex family preserving the exact minimal chiral/background-anomaly packet and a converse D5-varying family preserving D3 closure, or a direct exact joint atlas replacing the deformation test.
Local-3N	A source-complete family preserving the faithful local action, one active right-handed state per family, covariant support, complete local anomaly packet, D3 response, D4 history, and all matching data.

D5 row	Required new PCR-5E result
Explicit breaking	A universal target-bearing neutrino-matched, retained-transparent explicit-breaking constructor, or a complete source-family atlas proving exact admitted and solved-empty strata.
Spontaneous breaking	One common action, breaking carrier, vacuum orbit, stabilizer, matching, and continuation packet compatible with D1–D4, together with exact fibre/cardinality closure.
Remnant	The spontaneous-breaking data together with faithful quotient, operator congruence, global-form/line/defect data, and exact preservation under D3/D4 continuation.

19.5 Optional enlargement of the positive scope

The accidental and exact-global vectorlike-joint solved strata are complete at their announced target-and-comparison-qualified scope. The following are enlargements, not additional conditions required for the current endpoint:

1. classify every source chart in the full pre-frozen vectorlike-joint comparison classes and compute the full overlap groupoid;
2. extend the joint construction over the complete D3 RCF-finite atlas;
3. resolve the non-RCF finite-operational remainder;
4. determine the exact moduli type and isotropy rather than only continuum lower bounds;
5. update the ROC target if a later complete joint theorem forces several terminal roles or an irreducible higher-arity bearer.

19.6 Trace updates and formalization boundary

A future empirical or structural trace enters only through an explicit target update carrying the new trace, enlarged witness family and continuation monoid, source/physical evaluation, restriction to the prior target, and a proof of preservation, splitting, or elimination under the new certificate.

The supplementary formal records and consistency checker certify scope, provenance, schema, theorem references, and record identity. They do not prove semantic source completeness, one-apex physicality, or the mathematical authority of a target evaluator.

19.7 Subsequent analysis

Two routes are licensed:

1. close the parent blockers to obtain a full-neutrino PCR-5E finality certificate; or
2. begin PCR-6 strictly on one of the two completed vectorlike-joint downstream data, preserving the target-and-comparison tag and making no universal parent-neutrino claim.

The first route is required for the unrestricted physical-nature endpoint. The second tests the carrier/spectral question on a rigorously completed subcategory without weakening or overwriting the parent programme.

20 Conclusion

PCR-5E completes the simultaneous stage of the neutrino carrier target-extension programme at the scopes for which the requisite source and response structure is available. The analysis does not intersect marginal survivor sets. It constructs a target-indexed simultaneous family, derives the complete cross-relation system from semantic joint uses, forms a common source before physical evaluation, imposes one-apex physicality, and reconstructs the joint standing-carrier quotient through obstruction equalizers and groupoid codescent.

The resulting theorem has two levels. At parent scope, the unresolved boundaries remain explicit: complete-D4 branches require JLR-D4-1, D3-1.1 requires Global Fibre Completion, the full D3-1.3 finite-operational source space contains an unresolved remainder, and five D5 joint comparisons retain target-specific open conditions. These statuses are not eliminations and do not license conclusions about physical nonexistence.

Within the two D3-1.3/public-summary vectorlike-joint comparison classes, however, the simultaneous problem closes exactly. A common positive light spectral source, the source-realized vectorlike Dirac threshold family, the source-guarded standard-thermal D4 history, and the relevant D5 charge structure coexist on one physical apex. The bounded-accidental sector retains a nondegenerate accepted D2 amplitude interval, whereas the exact-global sector restricts D2 to its defined zero-amplitude subfibre. Variation of the retained heavy threshold produces continuum many oscillation-base classes; the independently authorized common D1/D4 scale direction produces continuum many enriched lifts above each fixed vectorlike-joint base class; and the D3-1.3 finite-unitary deficit family supplies a continuum complement. Consequently,

$$\emptyset \subsetneq \text{im } \rho_{J,VJ}^{\tau_5} \subsetneq Q_{\text{osc}}^{\text{SCIC}}, \quad \left| \text{im } \rho_{J,VJ}^{\tau_5} \right| \geq \mathfrak{c},$$

$$\left| Q_{\text{osc}}^{\text{SCIC}} \setminus \text{im } \rho_{J,VJ}^{\tau_5} \right| \geq \mathfrak{c}, \quad \left| Q_{J,VJ}^{\tau_5} \right| \geq \mathfrak{c}, \quad \tau_5 \in \{T_{\text{acc}}^{(4)}, T_{\text{glob}}\}.$$

Neither completed quotient is singleton, and the one-role terminal Role-Occupancy Closure diagnosis is therefore over-residue.

The standing-coordinate analysis also separates several structures that are easily conflated. D1 and D4 descend through the canonical minimal generator $G_{\beta\Sigma}$ while remaining distinct target faces. D2 and exact-global $B - L$ are obstruction-coupled rather than commonly generated. The decided D3 active-closure and D5 symmetry coordinates are independent. No common generator for all admitted neutrino phenomena and no emergent mixed-carrier selector follows from the present joint target.

PCR-5E therefore determines a non-singleton joint carrier geometry at two exact target-and-comparison-qualified scopes while preserving the unresolved parent theory as an explicit open boundary. The subsequent carrier-identity analysis may study the physical nature and minimal organization of the completed joint classes, but it must retain their target tags, SCIC distinctions, and unresolved complements. In particular, no later interpretation may convert the proper-continuum result into a singleton by representative choice or by importing information absent from the joint certificate.

two exact simultaneous vectorlike-joint sectors have proper-continuum base images, continuum complements, continuum enriched fibres, and non-singleton joint standing-carrier quotients, while the complete parent neutrino joint problem remains explicitly incomplete.

A Notation and status calculus

A.1 Core objects

Symbol	Meaning
T_{osc}	Frozen neutrino oscillation target.
N_{osc}	Concrete PCR-5B pointed standing-core groupoid.
$Q_{\text{osc}}^{\text{SCIC}} = \pi_0 N_{\text{osc}}$	Exact oscillation SCIC quotient.
Ξ_{osc}	Seven-channel retained standing packet.
$\tau = (\tau_3, \tau_4, \tau_5)$	PCR-5E target branch.
T_E^τ	Typed simultaneous target for branch τ .
S_E^τ	Conservative joint source category.
PhysMod_E^τ	One-apex joint physical realization groupoid.
N_E^τ	Joint standing-core groupoid after physical reduction and lawful quotient.
R_J^τ	Joint restriction functor to N_{osc} .

Symbol	Meaning
ρ_J^τ	Descended map on connected-component quotients.
$\mathcal{L}_J^\tau(q)$	Connected joint lift fibre over base class q .
$\mathcal{Q}_J^\tau$	Final joint SCIC quotient.
$\mathcal{P}_{\text{marg}}^\tau(q)$	Raw common-base homotopy pullback of marginal fibres.
$\mathcal{O}_E^\tau(q)$	Total obstruction/status groupoid.
$\Sigma_E^\tau(W)$	Complete joint continuation signature.
I_ν^τ	Phenomenon-to-standing-coordinate incidence relation.
$N_{E,W}$	No-repair source-sufficiency function.
$\text{MinSupp}_E(W)$	Minimal-support antichain.
$\mathcal{R}_E^\tau$	Target-derived joint role architecture.
$\Sigma_E^\tau/\Gamma_{E,\tau}^{\text{lawful}}$	Compatible role-assignment orbit object used by ROC.

A.2 Marginal labels

Label	Meaning
D1	Direct kinematic mass and endpoint spectral response.
D2	^{76}Ge LNV search response.
D3-1.1	Original heavy-state/active-closure/CLFV target.
D3-1.3	Quantitative successor active-closure target.
D4 public	Public-95 scalar summary plus standard-thermal source/history.
D4 complete	Complete correlated cosmological target, currently JLR-D4-1 NotReady.
$T_{\text{acc}}^{(4)}$	Accidental $B - L$ conservation through active dimension four.
$T_{\text{bg,min}}$	Minimal chiral background/anomaly packet.
$T_{\text{loc},3N}$	Faithful unbroken local $U(1)_{B-L}$ with one active RH neutrino per family.
T_{glob}	Exact global action and continuation-stable protector.
$T_{\text{exp}}[O]$	Target-bearing explicit-breaking target indexed by operator/process O .
$T_{\text{sp}}[q\Phi]$	Spontaneous-breaking target with faithful breaking charge.
$T_{\text{rem}}[\Gamma, q\Phi]$	Faithful remnant target.

A.3 Status values

Status	Meaning
N/A	Residual is inapplicable by frozen target/source typing.
Pass	Applicable residual evaluates positively.
Fail	Applicable residual evaluates negatively; a physical/source obstruction is proved.
Undef	Required in-scope value is not defined.
ROOS	Semantic point is meaningful but outside the declared response evaluator scope.
SU	A new source certificate or source grammar is required.
TU	A new target certificate is required.
NotReady[κ]	Named theorem, evaluator, atlas, overlap, continuation, or finality component is incomplete.
SolvedEmpty	Every source presentation is covered and has a certified applicable failure.
TargetQualified	A complete theorem holds only on the named weaker target or subcategory.
OverResidue	Live terminal carrier residue is larger than the target role set.
UnderResidue	Target roles cannot all be occupied.

Status	Meaning
IncompatibleResidue	Cardinality/eligibility may pass but no globally compatible assignment exists.
Licensed	Compatible assignment orbit is singleton.

A.4 Coordinate relations

Relation	Definition
IndependentCoordinate	Two target-faithful deformation directions exist, each varying one separator while holding the other fixed; no functional factorization in either direction.
CommonGenerator	Several separators factor through one target-authorized, standing-bearing, minimal joint invariant.
ObstructionCoupled	One separator constrains the passing values of another but no two-way common-generator factorization exists.
AlignedSelectionDistinct	Targets have the same or nested base support for distinct standing reasons.
EmergentJointSynergy	A standing invariant or indispensable higher-arity relation appears only on the simultaneous target.
NotEstablished	No positive theorem and no readiness gap requiring an unknown source chart at the announced subcategory.
NotReady	At least one required deformation, source, physical, matching, or closure component is missing.

B Target fingerprints and branch classification

B.1 Fingerprint schema

Every branch catalogue contains the following fields:

1. `branch_id`;
2. D3 target tag and process scope;
3. D4 target tag and evaluator status;
4. D5 target tag and target-specific parameters;
5. PCR-5B/B15 source/base version;
6. marginal source-extension identifiers;
7. applicability and acceptance-rule versions;
8. uncertainty/covariance object identifiers;
9. continuation monoids;
10. target/source update boundaries;
11. theorem authority status and parent/subcategory relation;
12. D3 successor-branch ARC authority (ARC-U1), permanent no-supersession, and no-backward-transport flags;
13. where a vectorlike-joint child is claimed, the E-S7 fixed-base foliation certificate and all seven base-packet equalities.

B.2 Canonical branch identifiers

The machine identifier is

$$E_D3\{11|13\}_D4\{PUB|COMP\}_D5\{ACC|BG|LOC|GLOB|EXP|SP|REM\}.$$

For example,

$$E_D313_D4PUB_D5GLOB$$

denotes the D3-1.3/public-summary/exact-global branch.

B.3 Branch classification table

Branch family	Target formed	Evaluator ready	Finality status
D3-1.1 / D4 complete / any D5	Yes	No	NotReady JLR-D4-1; D3 GFC also open.
D3-1.3 / D4 complete / any D5	Yes	No	NotReady JLR-D4-1.
D3-1.1 / D4 public / any D5	Yes	Yes at marginal level	NotReady D3-1.1 Global Fibre Completion.
D3-1.3 / D4 public / accidental	Yes, under ARC-U1	Yes	Target-and-comparison-qualified vectorlike-joint proper-continuum; E-S7 fixed-base continuum fibres; parent ROOS/unclosed.
D3-1.3 / D4 public / exact global	Yes, under ARC-U1	Yes	Target-and-comparison-qualified vectorlike-joint proper-continuum; E-S7 fixed-base continuum fibres; parent ROOS/unclosed.
D3-1.3 / D4 public / minimal background	Yes	Yes	NotReady D3–D5 background deformation.
D3-1.3 / D4 public / local-3N	Yes	Yes	NotReady local action/anomaly preservation.
D3-1.3 / D4 public / explicit	Yes	Yes	NotReady universal target-bearing transparent family.
D3-1.3 / D4 public / spontaneous	Yes	Yes	NotReady common action/-vacuum/stabilizer/matching.
D3-1.3 / D4 public / remnant	Yes	Yes	NotReady faithful quotient/-congruence/continuation.

B.4 ARC authority and fixed-base foliation fields

Every D3-1.3 branch catalogue carries the authority tuple

$$(\text{ARC-U1}, \text{AdditionalBranch}, \text{NoSupersession}_{1,1}, \text{NoBackwardTransport}).$$

Every vectorlike-joint child certificate additionally carries

$$\text{BaseLeaf}_{m_0, \lambda} = (q_{m_0, \lambda}, I_T^\circ, T^{\text{face}}, R^{\text{stand}}, \Delta^{\text{del}}, C^{\text{act}}, M^{\text{match}}, B^{\text{bdy}}, F^{\text{same}}),$$

with equality of all seven packet coordinates across the complete τ -leaf and injective enriched D1/D4 separation on that same leaf. Missing authority or a failed coordinate equality returns **NotReady**; it is never converted into elimination.

B.5 No-equivalence certificates

The following pairs have explicit non-equivalence certificates:

- D3-1.1 versus D3-1.3: acceptance semantics and process scope differ;
- D4 public versus complete: evaluator and covariance geometry differ;
- D5 accidental versus exact global: bounded operator predicate versus protector;
- D5 explicit versus spontaneous: active violation versus invariant pre-breaking source and vacuum orbit;
- D5 spontaneous versus remnant: breaking/stabilizer versus faithful quotient and operator congruence.

These certificates are used by E07 to reject false path equivalence.

C Complete residual normal form

This appendix gives the canonical residual schema used by the joint obstruction atlas. It is a typed scientific schema, not an implementation-specific table.

C.1 Residual record

Each residual record is

$$r = (\text{id}, \text{sector}, \text{domain}, \text{guard}, \text{equation}, \text{codomain}, \text{status}, \text{firstFail}, \text{update}, \text{provenance}).$$

A residual may refer to a primitive source equation, a physical existence condition, a covariance relation, an acceptance predicate, a continuation square, or a common-cone requirement.

C.2 Residual sectors

Sector	Representative residuals	Failure meaning
Base	Oscillation channel, response multiplicity, boundary, gauge, anomaly, vacuum, base matching	The retained PCR-4 physical core is not realized.
D1	Endpoint support, response, nuisance, acceptance, source envelope, transparency	Direct-kinematic target fails or source/response is ill-formed.
D2	Source amplitude, physical contraction, nuclear/detector frame, rate, acceptance, transparency	LNV response leg fails at source, physical, or detector/statistical level.
D3	Direct heavy response, active tester, universality, CLFV, process scope, matching	D3 target fails or is response-out-of-scope.
D4	Population, collision/decay, background, stress-energy, perturbations, summary band, likelihood	D4 source/history/response fails or complete evaluator is unavailable.
D5	Action, charge lattice, anomaly, protector, violation, breaking, stabilizer, remnant, matching	D5 target-specific source or physical condition fails.
Mass-population	Shared light masses, D1 measure, D4 mass sum/population	Marginal legs use inconsistent spectral source data.
LNV-charge	D2 amplitude versus D5 action/breaking/remnant	A target-bearing violation has the wrong charge status.

Sector	Representative residuals	Failure meaning
Threshold-kinetic	D3/D5 heavy states versus D4 history	An allegedly off-support state has an enabled D4 kinetic or stress-energy incidence.
Rep./anomaly	Shared charge/representation and anomaly ledger	Marginals use incompatible quantum numbers or anomaly records.
Vacuum/background	D4 background versus D5 breaking/stabilizer and common deformation branch	Marginal structures do not coexist on one vacuum/background.
Matching/continuation	Shared thresholds, EFT maps, RG transport, oriented matching, continuation squares	Marginal records cannot be transported on one physical continuation.
Higher arity	Three-way mass/charge/population, source/boundary/matching, or action/vacuum/continuation relations	Pairwise pass does not extend to the full relation.
Common cone	One physical apex and compatible projection arrows	Marginal witnesses are spliced from different physical objects.

C.3 Ordered first-failure rule

The default ordered first-failure priority is:

$$\begin{aligned} &\text{TargetTyping} \prec \text{SourceCoherence} \prec \text{Activity} \prec \text{BasePhysicality} \prec \text{MarginalPhysicality} \\ &\prec \text{CrossResidual} \prec \text{Acceptance} \prec \text{Continuation} \prec \text{CommonCone} \prec \text{Finality}. \end{aligned}$$

The order localizes the earliest dependency failure. It does not erase later failures; the full residual vector is retained in the ledger.

C.4 Solved-empty certificate

A fibre $\mathcal{L}_J^\tau(q)$ has a solved-empty certificate only if:

1. the source grammar covers every source presentation over q at the declared scope;
2. every source chart and overlap is dispositioned;
3. every chart has an applicable residual with status Fail;
4. no chart or overlap is Undef, ROOS, SU, TU, or NotReady;
5. the failure certificates are stable under lawful presentation changes and base transport.

C.5 Vectorlike-joint residual values

Residual	Accidental VJ certificate	Exact-global VJ certificate
D1/D4 mass identity	Pass by common M_{light}	Pass by common M_{light}
KATRIN scalar bound	Pass by $m_\beta \leq \Sigma_\nu \leq \Sigma_{95}^{\text{pub}}$	Same
D2 empirical acceptance	Pass for $z \in I_{76}$	Pass at defined zero $z = 0$
D2/D5 charge relation	Pass: source dimension above four	Pass: exact amplitude zero and protector intact
D3 active tester	Pass: $c_E^2 = 1 - \delta_E > 0.987$	Same
D3/D4 activity relation	Pass by exact source-guarded off-seed nullity	Same

Residual	Accidental VJ certificate	Exact-global VJ certificate
D4 kinetic saturation	Pass: exactly three light modes	Same
D5 target	Pass through bounded accidental predicate	Pass through faithful action/protector
D4/D5 vacuum covariance	Pass on common unbroken branch	Pass on common unbroken branch
Base packet along fixed- λ absolute-scale leaf	Pass in all seven PCR-5B coordinates by E-S7	Same
Base matching on conservative absolute-scale leaf	Pass by Theorems 10.5 and 10.6	Same
Common matching	Pass on one oriented matching packet	Pass on one oriented matching packet
Common cone	Pass by vectorlike-joint amalgamation	Pass by vectorlike-joint amalgamation

D Supplementary formal records and reproducibility

The accompanying supplementary materials provide formal representations of the target family, source grammar, simultaneous residual system, solved fibre geometry, standing-coordinate relations, cardinality results, Role-Occupancy data, and downstream PCR-6 interface. Their purpose is reproducibility and provenance. The mathematical authority of the paper remains in the definitions, propositions, and theorems of the main text.

D.1 Supplementary records

Record	Scientific content
Target classification	Twenty-eight parent branch fingerprints, the raw vectorlike-joint precursor classes, the source-qualified comparison classes, and all target-specific completeness boundaries.
Marginal target data	Versioned D1–D5 target, source, restriction, and authority information used by the joint construction.
Semantic saturation	The E-S1 target-facing trace, finite-incidence saturation, lawful transport structure, and update boundary.
Joint source structure	The activity-reduced source grammar, source reducts, cross-relation types, and D4 activity guards.
D3/D5 vectorlike realization	The common vectorlike source coefficients, H/U/L/process statuses, exact CLFV cancellation, D5 conditions, strict margins, and typed complement.
Simultaneous residual system	Residual sectors, status codomain, first-failure order, source-guarded D4 values, and solved vectorlike-joint values.
Joint fibre geometry	The two solved vectorlike-joint subgroupoids, parameter domains, continuum injections, overlap data, and typed complements.
Source composition	Source blocks, deletion results, minimal supports, and no-repair composition dispositions.
Standing-coordinate structure	Target-tagged separators, the D1/D4 common-generator factorization, D3/D5 independence, and D2/D5 obstruction relation.
Cardinality and Role-Occupancy	Image and quotient lower bounds, moduli type, continuum complement, joint role sets, lawful automorphisms, and over-residue conclusions.

Record	Scientific content
Base conservativity	The seven-channel fixed-base theorem, the strict-margin product region, and factorization of the frozen oscillation matching generators.
Downstream PCR-6 interface	The E-H1–E-H10 data for the two completed target-and-comparison-qualified sectors and the exact unresolved parent boundaries.

D.2 Consistency checks

The supplementary consistency checks verify that the distributed records preserve the branch identifiers, target separations, theorem dependencies, typed status distinctions, continuum lower bounds, fixed-base equalities, and downstream scope tags stated in the manuscript. They also verify the integrity of the supplementary files.

These checks are deliberately subordinate to the scientific argument. Successful verification does not establish semantic source completeness, physical existence of a common apex, completeness of the target witness family, or the external mathematical results used in the descent arguments. Those claims are established only by the theorem chain and its cited dependencies.

E Source and dependency index

The following support index records the manuscript title, role, and authority boundary used in PCR-5E. The bibliography keys are shown in parentheses.

ID	Support manuscript	PCR-5E role and authority boundary
S1	<i>Neutrino Carrier Target-Extension and Joint-Closure Mini-ARC: Detailed ARC Theorem Skeleton (miniARC)</i>	Original E01–E15 architecture and endpoint neutrality. Planning authority, not completed proof.
S2	<i>Neutrino Carrier Target-Extension Mini-ARC Refinement Addendum After PCR-5D1 (refine)</i>	E-R1–E-R3 coordinate diagnostics and finality conditions.
S3	<i>Functorial Target Extension of Standing-Carrier Quotients (pcr5c)</i>	Primary target/fibre/no-merger/composition/joint calculus.
S4	<i>Exact Standing-Carrier Stratification of the Neutrino Oscillation Realization Space (pcr5b)</i>	Frozen base atlas, B15 frames, groupoids, overlaps, and continuum quotient.
S5	<i>Direct Kinematic-Mass Target Extension of the Neutrino Standing-Carrier Space (d1)</i>	D1 source envelope, response, full-base continuum, and absolute support standing.
S6	<i>Lepton-Number-Violating Target Extension of the Neutrino Standing-Carrier Space (d2)</i>	D2 exact amplitude, defined zero, one-apex attachment, continuum fibres, and failure status.
S7	<i>Heavy-State, Active-Closure, and Charged-Lepton-Flavour Extension of the Neutrino Standing-Carrier Space (d311)</i>	Original D3-1.1 continuum survivor and explicit limits; global completion remains open.
S8	<i>Successor-Target Quantitative Active Closure of the Neutrino Standing-Carrier Space (d313)</i>	D3-1.3 proper image, deficit complement, RCF-finite closure, and ROOS remainder.
S8a	<i>PCR-5D3 Programme Separation Memorandum: THUCL-D3-1.1 Global Fibre Completion and the TU-D3-1.2/1.3 Successor-Target Line (d3sep)</i>	Permanent programme lock and explicit requirement for a later PCR-5E ARC update.
S8b	<i>PCR-5E D3 Successor-Branch ARC Target-Update Memorandum (arcupdate)</i>	ARC-U1 authority admitting 1.3 as an additional PCR-5E branch without superseding 1.1.

ID	Support manuscript	PCR-5E role and authority boundary
S9	<i>Cosmological Population and Transport Extension of the Neutrino Standing-Carrier Space (d4)</i>	Public-summary full-base continuum, standard-thermal source/background, D1/D3 tags, and JLR-D4-1 boundary.
S10	<i>Symmetry and Charge Extensions of the Neutrino Standing-Carrier Space (d5)</i>	Seven D5 targets, exact images, D3 comparison, bearer theorem, and NotReady rows.
S11	<i>Target-Fixed Physical Role Necessity for the Realized Neutrino Oscillation Interface (pcr1)</i>	Frozen carrier-neutral role complex; no microscopic or carrier-count import.
S12	<i>Complete Physical Interface of the Neutrino Propagation Role Complex (pcr2)</i>	Complete target-facing trace and shared-incidence semantics.
S13	<i>Source-First Exhaustion of the Physical Carrier Universe (pcr3)</i>	Source-first grammar, activity reduction, and candidate-completeness scope.
S14	<i>Simultaneous Physical Constraint Collapse of the Neutrino Carrier Universe (pcr4)</i>	One-object/common-cone physical survival and residual completeness.
S15	<i>Compatible Role-Occupancy Closure for the Neutrino Propagation Role Complex (pcr5)</i>	Internal/terminal separation and pre-extension over-residue diagnosis.
S16	<i>Standing-Carrier Upper Closure for the Neutrino Propagation Role Complex (pcr5a)</i>	Seven-channel packet, deletion standing, four marked classes, and packet-image identity.
S17	<i>Standing Carrier Identity Closure (scic)</i>	Unique same-scope continuation-signature carrier quotient and activation gate.
S18	<i>Role-Occupancy Closure under Admissibility (roc)</i>	Compatible assignment orbit criterion and target-derived role discipline.
S19	<i>A Fixed-Domain Exhaustion Theorem (fixed)</i>	Same-scope typing, no repair, canonical quotient, and no-fifth-locus discipline.
S20	<i>Constraint-Induced Physical Carrier Realization: Detailed ARC Plan (pcrarc)</i>	Programme order, anti-bias freeze, and completion architecture.
S21	<i>Mass-Matrix, Eigenstructure, and Ratio Rigidity over the Canonical Coupling Carrier (mgn5)</i>	Prevents mass matrices/eigenvalue labels from generating carrier standing.
S22	<i>Mixing-Angle and Phase Standing over Independently Standing Mass Carriers (mgn6)</i>	Boundary reconstruction and no mixing standing from basis notation.
S23	<i>Neutrino Ordering Prediction-Lock Closure under Observable Transport (aot8)</i>	Trace-versus-separation and no pre-lock ordering selector.
S24	<i>The Neutrino-Mass Low-Energy Exact-Character Endpoint (exactchar)</i>	Separate low-energy target and no selector import.
S25	<i>The Higgs Bridge Carrier: Kernel Consequences for Electroweak Mass-Response (higgs1)</i>	Premetric bridge/carrier priority and representative insufficiency.
S26	<i>Endpoint on Higgs Structural Identity (higgs2)</i>	Supplementary only.
S27	<i>No Primitive Higgs Scale (higgs3)</i>	Identity through carrier occupation; Yukawa/mass descriptions are downstream.
S28	<i>Higgs Vacuum Route Status (higgs4)</i>	Carrier-indexed scale and downstream metric projection.
S29	<i>Higgs Extension Classifier (higgs5)</i>	Route availability versus standing closure and target-update discipline.
S30	<i>The Higgs Domain Endpoint (higgs6)</i>	Same-scope extension preservation and extra-scalar nonidentity.
S31	<i>Carrier Identity versus Spectral, Conjugation, Closure, and Transport Standing (handoff)</i>	No residual primitive Higgs attribute outside carrier architecture.
		D1–D4 transition data and pre-D5 carrier/spectral authority boundary.

ID	Support manuscript	PCR-5E role and authority boundary
S32	<i>PCR-5E Joint Neutrino Physics Carrier Closure: Detailed Theorem Skeleton</i> (pcr5eskel)	Detailed specification of E00–E15, E-R1–E-R3, the branch matrix, and E-H1–E-H10. Planning authority only; the proofs are supplied in the present manuscript.

F Theorem dependency index

F.1 Main theorem ladder

ID	Theorem	Principal imports	Primary output
E00	Target/version/import lock	Mini-ARC, theorem plan, D3/D4/D5 downstream interfaces, fixed-domain	Branch certificate and no-laundering rule.
ARC-U1	D3 successor-branch ARC authorization	D3 programme-separation memorandum and successor target certificate	Additional tagged 1.3 PCR-5E branch with permanent 1.1 preservation.
E-S1	Semantic joint-use saturation	PCR-2/PCR-3/PCR-4 trace completeness	Complete finite target-active cross grammar.
E-S2	Joint source normal form	E-S1 and marginal source normal forms	Complete activity-reduced joint source category.
E-S3	Raw VJ precursor freeze	D5 exact-response source grammar, D1/D2/D4 downstream interfaces	Pre-response raw target-and-comparison fingerprints.
E01	Joint-target formation	PCR-5C target category, marginal targets	Typed target colimit and cross signature.
E02	Common-base pullback	PCR-5B/B15, marginal restrictions	Raw common-base tuple groupoid.
E-S4A	Uniform vectorlike D3/D5 strict-margin product box	D3 finite response registries, D5 exact-response source, compact light interval	Cartesian Pass region with positive activity and fixed routing.
E-S4	D3-1.3/D5 common source landing	D3 H/U/L/process target, D5 exact-response family	Complete same-source marginal bridge and total typed status decomposition.
E-S5	Source-qualified VJ freeze	E-S3 and E-S4	Pre-joint target-and-comparison child certificates.
E03	One-point simultaneous survival	PCR-3, PCR-4, marginal physical downstream interfaces	Joint source and physical groupoids.
E04	Marginal-to-joint obstruction	PCR-4 status semantics, cross residuals	Total obstruction atlas and solved-empty rule.
E05	Joint restriction/no merger	PCR-5B, SCIC, transparency	Restriction maps and no-merger theorem.
E06	Exact joint fibres/codescent	PCR-5B/PCR-5C atlas method	Joint fibre atlas and quotient/decomposition.
E07	Path independence	PCR-5C composition/path equivalence	Endpoint-equivalent path certificate.
E08	Joint source synergy	PCR-5C source-sufficiency calculus	Minimal-support antichain and dispositions.
E09	Emergent mixed selection	Marginal supports, cross residuals	NotEstablished/NotReady status.

ID	Theorem	Principal imports	Primary output
E10	Base selection	Joint lift support	Branch-specific image/readiness matrix.
E11	Enriched singleton	PCR-5C full selection, SCIC	Necessary and sufficient singleton criterion.
E12	Cardinality/moduli	Codescent atlas	Mixed-continuous VJ classification and parent readiness.
E13	Trace versus structure	SCIC route exhaustion, AOT	No-selector and update theorem.
E14	Realization licensing	ROC, joint role architecture	VJ over-residue and parent NotReady.
E15	PCR-6 downstream interface	Entire E chain	E-H1–E-H10 solved-subcategory and parent certificates.
E-R1	Standing-coordinate incidence	D-series separators	Invariant incidence object.
E-R2	Independent factorization	PCR-5C factorization, cross equations	Conditional D3/D5 factorization and full-product failure.
E-S6	D1–D4 canonical common generator	Complete D1/D4 continuation traces	Universal minimal common-response quotient.
E-S7A	Frozen oscillation matching-generator factorization	PCR-5A primitive matching generators, PCR-5B normal forms, E-S4A product box	Complete matching channel factors through the exact composite base reduct.
E-S7	Vectorlike-joint absolute-scale base conservativity	PCR-5B seven-channel packet, E-S4/E-S5 source family, D1/D4 exact reducts	Fixed- (m_0, λ, s_E) , τ -indexed continuum inside one base SCIC fibre with enriched scale splitting.
E-R3	Common generator/synergy	D1–D5 diagnostics	D1/D4 common generator, D2/D5 obstruction, D3/D5 independence.

F.2 New local scientific theorems

Theorem	New PCR-5E content
Uniform strict-margin product box	Proves a Cartesian solved region over the full positive light interval, a nondegenerate threshold interval, and a positive activity interval; no projection-only or $s = 0$ topology argument remains.
Vectorlike source landing	Gives one exact D3-1.3/D5 source, complete H/U/L/process status, exact CLFV cancellation, retained base, matching, and target-faithful threshold family.
Frozen matching-generator factorization	Audits every path, retained-resolution, orientation, rank, threshold, continuation/failure, and provenance generator of M^{match} and proves exact composite-base-reduct factorization.
Absolute-scale base foliation	Proves all seven base-packet channels are constant under τ at fixed intrinsic (m_0, λ, s_E) , while D1/D4 enriched traces separate the points.
VJ total status decomposition	Partitions the pre-frozen source scope into Pass, Fail, N/A, ROOS, source-update, and target-update strata without survivor-defined target formation.
D1–D4 mass-response factorization	Proves a common response quotient and KATRIN acceptance from the D4 positive mass-sum interval.

Theorem	New PCR-5E content
Exact-global D2 zero subfibre	Derives the physical D2 zero from the exact-global protector.
Accidental D2 compatibility	Shows higher-dimensional LNV can vary while bounded active $d \leq 4$ accidental standing remains fixed.
D3–D4 source-guarded relative extension	Proves exact three-light-mode D4 history with every vectorlike-heavy D4 activity guard false; no mass or small-coupling nullity is used.
Vectorlike-joint one-apex amalgamation	Constructs the simultaneous D1–D5 physical objects on the accidental and exact-global solved strata.
Vectorlike-joint proper-continuum closure	Proves continuum image, continuum complement, continuum fibres, and non-singleton quotient.
D1–D4 canonical common generator	Proves the universal coarsest-factor property of $G_{\beta\Sigma}$.
Full-product factorization failure	Shows why coordinate independence does not yield a five-leg Cartesian fibre.
Parent PCR-5E NotReady theorem	Exhausts all twenty-eight branches and names at least one blocker in each.

F.3 Location of principal results

Result	Location
Target family and semantic saturation	Sections 2, B and 4.
Marginal target data	Section 3.
D3/D5 vectorlike realization	Section 8.
Joint source theory	Section 5.
Residual and obstruction structure	Sections C and 6.
Vectorlike-joint solved physical families	Section 10.
Standing-coordinate incidence and common generator	Section 12.
Source composition	Section 13.
Selection and cardinality	Sections 11 and 14.
Role-Occupancy and realized-trace status	Section 15.
Global endpoint theorem	Section 16.
Downstream PCR-6 interface	Section 17.
Robustness and counterexample analysis	Section 18.
Conditions for scope revision	Section 19.

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